**ON A/C ALL

21–21–00–001

AIR DISTRIBUTION, FLIGHT COMPARTMENT

Introduction

Conditioned air is supplied to the flight compartment to maintain a comfortable environment. Conditioned air for the flight compartment is also used for side window demisting and aircraft pressurization.

General Description

Air supply to the flight compartment is ducted from the air conditioning pack, through the rear pressure bulkhead, to the left side of the splitter duct. From the splitter duct, the flight compartment air is ducted along the right side of the aircraft below the cabin floor. Before reaching the flight compartment, the distribution system also supplies conditioned air to the aft baggage compartment inlet, forward lavatory gasper and the forward flight attendant's gasper.

At the flight compartment bulkhead, a wye duct splits the airflow into two individual but identical distribution systems, one for the pilot side and the other for the copilot side. In the flight compartment, the distribution system has lower level and upper level outlets. The upper level outlets are demist nozzles for the pilot's and copilot's side windows. The lower level outlets include a foot warming piccolo tube (near the rudder pedals), a fixed grille near knee height and a large torso gasper. Flow control levers located near the window sill allow the pilot and copilot to control the flow of air into the flight compartment. A small manually controlled gasper at window height is also supplied.

The Environmental Control System (ECS) Electronic Control Unit (ECU) monitors a supply duct temperature sensor, an overtemperature switch and a flight compartment temperature sensor as part of the control loop (refer to SDS 21–61–00).

Detailed Description

Refer to Figures 1 and 2.

Refer to Figure 3.

The air conditioning pack supplies conditioned air through its supply ducting to a fitting in the rear pressure bulkhead. The fitting is a 7 in. (177.8 mm) (diameter aluminum tube divided down the middle (bifurcated). The left side of the bifurcation supplies flight compartment air while the right side supplies cabin air. Forward of the pressure bulkhead, the flight compartment and cabin flows are separated by the splitter duct. The splitter duct is a semi–rigid bifurcated duct made from silicone impregnated glass cloth. It has a 7 in. (177.8 mm) diameter inlet and two outlets. A 4 in. (101.6 mm) outlet on the left side supplies flight compartment air and a 7 in. (177.8 mm) outlet on the right side supplies cabin air.

The 4 in. (101.6 mm) supply duct routes the conditioned air from the splitter duct to the flight compartment bulkhead. The duct is located below the cabin floor on the right side of the aircraft. In addition to supplying the flight compartment, the 4 in. (101.6 mm) diameter duct supplies the aft baggage compartment inlet and the forward lavatory and flight attendant gaspers.

A 2 in. (50.8 mm) diameter duct branches from the main 4 in. (101.6 mm) diameter flight compartment duct at station X743 and supplies the aft baggage compartment inlet. Two solenoid valves are installed; one at the inlet to the baggage compartment and the other at the outlet grille. These solenoid valves control the ventilation of the baggage compartment (refer to SDS 26–12–00).

Aft of the flight compartment bulkhead at station X–18, a 1.5 in. (38.1 mm) duct branches off from the main 4 in. (101.6 mm) duct. This duct then divides into a .75 in. (19.05 mm) duct which supplies the forward lavatory gasper and an 1 in. (25.4 mm) duct which supplies the forward flight attendant gasper.

On aircraft with Modsum 4–459190 incorporated, between the stations X564.500 and X747.602, the upper outlet duct assembly, the air systems orifice assembly (2.00 in. (50.8 mm) diameter duct), and the galley – flight attendant supply duct assembly are removed. The aft baggage compartment duct assembly and aft baggage compartment inlet and outlet are installed.

Refer to Figures 4 and 5.

Aft of the flight compartment bulkhead (station X39) on the right side of the aircraft, , the 4 in. (101.6 mm) diameter supply duct divides into two 3 in. (76.2 mm) diameter ducts. These 3 in. (76.2 mm) ducts penetrate the bulkhead and supply the left and right side flight compartment distribution systems. The left and right side distribution systems in the flight compartment are identical to each other.

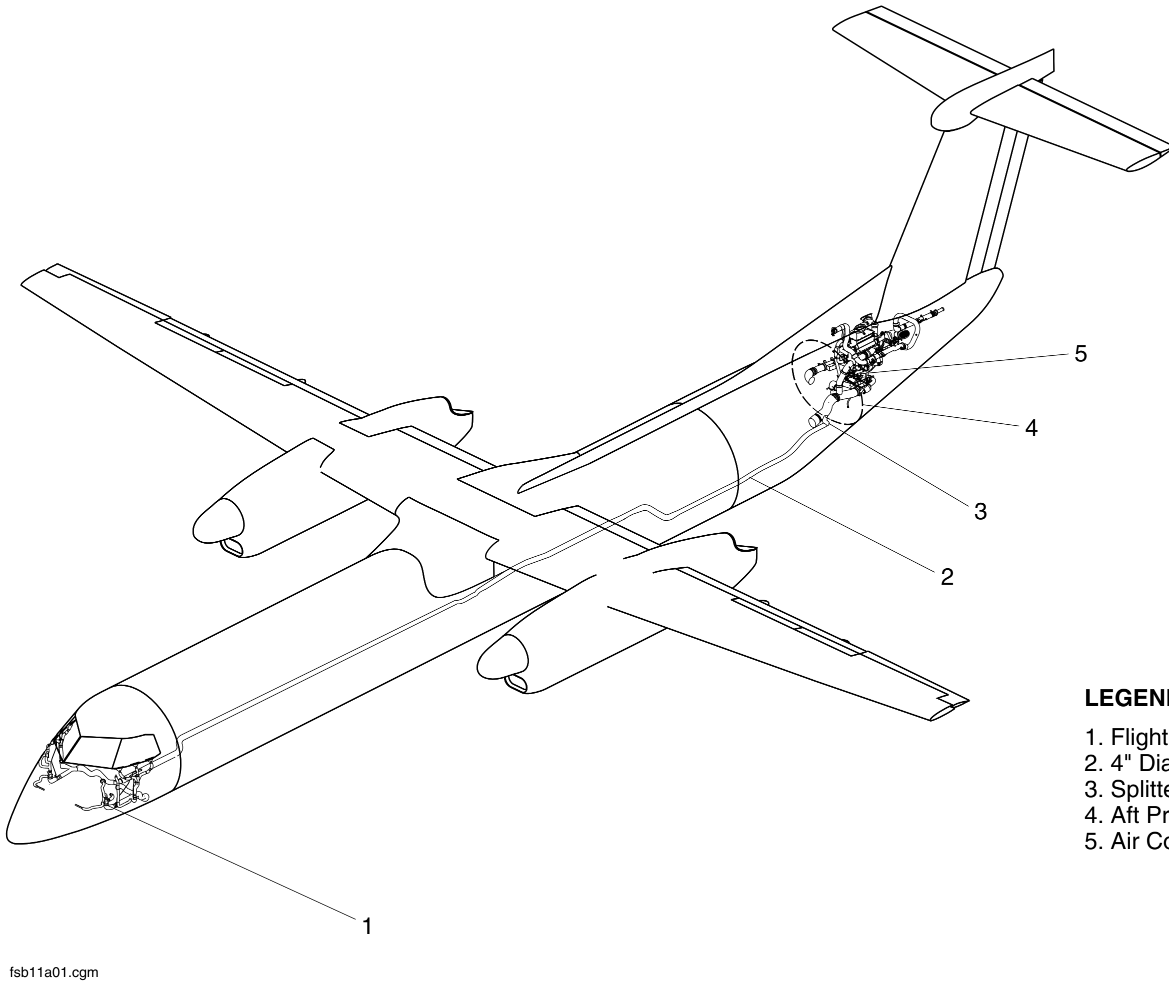
Refer to Figures 6 and 7.

Forward of station X54, a 2 in. (50.8 mm) diameter duct branches off from the 3 in. (76.2 mm) duct to supply the side window demist system. A manual flow control valve is installed in this 2 in. (50.8 mm) duct at floor level. A flow control lever located at shoulder height regulates the quantity of air flowing through the valve. The airflow from the 2 in. (50.8 mm) control valve is then directed to the side windows through three demist nozzles installed at the window sill level.

Between stations X54 and X69.4, a 1 in. (25.4 mm) diameter duct branches off from the 3 in. (76.2 mm) supply duct. This 1 in. duct supplies a shoulder height small gasper. The small gasper must be controlled at its outlet since the 1 in. (25.4 mm) supply duct is not regulated by the flow control valve.

A 3 in. (76.2 mm) diameter flow control valve is installed below the flight compartment floor forward of station X69.4. This valve controls the flow of air to the torso gasper, fixed sidewall grille duct and the foot warming piccolo tube.

A dedicated flow control lever located at shoulder height regulates the quantity of air flowing through the 3 in. (76.2 mm) valve to these outlets.



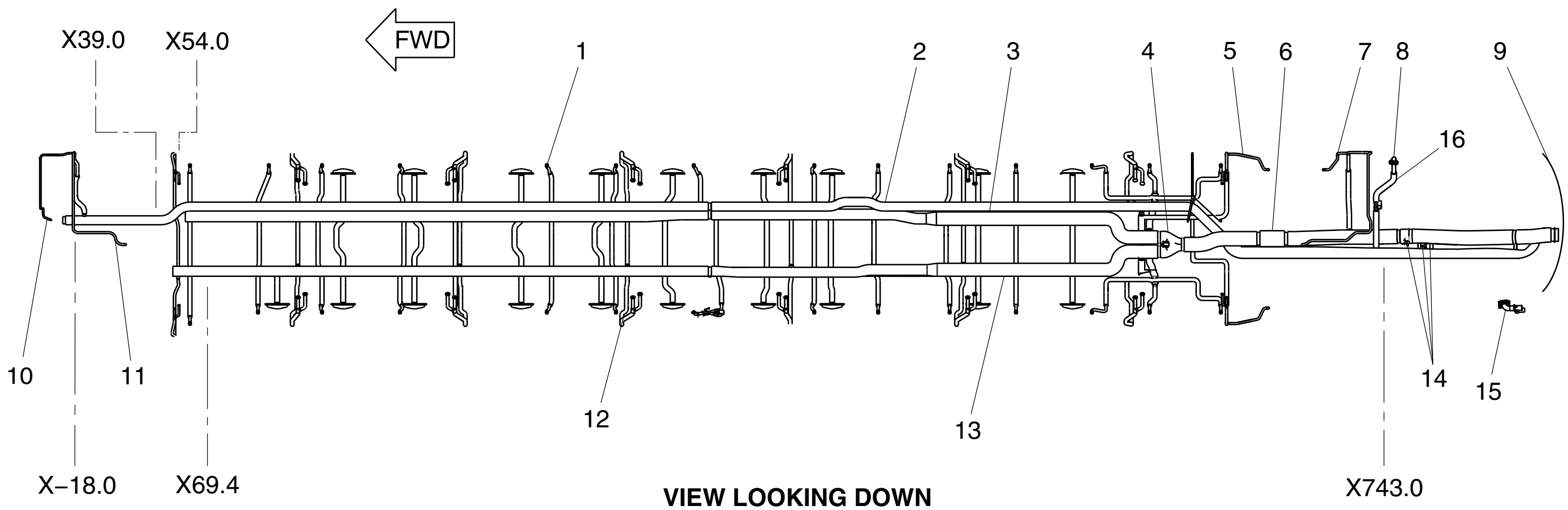
LEGEND

1. Flight Compartment Distribution System.
2. 4" Diameter Supply Duct.
3. Splitter Duct.
4. Aft Pressure Bulkhead.
5. Air Conditioning Pack.

FLIGHT COMPARTMENT AIR SUPPLY
Figure 1

LEGEND

1. Dado riser (typical).
2. Flight compartment supply duct.
3. Lower cabin distribution duct.
4. Distribution damper.
5. Aft galley gasper.
6. Noise treatment duct.
7. Aft flight attendant gasper.
8. Aft baggage compartment inlet.
9. Rear pressure bulkhead.
10. Forward lavatory gasper.
11. Forward flight attendant gasper.
12. PSU gasper riser (typical).
13. Upper cabin distribution duct.
14. Temperature sensors/switches.
15. Aft baggage compartment outlet.
16. Aft baggage compartment duct.



VIEW LOOKING DOWN
ON AIRCRAFT WITHOUT MODSUM 4–459190

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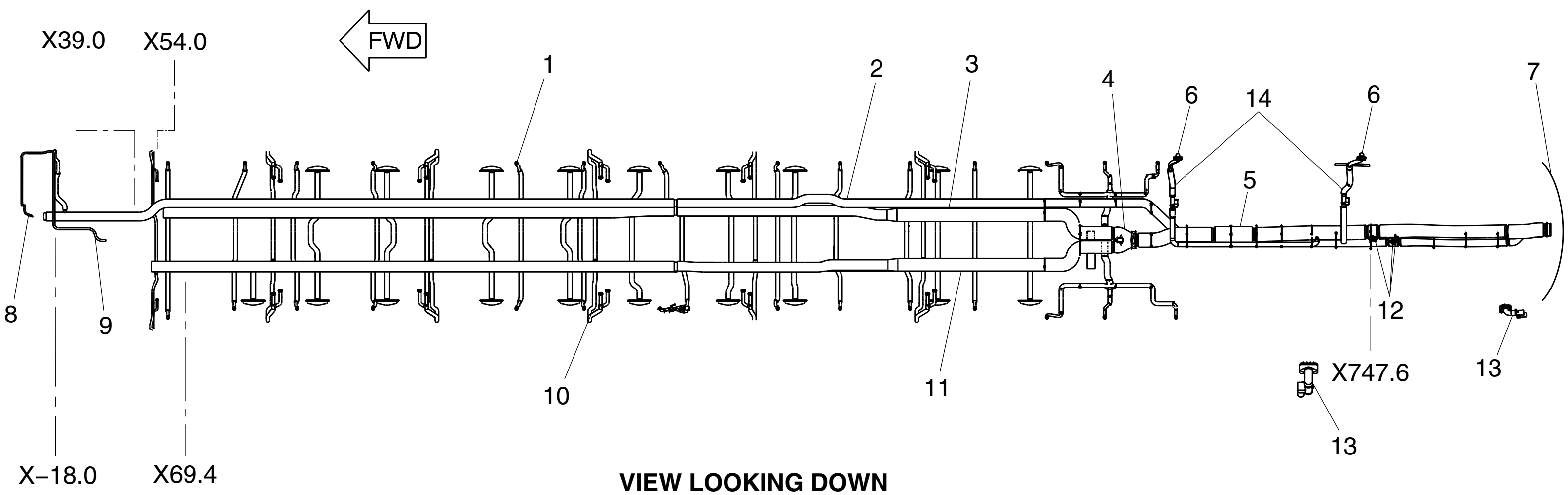
Flight Compartment Air Supply Detail
Figure 2 (Sheet 1 of 2)

LEGEND

1. Dado riser (typical).
2. Flight compartment supply duct.
3. Lower cabin distribution duct.
4. Distribution damper.
5. Noise treatment duct.
6. Aft baggage compartment inlet.
7. Rear pressure bulkhead.
8. Forward lavatory gasper.
9. Forward flight attendant gasper.
10. PSU gasper riser (typical).
11. Upper cabin distribution duct.
12. Temperature sensors/switches.
13. Aft baggage compartment outlet.
14. Aft baggage compartment duct.



VIEW LOOKING DOWN ON AIRCRAFT WITH MODSUM 4−459190

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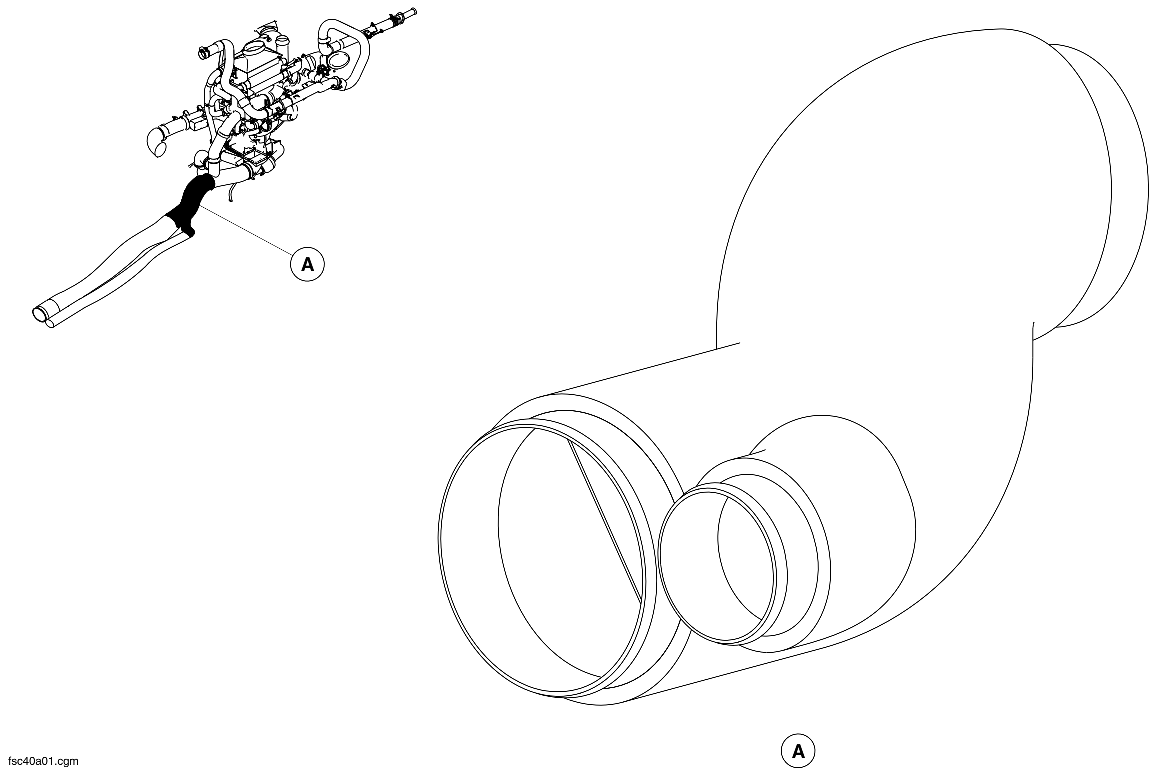
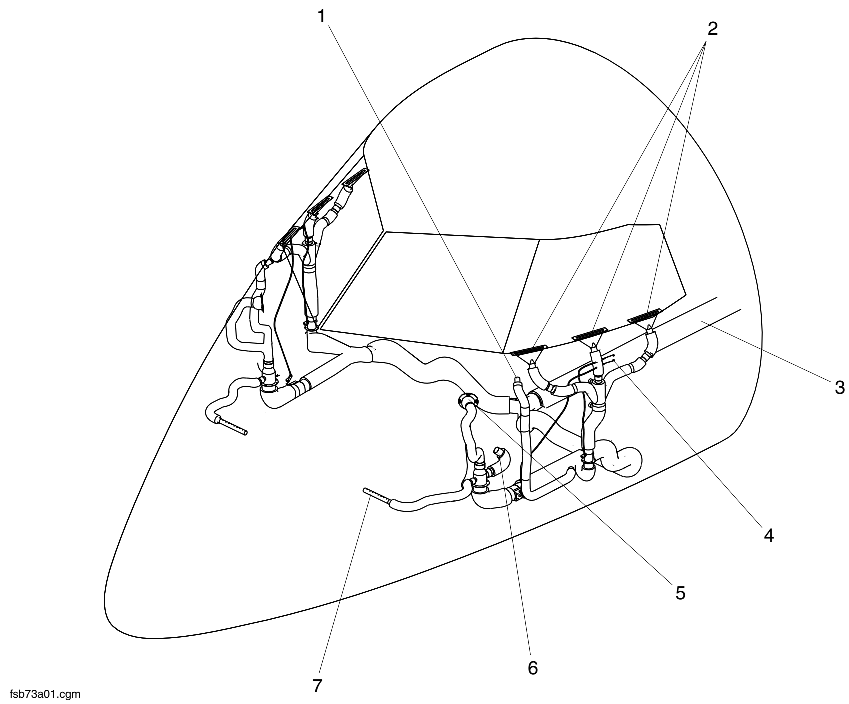
Flight Compartment Air Supply Detail
Figure 2 (Sheet 2 of 2)



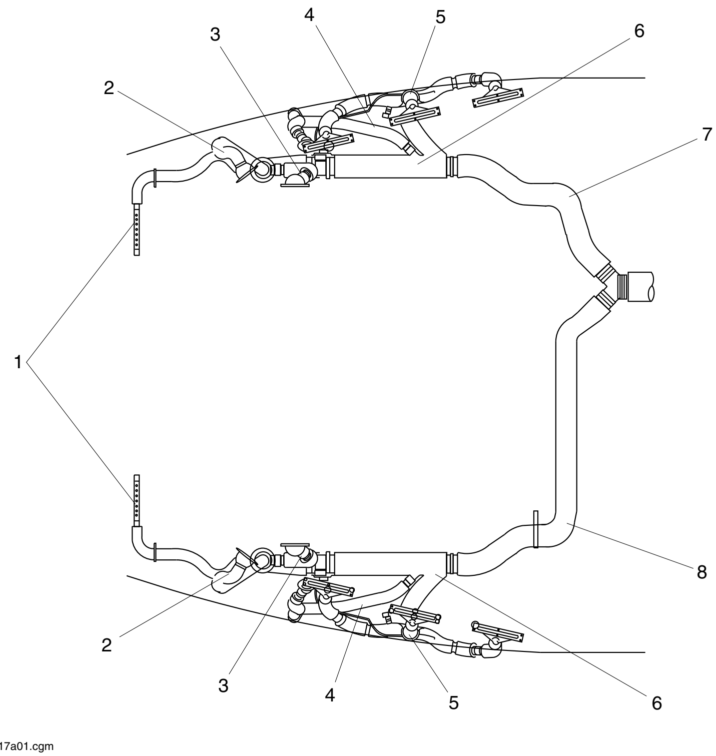
SPLITTER DUCT
Figure 3



LEGEND

1. Small Gasper.
2. De–Mist Nozzles.
3. 4" Diameter Supply Duct.
4. Flow Control Levers.
5. Flex Torso Gasper.
6. Side Grille Gasper.
7. Foot Warming Tube.

FLIGHT COMPARTMENT AIR DISTRIBUTION PAGE 1
Figure 4



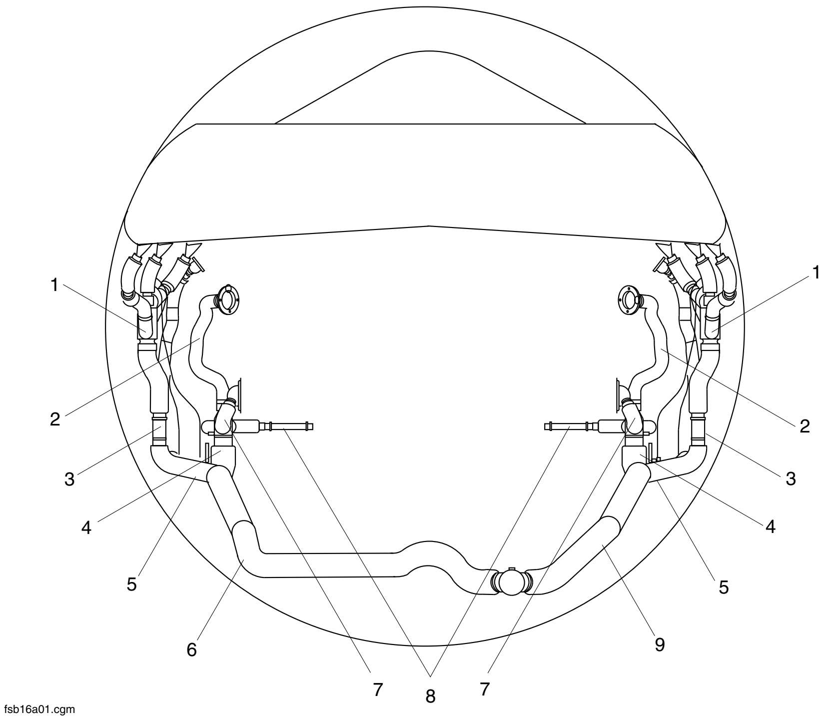
LEGEND

1. Foot Warm Vented Ducts.
2. Flex Torso Gasper Duct.
3. Grille Side Flex Duct.
4. Small Gasper Duct.
5. Trident Duct.
6. Multi−Port Supply Duct.
7. Aft Main Supply R/H.
8. Aft Main Supply L/H.

NOTE

View looking down.

FLIGHT COMPARTMENT AIR DISTRIBUTION PAGE 2
Figure 5



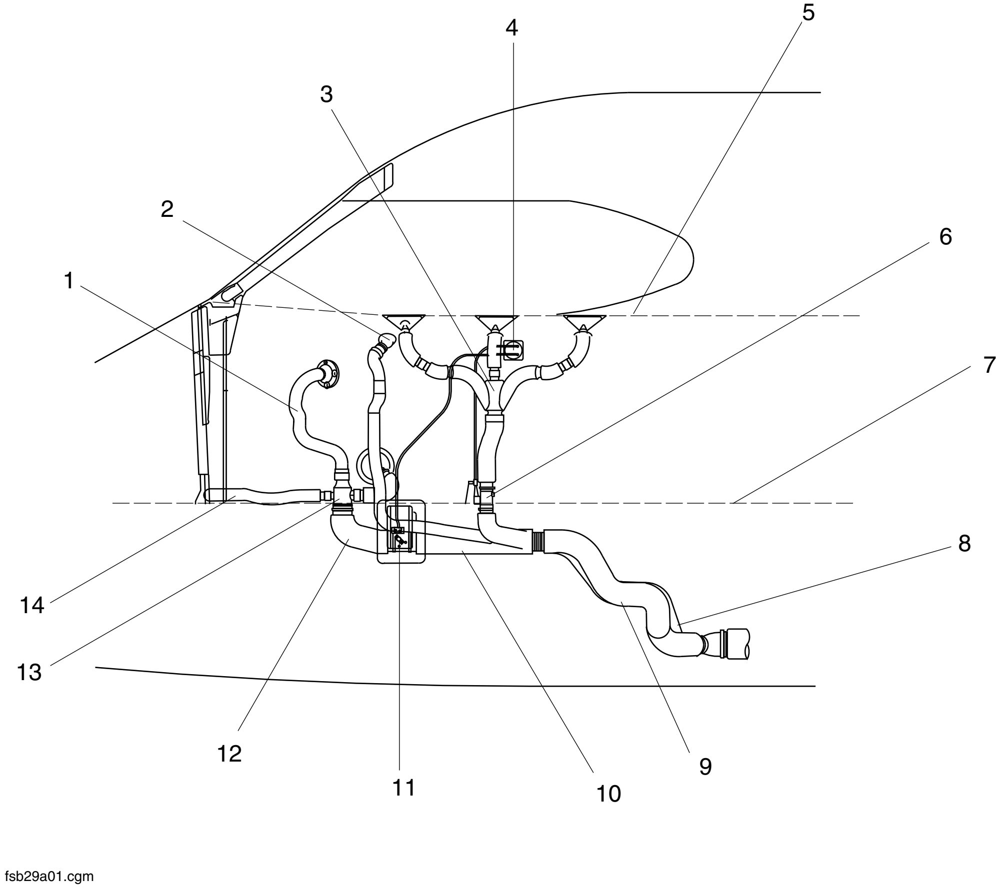
LEGEND

1. Trident Duct.
2. Flex Torso Gasper Duct.
3. Damper Valve.
4. Forward Supply Duct.
5. Under Floor Take−Off Duct.
6. Aft Main Supply Duct L/H.
7. Grille Side Flex Duct.
8. Foot Warm Vented Duct.
9. Aft Main Supply Duct R/H.

NOTE

View looking forward.

FLIGHT COMPARTMENT AIR DISTRIBUTION PAGE 3
Figure 6



LEGEND

1. Flex Torso Gasper Duct.
2. Small Gasper.
3. Trident Duct.
4. Flow Control Levers.
5. Sill Level.
6. Damper Valve.
7. Flight Deck Floor.
8. Aft Main Supply Duct R/H.
9. Aft Main Supply Duct L/H.
10. Multi–Port Supply Duct.
11. Shut–Off Valve.
12. Forward Supply Duct.
13. Stub Duct.
14. Foot Warm Supply Duct.

NOTE

View looking starboard.

FLIGHT COMPARTMENT AIR DISTRIBUTION PAGE 4
Figure 7

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21–22–00–001

AIR DISTRIBUTION, CABIN

Introduction

Conditioned air is supplied to the cabin to maintain a comfortable environment for the passengers and crew. Conditioned air is also used for aircraft pressurization.

General Description

Refer to Figure 1.

Air supply to the cabin is ducted from the air conditioning pack into the fuselage at the centre of the rear pressure bulkhead. The air is then ducted under the baggage compartment floor, where it splits into an upper and lower supply duct for each side of the fuselage. The lower duct supplies air to the dado panel vents, while the upper duct supplies air through sidewall risers to the overhead vents and gasper system. A distribution damper is set automatically depending on the cabin duct temperature. During heating operations, the distribution damper directs 70% of the warm air to the lower vents and 30% to the overhead vents. When cooling, 70% of the cool air is directed to the overhead vents and gaspers, and 30% to the lower vents. During standard temperatures, 50% of the air is directed to the overhead vents and 50% to the lower vents.

The baggage compartment has solenoid controlled inlet and outlet valves (ventilation control valves), which close when smoke is detected in the compartment and or when electrical power is lost (refer to SDS 26–12–00).

The Environmental Control System Electronic Control Unit (ECU) monitors a supply duct temperature sensor, an overtemperature switch and a cabin temperature sensor as part of the control loop (refer to SDS 21–61–00).

The air distribution, cabin and baggage compartment has the following subsystems:

- Damper, Distribution (21–22–06)
- Fan, Recirculation (21–22–16)
- Switch, Recirculation Fan (21–22–21)
- Duct, Splitter–Distribution (21–22–41)
- Duct, Distribution–Noise Treatment (21–22–46)

Detailed Description

Refer to Figures 2 and 3.

The air conditioning pack supplies conditioned air through its supply ducting to a fitting in the rear pressure bulkhead. The fitting is a 7 in. (177.8 mm) diameter aluminum tube divided down the middle (bifurcated). The left side of the bifurcation supplies flight compartment air while the right side supplies cabin air. Forward of the pressure bulkhead, the flight compartment and cabin air flows are separated by the splitter duct. The splitter duct is a semi–rigid bifurcated duct, with a single inlet and two outlets.

Forward of the splitter duct, at station X763, there is an aluminum section of the cabin supply ducting. The zone supply overtemperature switch and the two zone supply temperature sensors are located in this section of the ducting (refer to SDS 21–61–00).

Refer to Figures 4 and 5.

After supplying the aft galley gasper and the aft flight attendant gasper, the 7 in. (177.8 mm) cabin supply duct divides in two at the distribution damper. The left side of the damper supplies the upper cabin distribution duct which is routed under the cabin floor on the left side of the aircraft. The right side of the damper supplies the lower cabin distribution duct which is routed under the cabin floor on the right side of the aircraft. The upper cabin distribution duct supplies the Passenger Service Unit (PSU) gaspers and the sidewall downwash and ceiling upwash vents. The lower cabin distribution duct supplies the dado panels.

The right digital channel of the ECU controls the distribution damper. The ECU uses the signal from the cabin zone supply temperature sensor to determine which mode to apply to the distribution damper. In the heating mode (less than 55 °F (12.8 °C)), the distribution damper sends 70% of the air to the lower cabin distribution duct (dado panels) and 30% of the air to the upper cabin distribution duct (PSU gaspers). In the cooling mode (greater than 95° F (35 °C)), the distribution damper reverses the proportions. Between 55 and 95 °F (12.8 and 35 °C), the distribution damper divides the air equally between the upper and lower cabin distribution ducts.

Refer to Figure 6.

The upper cabin distribution duct has a 6.5 in. (165.1 mm) main supply duct which reduces to a 5 in. (127 mm) diameter duct. The 2 in. (50.8 mm) diameter risers which branch off the main duct, supply air to the large PSU bins.

Refer to Figure 7.

The 1.25 in. (31.75 mm) risers supply air to the smaller bins located at the extreme forward and aft end of the cabin interior. Each riser divides into two smaller (wye) ducts of 0.9 in. (22.9 mm) before entering the PSU. Forward duct supplies the sidewall downwash/ceiling upwash chamber, while the aft duct supplies the PSU gasper chamber.

On aircraft with ModSum 4–458928 OR 4–190608 incorporated, at Sta. X49.50 on the RH side, there are two parallel risers going to the feeder duct. The 1.25 in. (31.75 mm) diameter riser reduces to 1 in. (25.4 mm) and then again increases to 1.5 in. (38.1 mm). Forward duct supplies the sidewall downwash/ceiling upwash chamber, while the aft supplies the PSU gasper chamber.

Refer to Figure 14.

On aircraft with Modsum 4–459190 incorporated, between the stations X564.500 and X747.602, the upper outlet duct assembly, the air systems orifice assembly (2.00 in. (50.8 mm) diameter duct) and the galley – flight attendant supply duct assembly are removed. The aft baggage compartment duct assembly and aft baggage compartment inlet and outlet are installed.

Refer to Figures 8 and 9.

The sidewall downwash refers to the air that flows down the sidewall of the cabin interior. This air flows out of small holes located along the bottom of the downwash/upwash chamber behind the PSU bin. The ceiling upwash refers to the air that flows up towards the ceiling of the cabin interior. This air flows out of upward facing ducts connected to each end of the downwash/upwash chamber. These ducts are located between the PSU bins. Air entering the PSU gasper chamber supplies the individual passenger controlled gaspers.

The lower cabin distribution duct has a 6.5 in. (165.1 mm) main supply duct which reduces to a 5 in. (127 mm) diameter duct. Small (1.5 in. (38.1 mm)) risers branch off the main duct to supply the dado panel (elbow) inlets. The shape of the elbow inlet causes the air to flow fore and aft into the dado panel which forms a chamber for the dado panel outlets. The dado panel outlets are a fixed angle outlet (30 degree towards the floor) with adjustable flow control louvres.

Damper, Distribution (21–22–06)

Refer to Figure 4.

The distribution damper is an electronically controlled and electrically actuated valve with a 7 in. (177.8 mm) diameter inlet and two 6.5 in. (165.1 mm) diameter outlets. It is installed just aft of station X604.5.

The right digital channel of the ECU automatically sets the distribution damper depending on the cabin supply duct temperature. During heating, 70% of the warm air goes to the lower vents and 30% to the overhead vents. During cooling, 70% of the cool air goes to the overhead vents and gaspers, and 30% to the lower vents. During standard temperatures, 50% of the air is directed to the overhead vents and 50% to the lower vents.

The right digital channel of the ECU controls the electric motor of the distribution damper valve. If the right digital channel or the electric motor fails, the damper valve will remain in its last position. Two position switches in the damper valve send discrete signals to the ECU indicating whether the valve is in the full warm of the full cool position.

Fan, Recirculation (21–22–16)

Refer to Figure 10.

The recirculation fan, located on the right side of the air conditioning pack, draws cabin air through the recirculation headers just above the floor. The air is routed under the floor then to the rear pressure bulkhead, where it is mixed with pack conditioned air. The recirculation ducting contains a check valve that prevents reverse flow through the fan in flight or during pack operation. The recirculation fan switch controls the on/off operation of the recirculation fan.

Switch, Recirculation Fan (21–22–21)

Refer to Figures 11 and 12.

The recirculation fan switch is located on the AIR CONDITIONING control panel in the flight compartment. The switch controls the on/off operation of the fan through the ECU. When the switch is selected to the RECIRC position, the fan starts at slow speed to reduce initial current draw, then switches to high speed. The fan operates at slow speed during single pack operation.

Duct, Splitter–Distribution (21–22–41)

NOTE

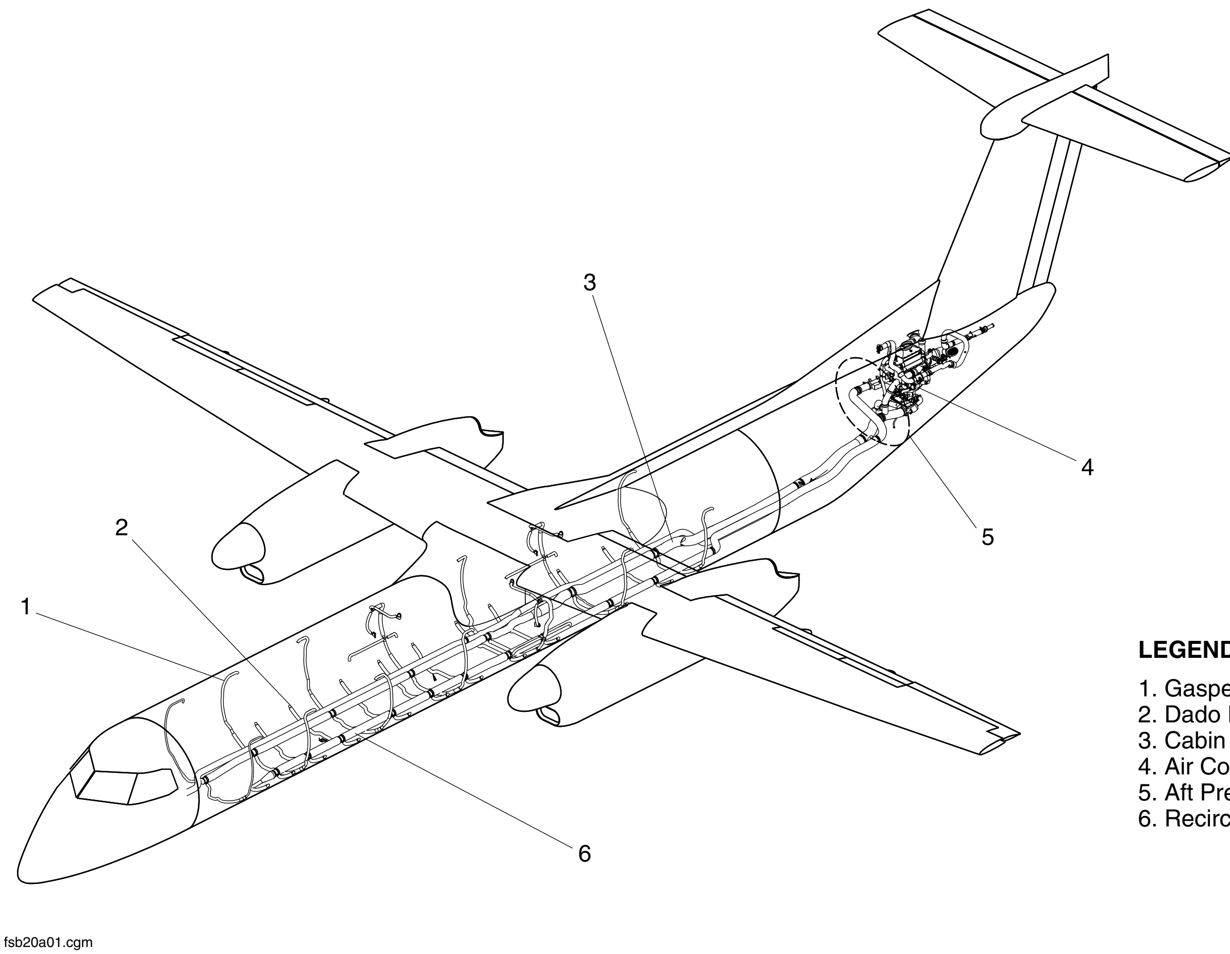
Refer to Figure 4.

The splitter duct is installed in the aft baggage compartment below the floor panels and behind the canted frame bulkhead panel. The flight compartment and cabin air flows are separated by the splitter duct. The splitter duct is a semi–rigid bifurcated duct made from silicone impregnated glass cloth. It has a 7 in. (177.8 mm) diameter inlet and two outlets. A 4 in. (101.6 mm) outlet on the left side supplies air to flight compartment and a 7 in. (177.8 mm) outlet on the right side supplies air to cabin.

Duct, Distribution–Noise Treatment (21–22–46)

Refer to Figure 13.

The distribution noise treatment duct reduces the overall cabin noise coming from the Air Cycle Machines (ACMs). The duct is a perforated aluminum tube surrounded by an insulating material and silicone impregnated glass cloth. The duct is installed in the cabin supply ducting between station X664.5 and X683.5.



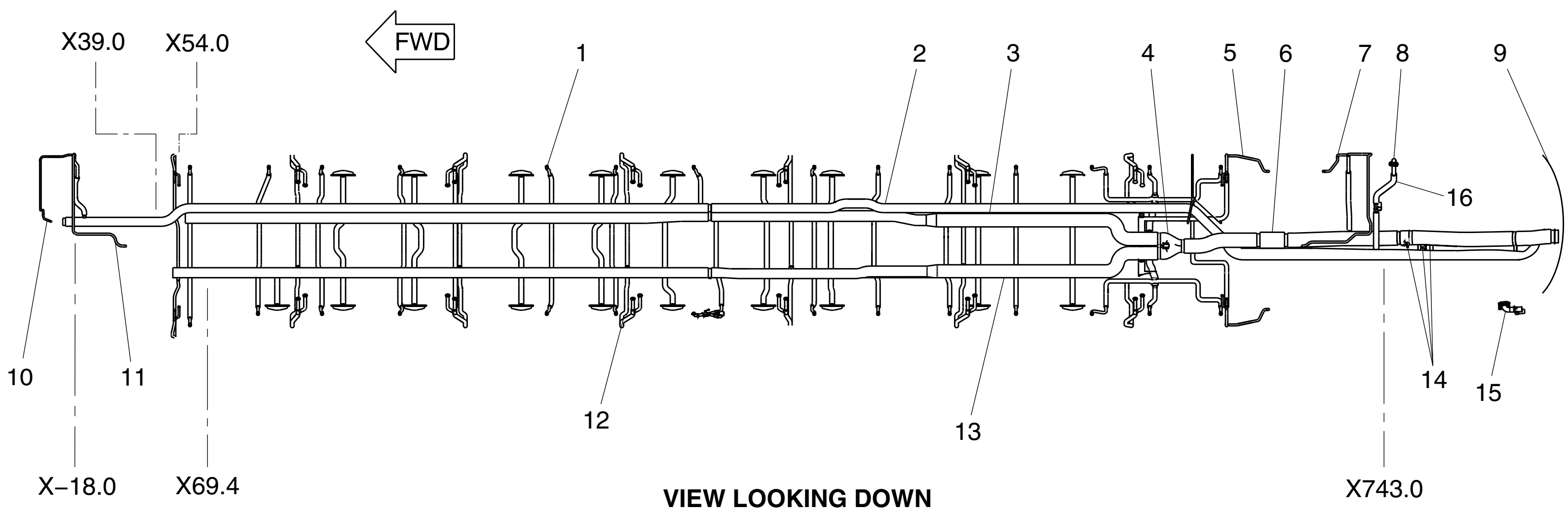
LEGEND

1. Gaspers.
2. Dado Panels.
3. Cabin Supply Duct.
4. Air Conditioning Pack.
5. Aft Pressure Bulkhead.
6. Recirculation Duct.

CABIN AIR DISTRIBUTION
Figure 1

LEGEND

1. Dado riser (typical).
2. Flight compartment supply duct.
3. Lower cabin distribution duct.
4. Distribution damper.
5. Aft galley gasper.
6. Noise treatment duct.
7. Aft flight attendant gasper.
8. Aft baggage compartment inlet.
9. Rear pressure bulkhead.
10. Forward lavatory gasper.
11. Forward flight attendant gasper.
12. PSU gasper riser (typical).
13. Upper cabin distribution duct.
14. Temperature sensors/switches.
15. Aft baggage compartment outlet.
16. Aft baggage compartment duct.



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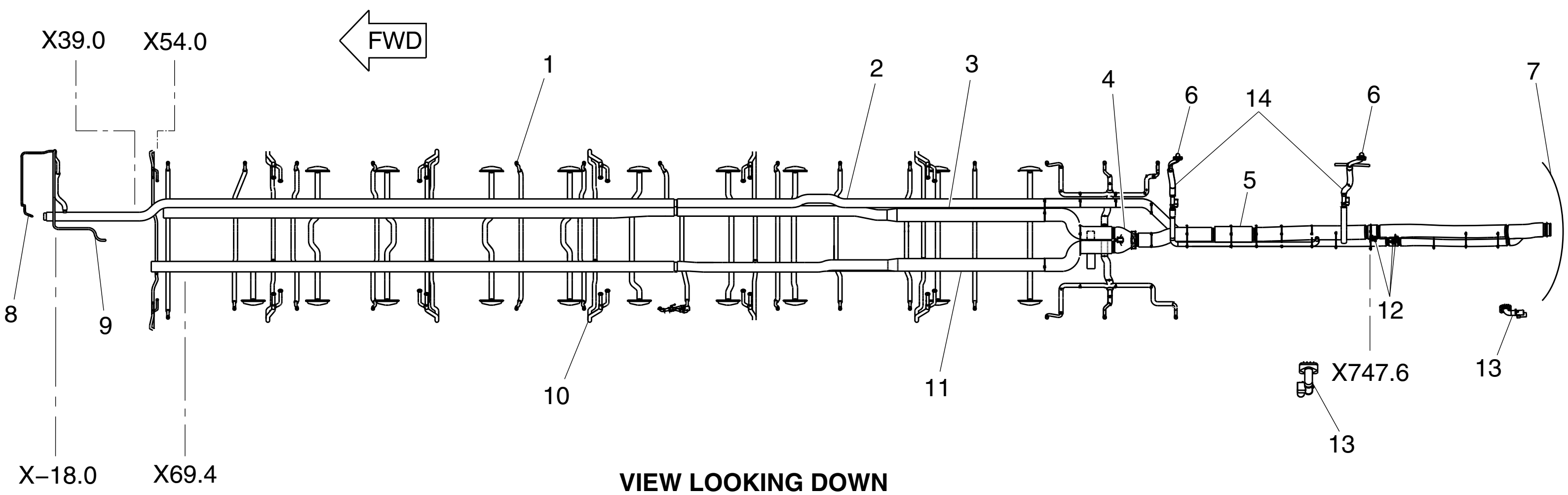
Cabin Air Distribution Detail Page 1
Figure 2 (Sheet 1 of 2)

LEGEND

1. Dado riser (typical).
2. Flight compartment supply duct.
3. Lower cabin distribution duct.
4. Distribution damper.
5. Noise treatment duct.
6. Aft baggage compartment inlet.
7. Rear pressure bulkhead.
8. Forward lavatory gasper.
9. Forward flight attendant gasper.
10. PSU gasper riser (typical).
11. Upper cabin distribution duct.
12. Temperature sensors/switches.
13. Aft baggage compartment outlet.
14. Aft baggage compartment duct.



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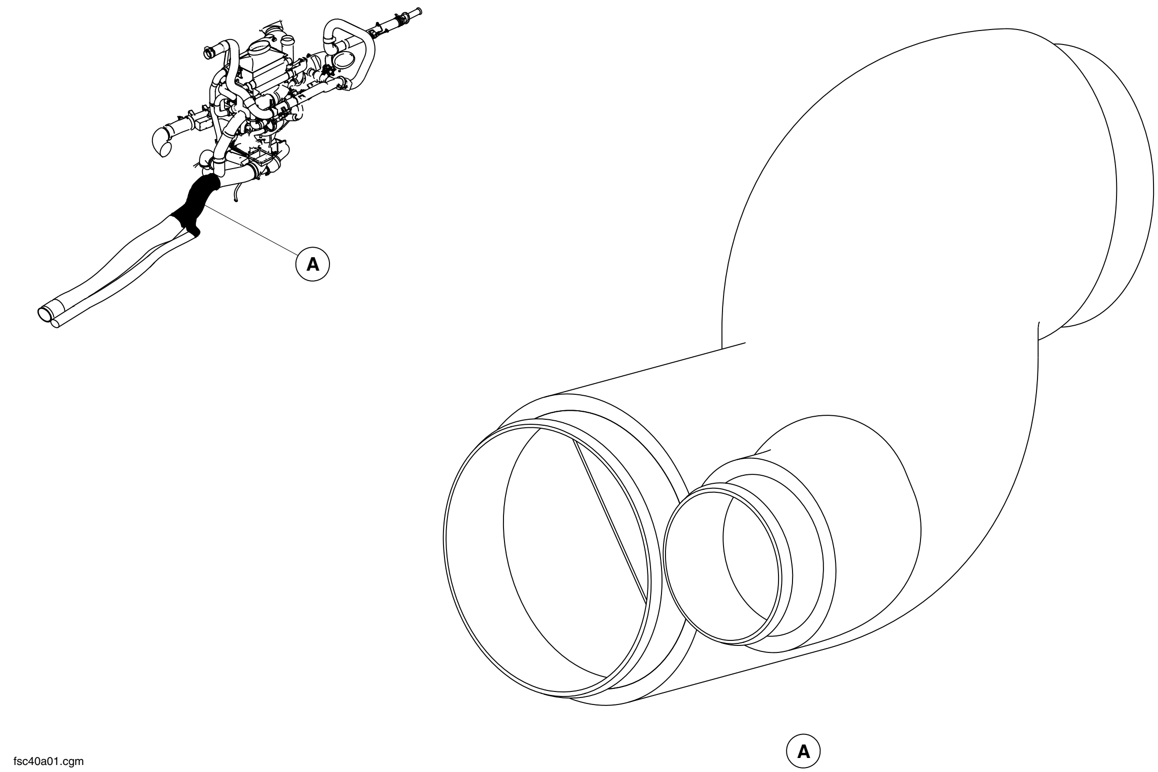
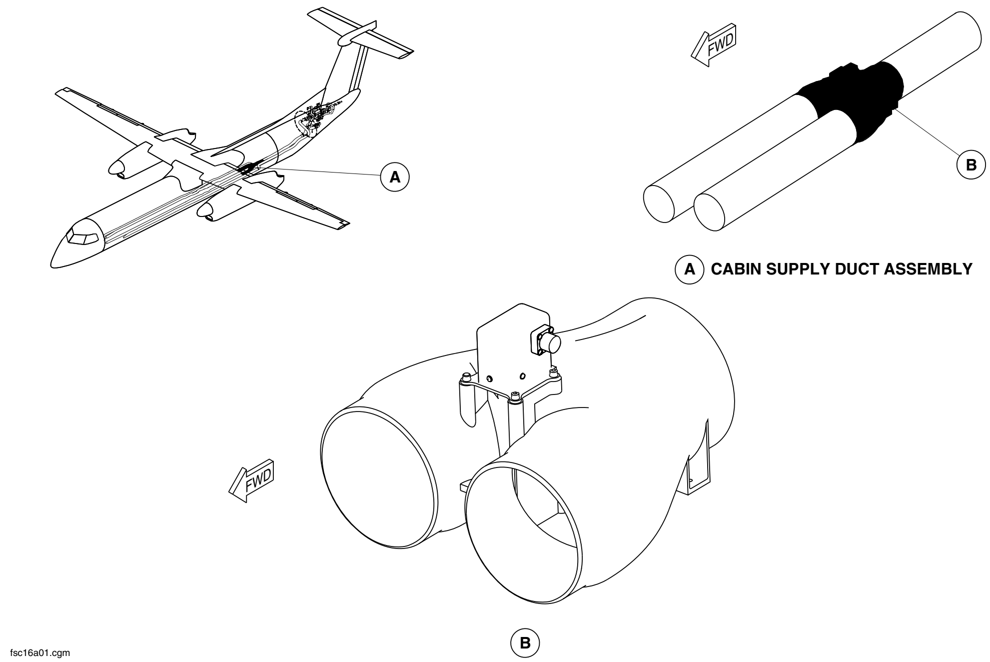
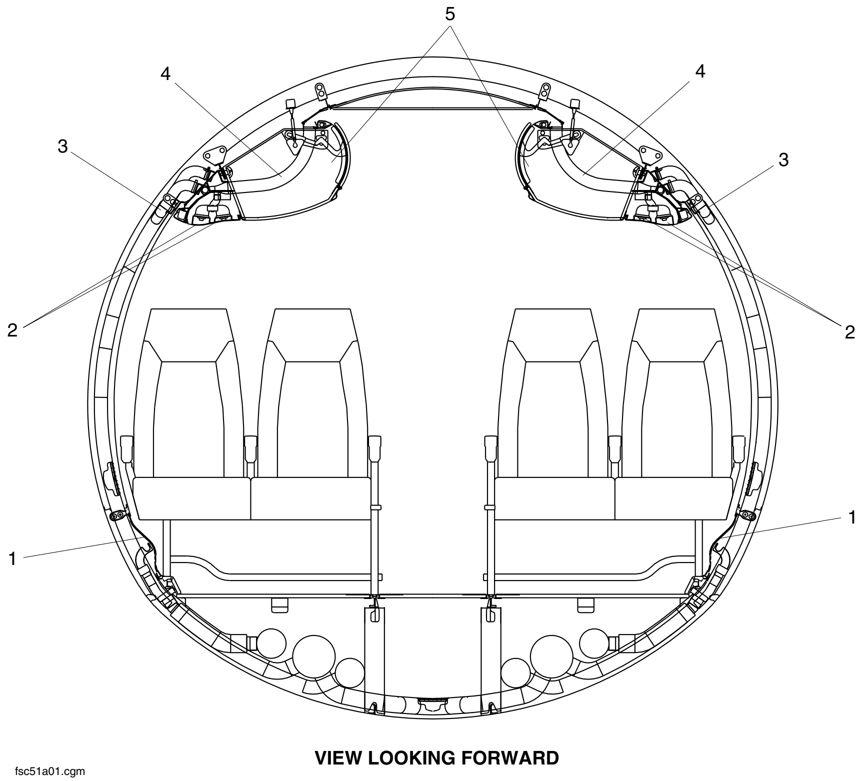
Cabin Air Distribution Detail Page 1
Figure 2 (Sheet 2 of 2)



SPLITTER DUCT
Figure 3



Distribution Damper
Figure 4



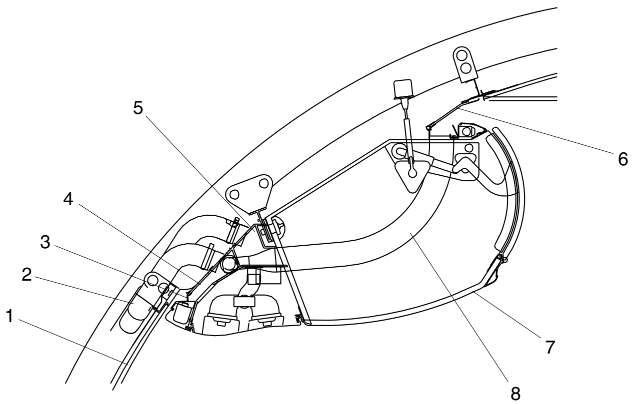
LEGEND

1. Dado Panel.
2. PSU Gaspers.
3. Wye Duct.
4. Ceiling Upwash Duct.
5. Overhead Bins.

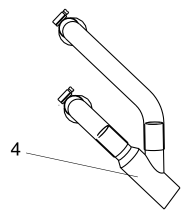
CABIN AIR DISTRIBUTION DETAIL PAGE 2
Figure 5

LEGEND

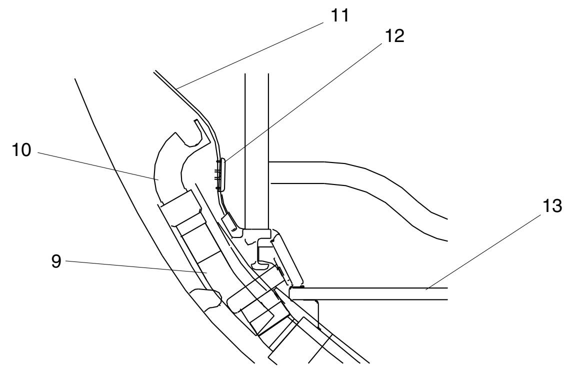
1. Sidewall Panel.
2. Wye Duct.
3. Sidewall Downwash Outlet.
4. Downwash/Upwash Chamber.
5. Gasper Chamber.
6. Ceiling.
7. Overhead Bin.
8. Ceiling Upwash Duct.
9. 1.5" Riser Duct.
10. Dado Panel (Elbow) Inlet.
11. Dado Panel.
12. Dado Panel Outlets.
13. Floor.



CEILING AIR DISTRIBUTION DETAIL



WYE DUCT DETAIL



LOWER AIR DISTRIBUTION DETAIL

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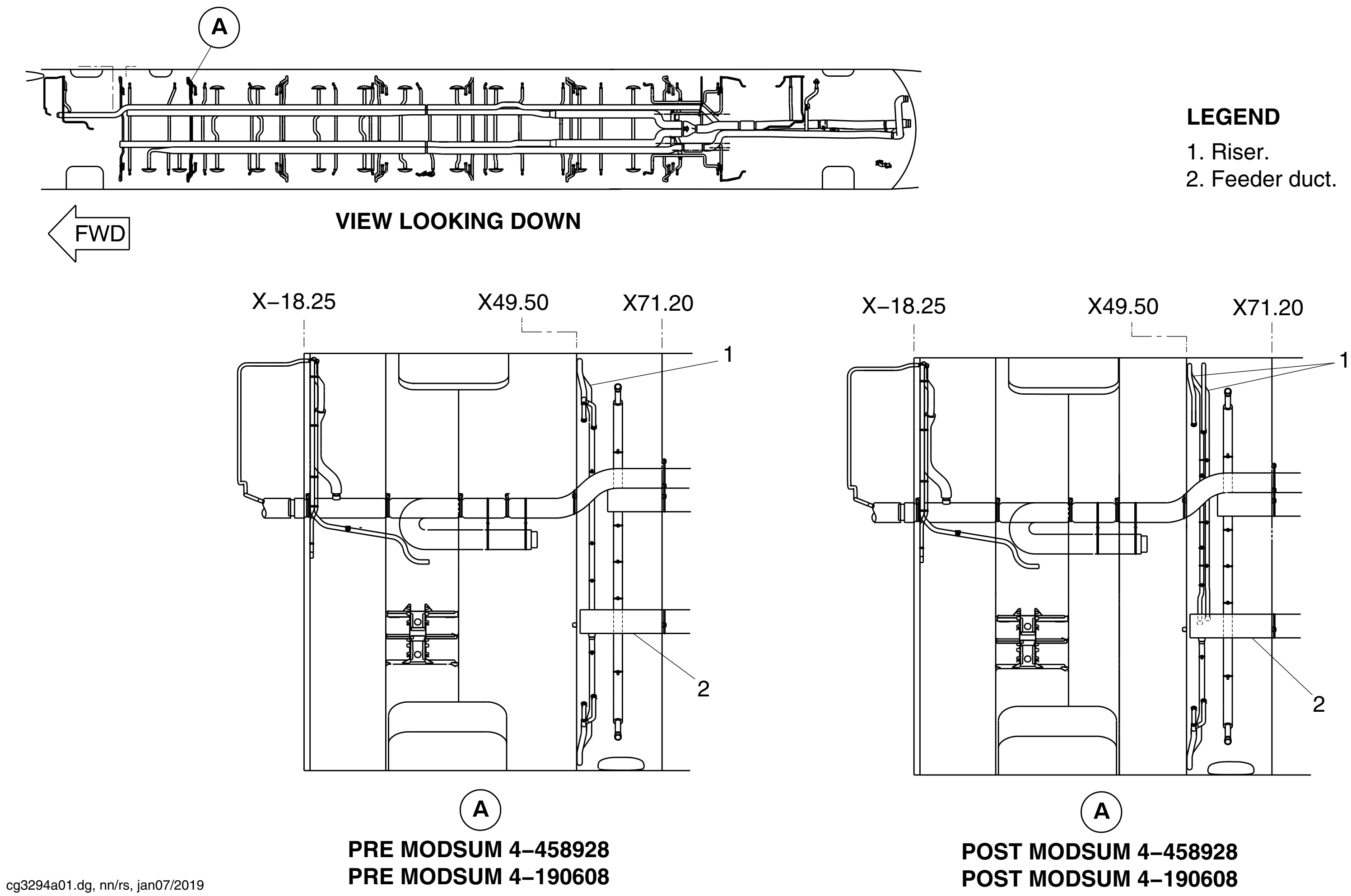
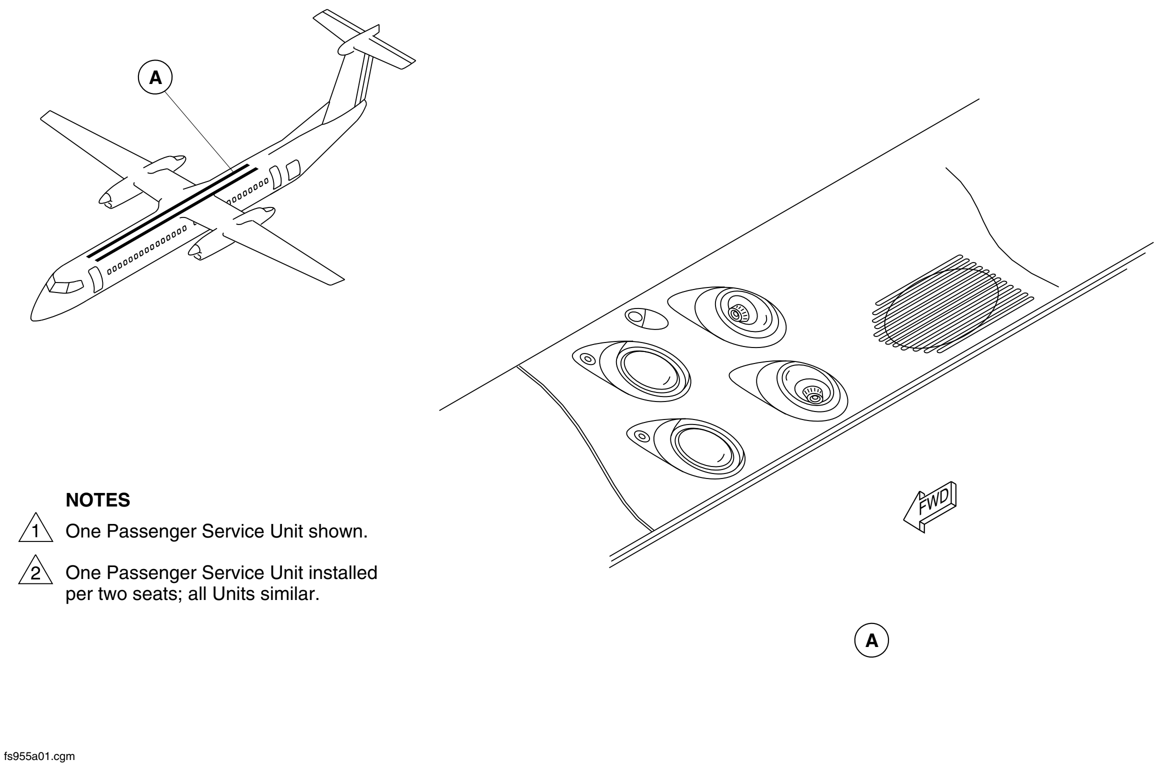
CABIN AIR DISTRIBUTION DETAIL PAGE 3
Figure 6



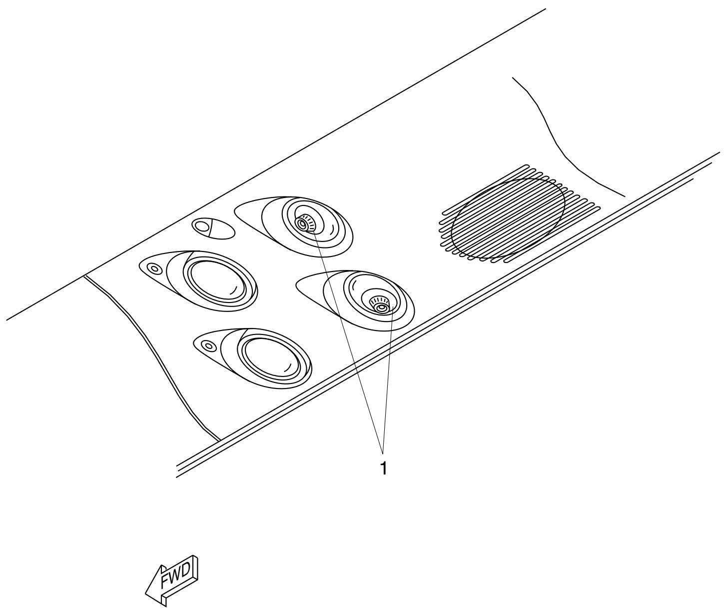
LEGEND

1. Riser.
2. Feeder duct.

Cabin Distribution System
Figure 7



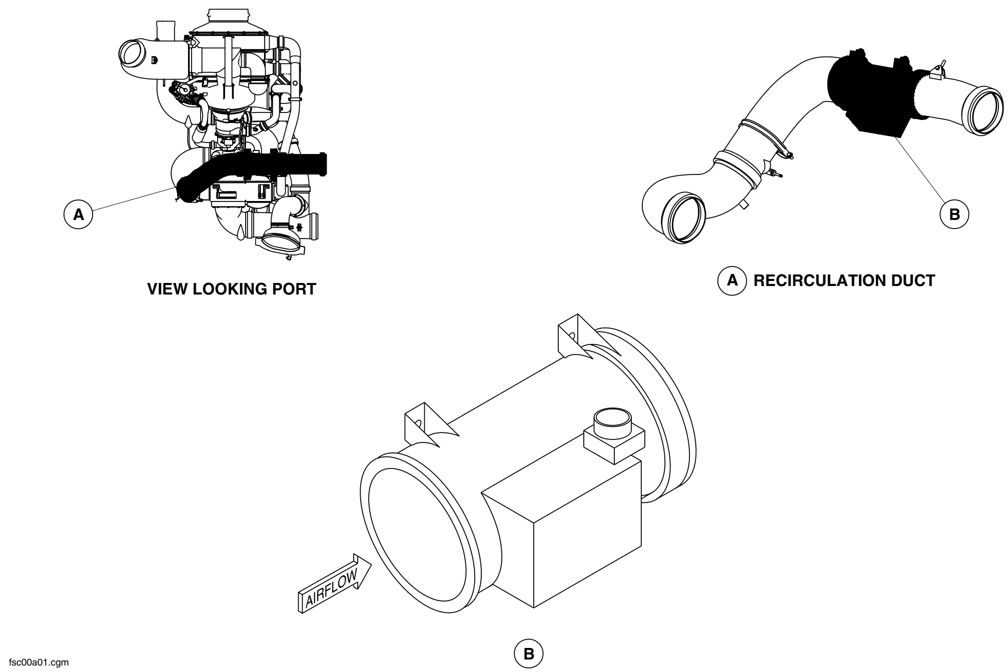
PASSENGER SERVICE UNIT (PSU) LOCATOR
Figure 8



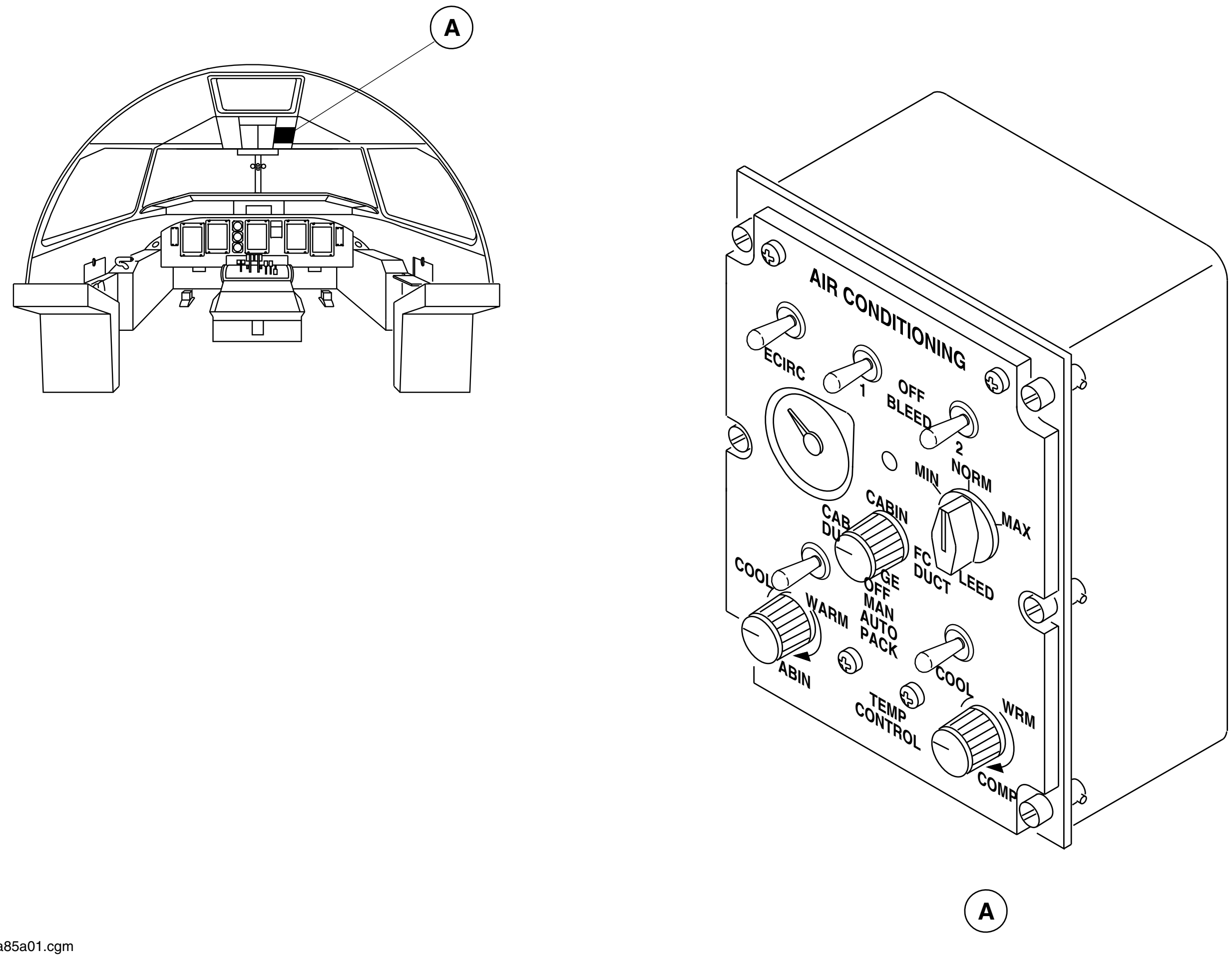
LEGEND

1. Gaspers.

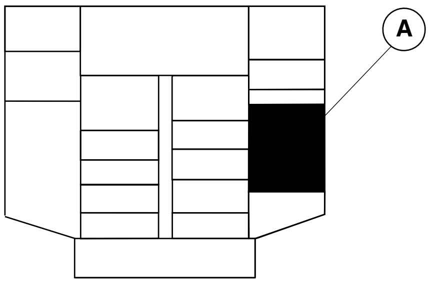
PASSENGER SERVICE UNIT (PSU) DETAIL
Figure 9



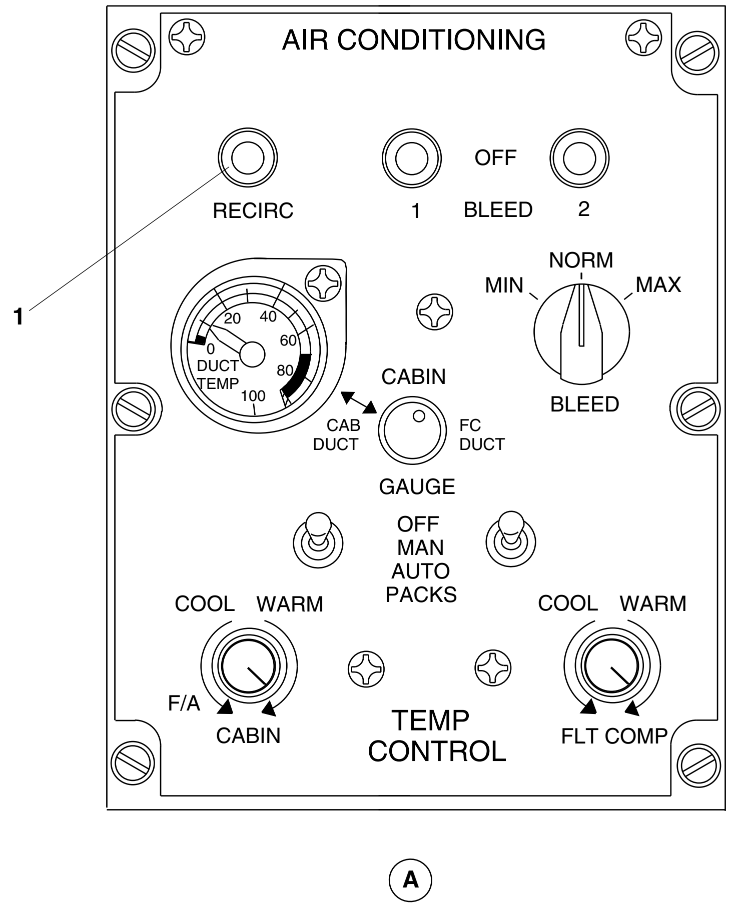
RECIRCULATION FAN
Figure 10



AIR CONDITIONING CONTROL PANEL LOCATOR
Figure 11



OVERHEAD CONSOLE



LEGEND

1. Recirculation Fan Switch.

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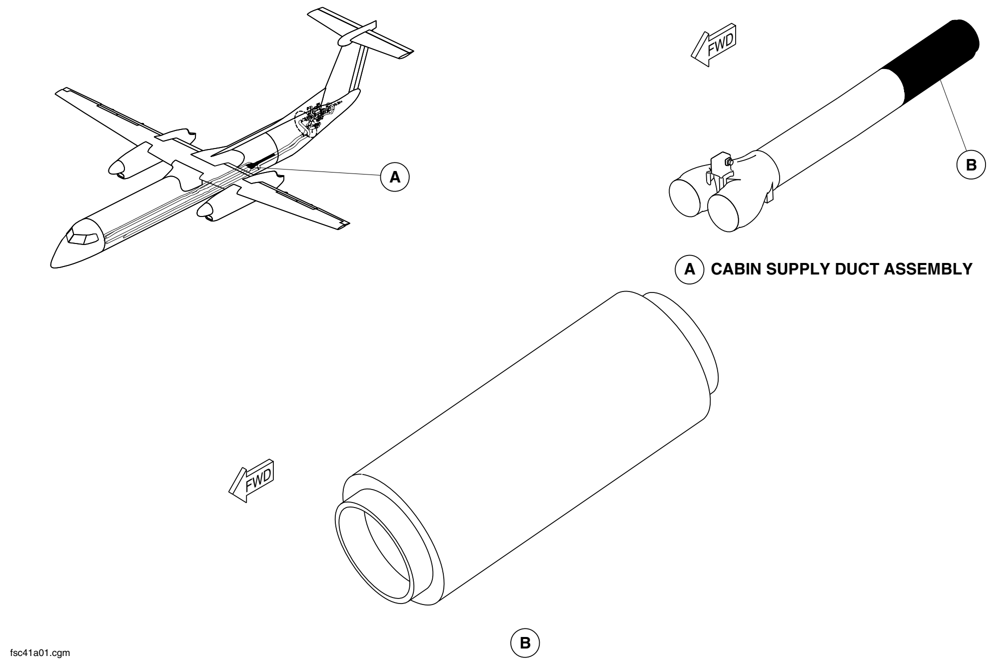
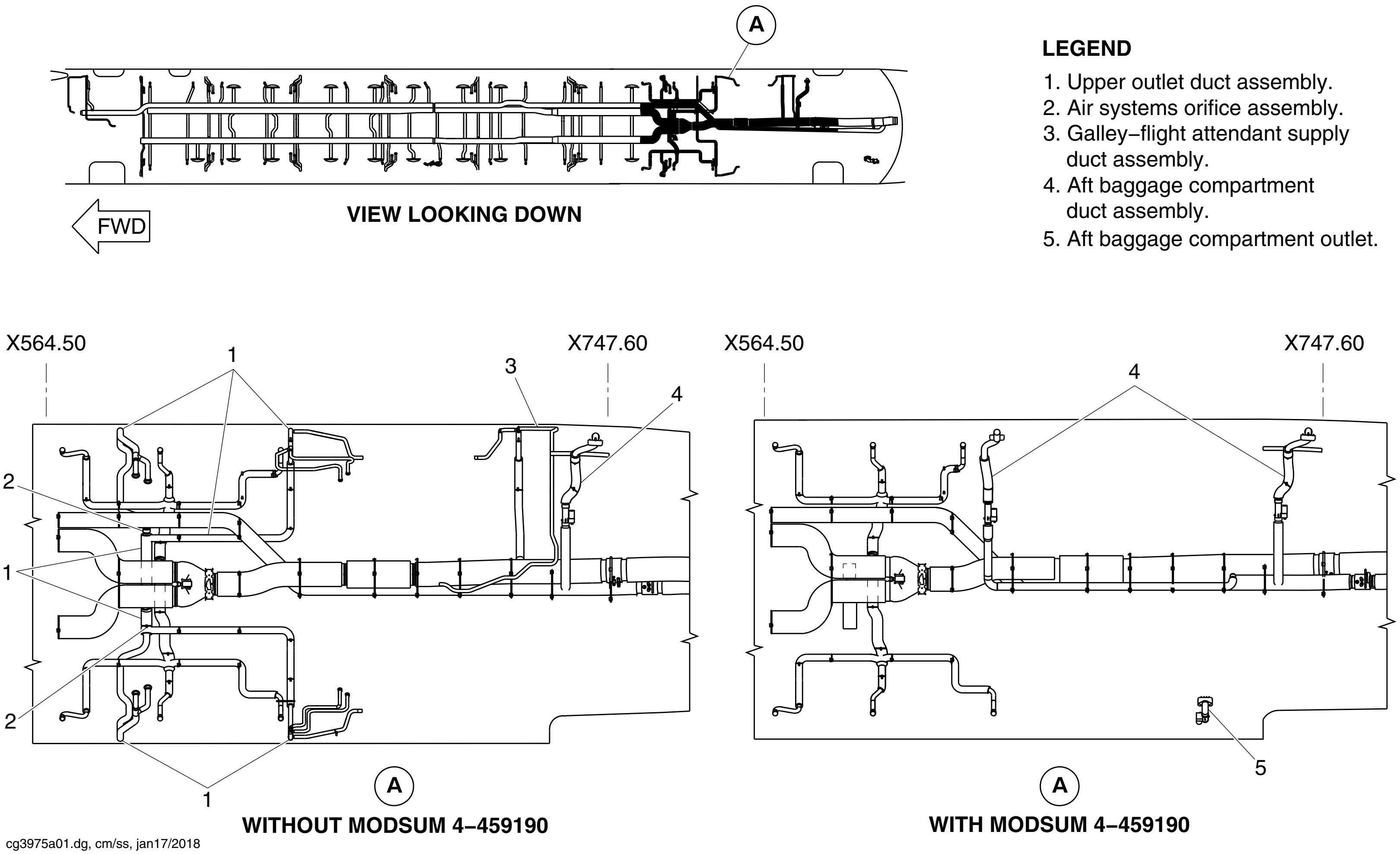
AIR CONDITIONING CONTROL PANEL / RECIRCULATION SWITCH
Figure 12



NOISE TREATMENT DUCT
Figure 13



LEGEND

1. Upper outlet duct assembly.
2. Air systems orifice assembly.
3. Galley–flight attendant supply duct assembly.
4. Aft baggage compartment duct assembly.
5. Aft baggage compartment outlet.

Cabin Distribution System
Figure 14

**ON A/C ALL

21–23–00–001

GROUND CONNECTION

Introduction

The ground connection provides a way of supplying conditioned air to the Environmental Control System (ECS) when the aircraft is on the ground.

General Description

An external source of conditioned air is attached to the ground connection. Conditioned air flows through the ground connection and its ducting to the air conditioning supply ducts. The ground connection system has the following components:

– Connection, Ground (21–23–01)

– Check Valve, Ground Connection (21–23–06)

– Ducts, Ground Connection (21–23–11)

Detailed Description

Refer to Figure 1.

An external air supply source is connected to the aircraft ground connection. Ducts carry the conditioned air from the ground connection point to the cabin and flight compartment supply ducts. The ground connection system has a check valve located in the ducting. The check valve is held in the open position by the air pressure from the external air supply source. ECS supply system ducting carries the conditioned air to the cabin and flight compartments. The check valve is held in the closed position by a spring when there is no pressure.

Connection, Ground

Refer to Figures 2 and 3.

The ground connection is a duct accessed through the external air conditioning door. The door is circular shaped, hinged and sealed, located at the rear fuselage right hand side. The connection point has a cover to prevent outside contamination from entering the system. The ground connection duct has a check valve to prevent system pressure discharging overboard when the ECS is operating using engine bleed air.

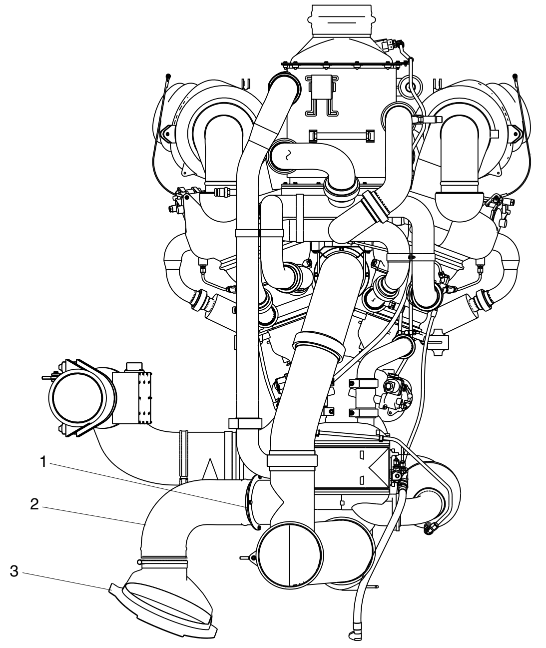
Check Valve, Ground Connection

Refer to Figure 4.

The check valve is a 5 in. (127 mm) diameter, aluminum, dual flapper valve. The check valve is installed in the ducting between the ground connection point and the air conditioning packs. During flight, the check valve prevents pressurized air from the air conditioning packs or the emergency ram air system from venting overboard. When there is no pressure available, the check valve is held closed by a spring to prevent dirt from entering the system.

Ducts, Ground Connection

The ground connection ducts carry conditioned air from an external supply source to the air conditioning supply ducts for the flight and cabin compartments.

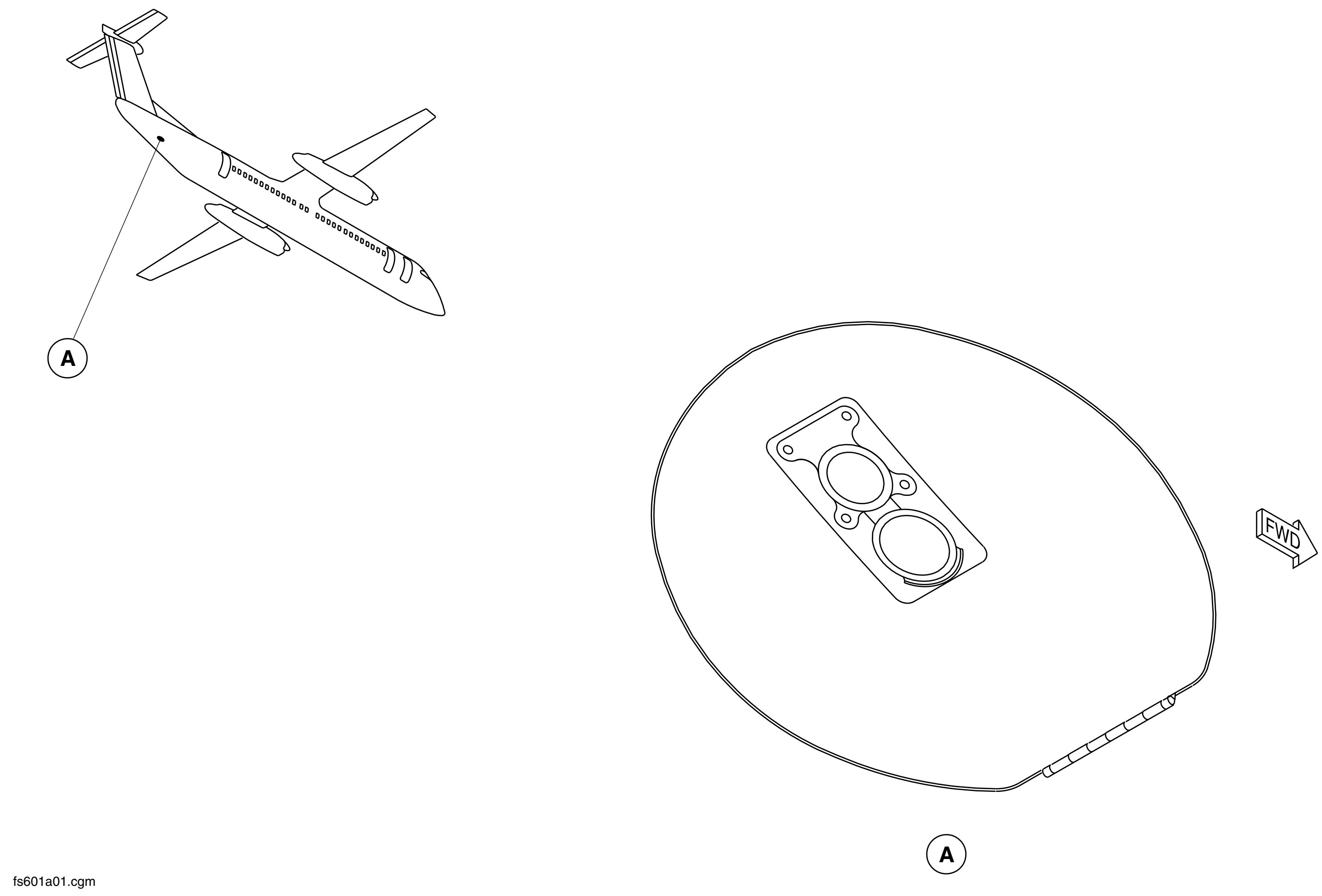


LEGEND

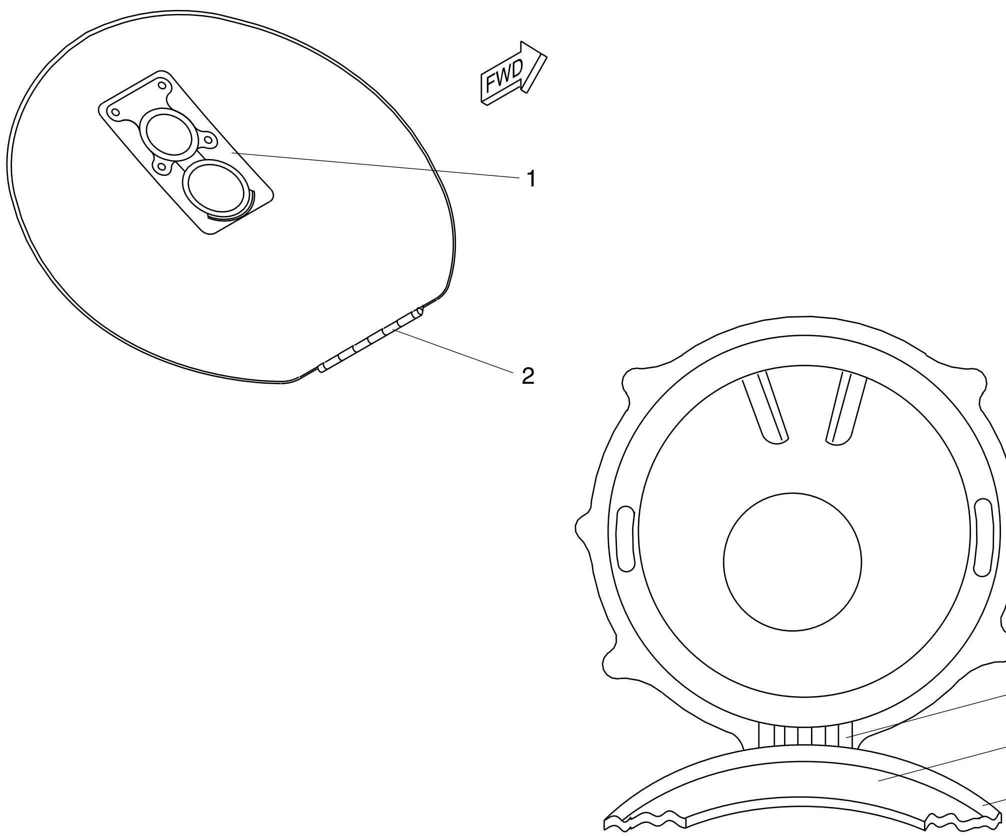
1. Ground Connection Check Valve.
2. Ground Connection Duct.
3. Ground Air Connection.

NOTE
View looking aft.

AIR CONDITIONING PACK / GROUND CONNECTION
Figure 1



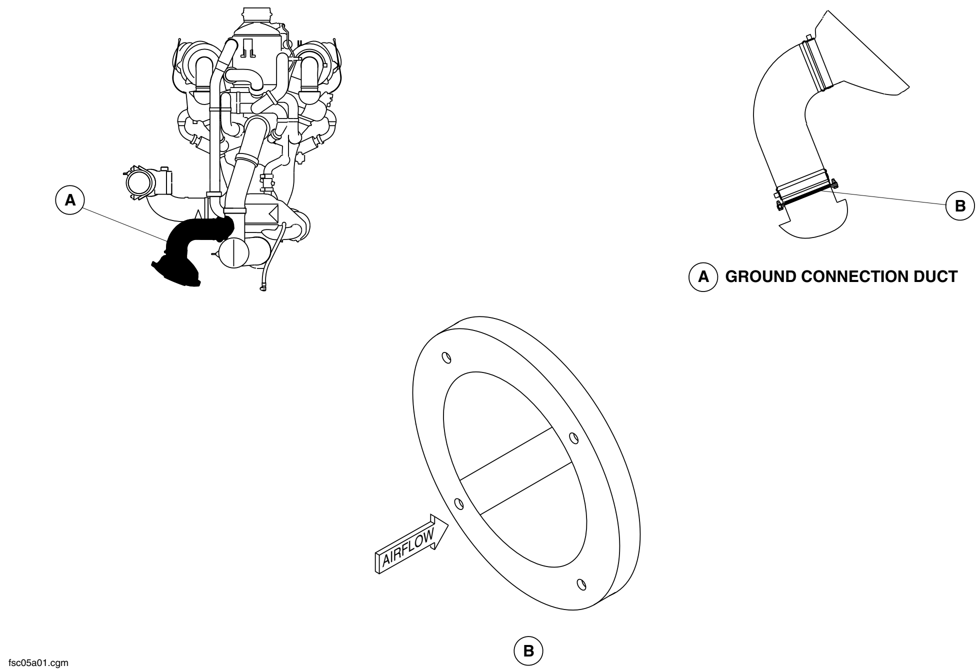
GROUND CONNECTION LOCATOR
Figure 2



LEGEND

1. Latch.
2. Hinge.
3. Gasket.
4. Door.

GROUND CONNECTION DETAIL
Figure 3



GROUND CONNECTION CHECK VALVE
Figure 4

**ON A/C ALL

21–24–00–001

RAM AIR

Introduction

The emergency ram air system supplies unconditioned air to the flight and cabin compartments in the event of a failure of the air conditioning packs.

General Description

A series of ducts carries the air from the NACA vents at the front base of the dorsal fin to the flight and cabin compartments. The emergency ram air system has a check valve located in the ducting. This valve prevents pressurized air from the air conditioning packs from venting out the NACA vents in the dorsal fin.

– Check Valve (21–24–01)

– Ducts, Ram Air (21–24–06)

Detailed Description

There are two functions of the ram air system:

– Supply cooling ram air to the dual heat exchanger (refer to SDS 21–51–00)

– Supply emergency ram air to the cabin and flight compartments in the event of a failure of the air conditioning packs.

Refer to Figure 1.

In flight, ram air enters the aircraft through the NACA vents at the base of the dorsal fin. It flows through the ram air ducting to the dual heat exchanger where it cools the bleed air entering the air conditioning packs (refer to SDS 21–51–00). If there is no air flow from the air conditioning packs, the pressure from the ram air is enough to overcome the spring in the emergency ram check valve. The check valve will open allowing cooling ram air to flow directly into the cabin and flight compartment supply ducts.

Refer to Figure 2.

During ground operations when there is no ram effect, the airflow through the NACA is insufficient for cooling. The Air Cycle Machine (ACM) fans continue to draw in air through the ram air inlet creating a negative pressure in the ram inlet. This negative pressure opens a rectangular–shaped ram inlet check valve to let in additional cooling air.

Emergency Ram Air Check Valve

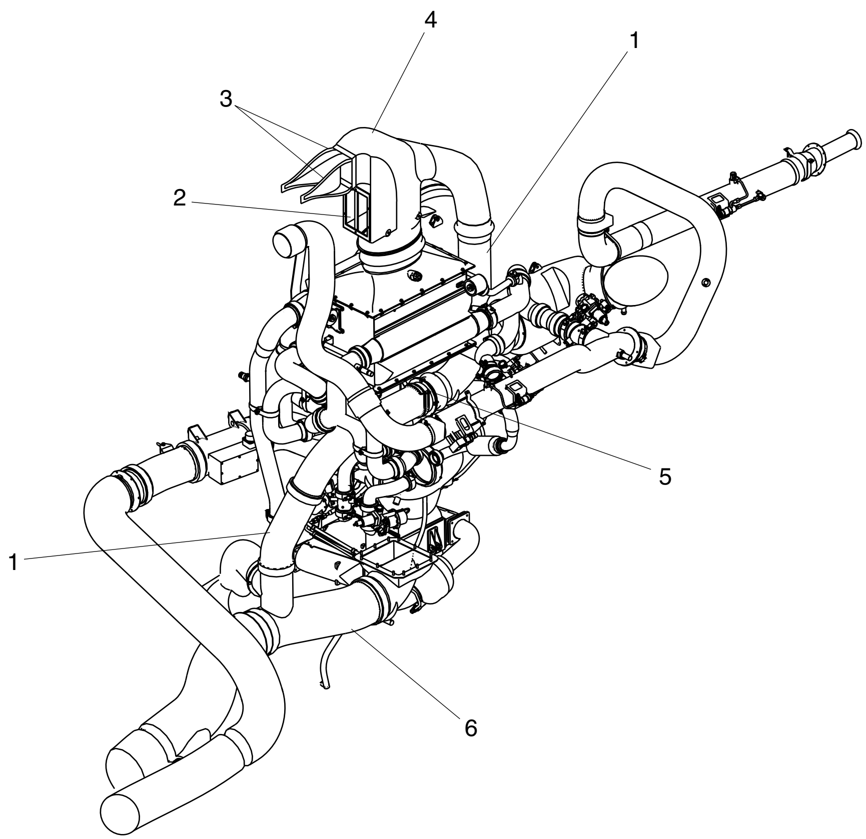
Refer to Figure 3.

The emergency ram air check valve is a circular, 5 in. (127 mm) diameter, dual flapper valve. It is spring–loaded to the closed position. The valve is installed between the upper and mid emergency ram air ducts.

Ducts, Ram Air

A series of ducts carries the emergency ram air to the flight and cabin compartments. The emergency ram air ducting starts at the aft side of the ram air inlet duct. The air is ducted down and then forward between the two ACM's to the front of the air conditioning pack. The ram air ducting then joins the conditioned air supply duct at a tee connection just upstream of the splitter duct. Thus the emergency ram air is distributed to the cabin and the flight compartments.

The ground connection ducting joins the emergency ram air ducting just upstream of the tee connection to the conditioned air supply duct.



LEGEND

1. Emergency Ram Air Duct.
2. Ram Inlet Check Valve.
3. Naca Vents.
4. Ram Air Inlet Duct.
5. Emergency Ram Air Check Valve.
6. Conditioned Air Supply Duct.

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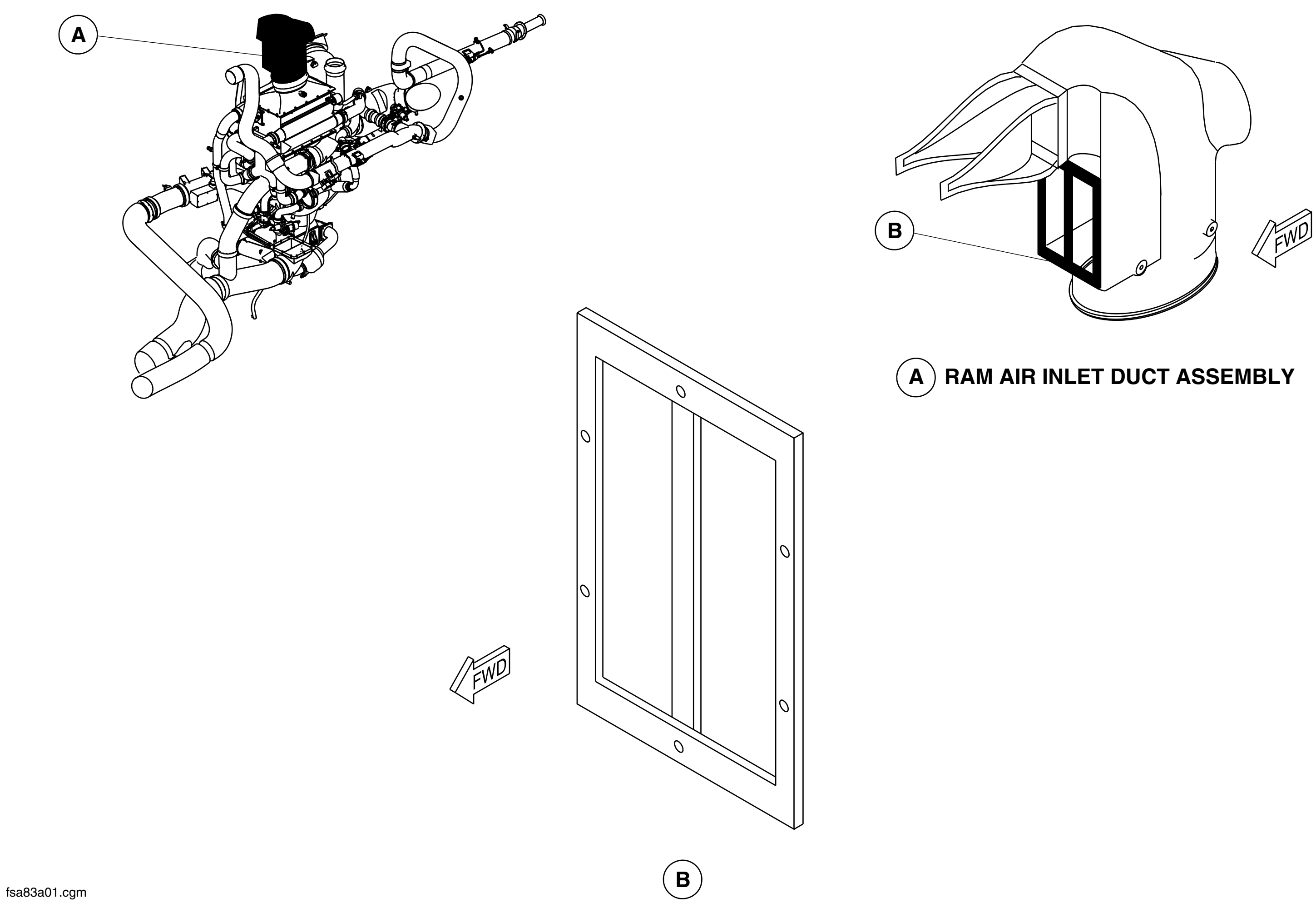
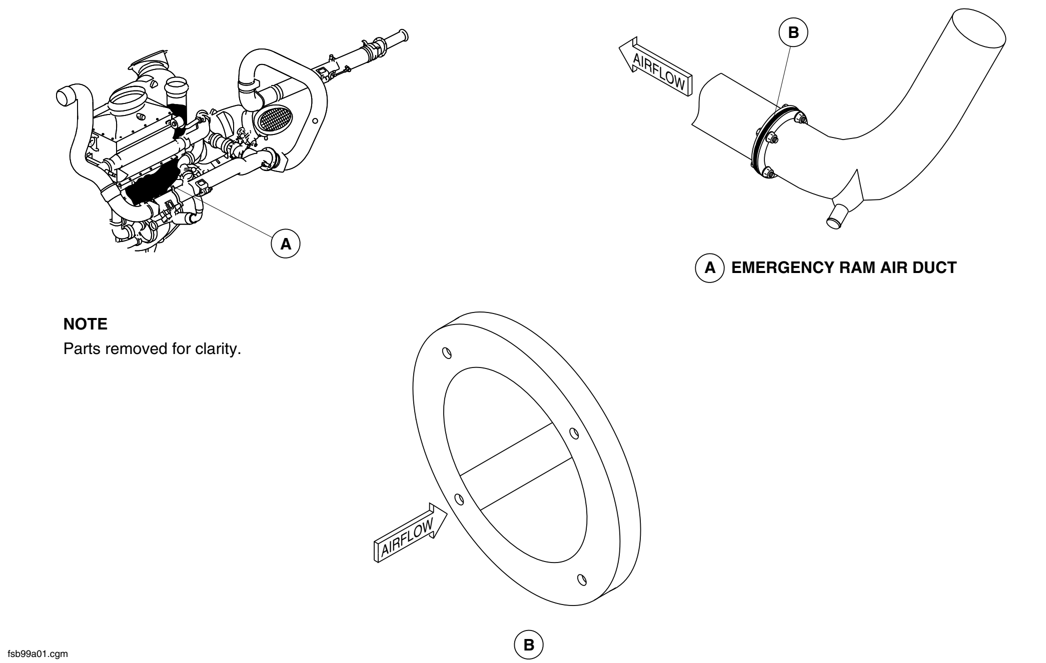
EMERGENCY RAM AIR SYSTEM
Figure 1



RAM INLET CHECK VALVE
Figure 2



EMERGENCY RAM AIR CHECK VALVE
Figure 3

**ON A/C ALL

21–25–00–001

AVIONICS COOLING

Introduction

The avionics cooling system has two independent duct networks and three fans. The system removes heat from the avionics rack, the wardrobe rack and the five liquid crystal displays. This improves the reliability and availability of the avionics systems.

General Description

Refer to Figures 1 and 2.

The aircraft has an extraction type cooling system for the avionics and liquid crystal displays (LCDs). Control of the system is automatic and requires no pilot action for both normal and abnormal operation. The cooling system has three identical fans, each of which can supply half the required cooling flow. Only two of the three fans are required to be operational for dispatch. The electrical and ducting systems have been designed so that single failures do not result in the loss of all the displays.

The avionics cooling system has the following components:

- Sensor, Zone Temperature (21–25–06)
- Switches, Zone Temperature (21–25–09)

Detailed Description

Refer to Figures 3 and 4.

The avionics cooling system has dual redundant duct assemblies. Each duct assembly has a cooling fan and branches that are connected to:

- 5 Liquid Crystal Displays (LCDs)
- Avionics Rack
- Wardrobe Rack.

Refer to Figures 5 , 6 and 7.

One of the duct assemblies is routed to the instrument panel from the pilot side, and the other assembly is routed from the copilot side. The duct assembly on the copilot side also has the standby fan #3 attached. The hot air is exhausted under the floor behind the flight compartment.

The cooling system removes the hot air produced by the five LCDs in the instrument panel, the avionics rack and the wardrobe rack. Fan #1 and fan #2 are running whenever the electrical power is applied to the aircraft dc main bus. If either of these fans fails, the standby fan #3 automatically starts operating. Each fan operating at high speed supplies one half of the required air flow. When the standby fan takes the place of the failed fan, the cooling system becomes fully operational again.

Because of the location and the high density design of the LCD displays, the operational time of the equipment without forced air cooling is limited to 60 minutes. When the internal temperature increases, the displays will reduce internal power dissipation by reducing the screen brightness to 60%. This reduction in screen brightness increases display availability. If there is the possibility of a misleading or total loss of display due to internal temperature increases, the display unit will shut down.

In case of electrical power failure, when only the battery power is available, fan #1 operates at low speed and fan #2 turns off. If fan #1 is not available, fan #2 operates at low speed. The standby unit fan #3 is not available when the main buses are not powered. This operational mode is capable of supplying the minimum air flow required for the LCDs to operate at reduced brightness. With minimum air flow and reduced brightness, the LCDs will operate with higher internal temperatures, but below the automatic shutoff threshold.

Refer to Figures 8 and 9.

The avionics cooling fan has an electronically controlled, two speed, brushless DC motor. Each fan has a built–in electronic low speed warning detector (LSWD) which sends a signal to the input/output module (IOM) when the fan is rotating slower than 8000 rpm. The IOM uses the low speed warning signal to activate the standby fan if there is a failure of fan #1 or fan #2. The IOM continuously monitors the fans for signals from the LSWD. Both IOM 1 and IOM 2 monitor all three fan fail outputs. Failure of any fan is recorded in the central diagnostic system (CDS) for further maintenance action. There is no indication of a single fan failure to the flight crew. If the failure of two fans occurs while on the ground, an advisory message is sent to the engine display (ED) to inform the pilot that dispatch is not permitted (Refer to Figure 10.). At the same time, the IOM turns on the AVIONICS caution light on the caution and warning panel. If the aircraft is in the air when a double fan failure occurs, the AVIONICS caution light does not come on and an advisory message is sent to the ED.

Refer to Figure 9.

A fan speed control unit is installed on the side wall of each of the three avionics cooling fans. At start–up, electrical power is supplied to all three fans. If there is no fan fail signal from the LSWD, fan #3 will shut down and fan #1 and fan #2 will operate in high speed mode (HSM). This is to ensure that a hidden failure does not exist in any of the three fans.

When dc generated power is available and the aircraft is in flight (i.e., no weight on wheels signal), the fan speed control unit operates as follows:

- Three fans available, two fans operating at HSM
- Two fans available, two fans operating at HSM
- One fan available, one fan operating at HSM.

If only battery power is available, the system automatically selects an operational fan (fan #1 or fan #2) based on signals from the LSWD. The selected fan operates from battery power at low speed mode (LSM). A fan operating at LSM can supply enough airflow to meet the avionics and LCD cooling requirements while still meeting the battery loading requirements. The standby fan #3 is not available when operating on battery power.

Sensor, Zone Temperature (21–25–06)

Refer to Figure 11.

The flight compartment zone temperature sensor is mounted on the inlet housing assembly installed on the co–pilot's rudder–pedal cover assembly. The flow of the air past duct inlet draws flight compartment air over the temperature sensor. This makes the zone temperature sensor to measure accurate ambient air temperature in the flight compartment.

The electronic control unit (ECU) uses the feedback temperature signals from the zone temperature sensor to regulate conditioned air temperature within the flight compartment. The zone temperature sensor is a dual thermistor sensor.

Switches, Zone Temperature (21–25–09)

Refer to Figure 11.

The ducting for each avionics cooling fan has a connection to one of two zone temperature switches located under the LCDs. These switches inhibit the fan operation on the ground, when the flight compartment temperature is below 41 °F (5 °C). This allows the LCD internal heater to operate during cold day starts without interference from the fans. The temperature switches are disabled by a relay if the WOW signal indicates that the aircraft is airborne. Each zone temperature switch has a second section which sends a signal to the IOM. The signal causes the IOM to inhibit the recording of a fan failure by the CDS when the temperature is below 50 °F (10 °C).

System Operation

All three fans have dual redundant power supply connections and power selection is controlled by relays. Relay 2126–K1 controls power to fan #1 and relay 2126–K1 controls power to fans #2 and #3. When 2126–K1 is energized by the left main bus, fan #1 is then powered by the left main bus. If power from the left main bus is lost, 2126–K1 is de–energized and fan #1 is powered from the left essential bus. Power from the right main bus energizes relay 2126–K3, which connects power from the bus to fan #2. If power from the right main bus is lost, 2126–K3 is de–energized and fan #2 connects to the right essential bus. Fan #3 receives power from the right main bus through relay 2126–K3, but if the right main bus is lost, de–energizing 2126–K3, fan #3 receives power from the left main bus.

Each fan has an ON/OFF control input and a HI/LO control input. If the ON/OFF input is a grounded circuit, the fan stops running and if the input is an open circuit, the fan will run at high speed. An open HI/LO control input circuit will cause the fan to operate at a high speed and a grounded input causes the fan to operate at a low speed. Each fan has one fan fail output for monitoring. A ground output signal indicates normal operation and an open output signal indicates a fan fail.

An ON/OFF line connects each fan to one of the two zone temperature switches. These switches inhibit fan operation on the ground when the cockpit temperature is below 41 °F (5 °C) so the fans do not interfere with the operation of the LCD display internal heater during cold day starts. Both switches have an output set at 41 °F (5 °C) and one at 50 °F (10 °C). When the temperature rises above each of these temperatures, the switch becomes an open circuit. The difference in output signals allows for sensor tolerances.

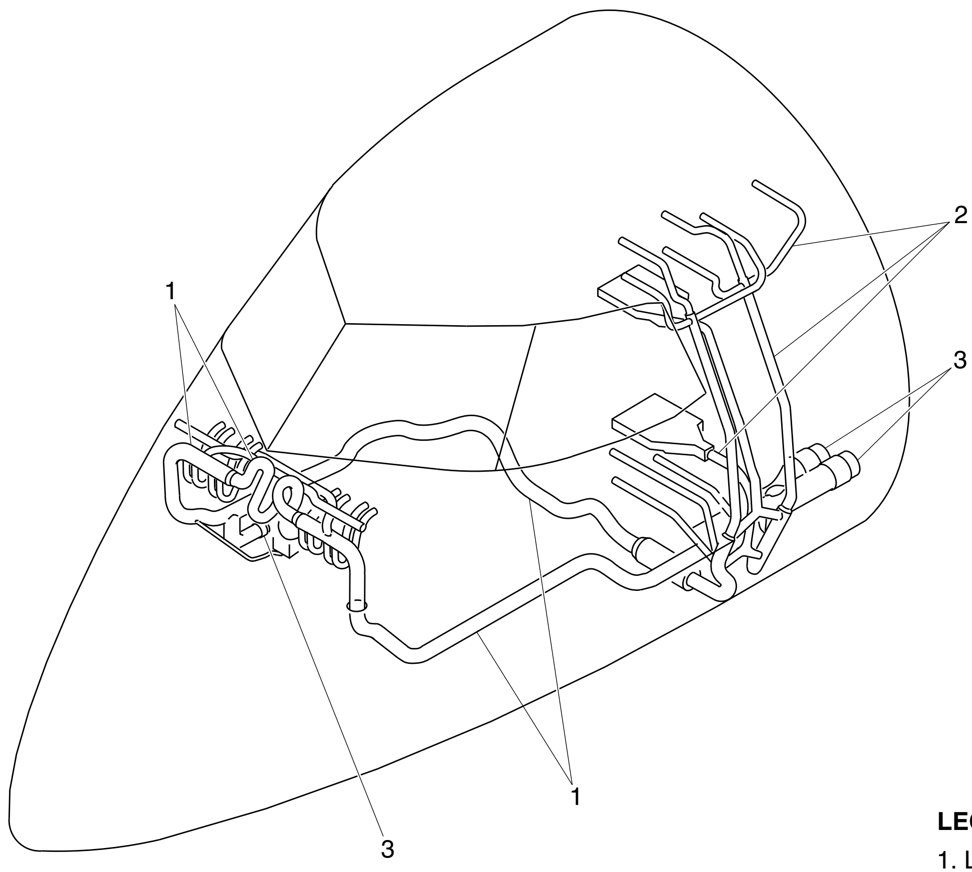
The landing gear system supplies a WOW signal to the cooling system through relay 2126–K5. This signal inhibits the temperature sensors in flight. When the WOW input changes to flight mode, relay 2126–K5 de–energizes and both switch outputs are disconnected.

IOM modules of the avionics receive the fan fail outputs of each fan and record the failures in the central diagnostic system. These cooling system connections are for monitoring purposes only and do not affect the cooling system logic.

During normal operation, power is supplied by the left and right main buses to relays 2126−K1, 2126−K3 and 2126−K4. Relay 2126−K1 connects fan #1 to the left main bus and 2126−K3 connects fan #2 to the right main bus. With power applied to relays 2126−K1 and 2126−K4, contacts through these relays become open circuits which set fan #1 and fan #2 to high. When a fan runs above the speed threshold, the fan fail output becomes shorted to ground. Similarly, grounding of the fan #2 fail output will energize relay 2126−K2. Fan #3 ON/OFF control is turned off when a grounded circuit input is received through relay 2126−K2 and the fan #1 fail output.

If fan #1 fails, the fan #1 output circuit becomes open and fan #3 is activated. If fan #2 fails, relay 2126−K2 is de−energized, which activates fan #3.

Loss of both main buses de−energizes relays 2126−K1, 2126−K3 and 2126−K4. Fan #1 then receives power from the left essential bus but with the HI/LO control input grounded, which puts fan #1 in the LSM. The LSM operation is used to minimize electrical power consumption. Fan #1 fail output is grounded when fan #1 is running in LSM. Through CR1 and relay 2126−K4, the fan #2 ON/OFF control is grounded and turns fan #2 off. With both main buses unavailable, fan #3 is not powered. If fan #1 fails, the fan #1 fail switch becomes open circuit which activates fan #2 through CR1 and relay 2126−K4. Fan #2 becomes powered by the right essential bus through relay 2126−K3. Because relay 2126−K4 is de−energized, fan #2 runs at low speed.



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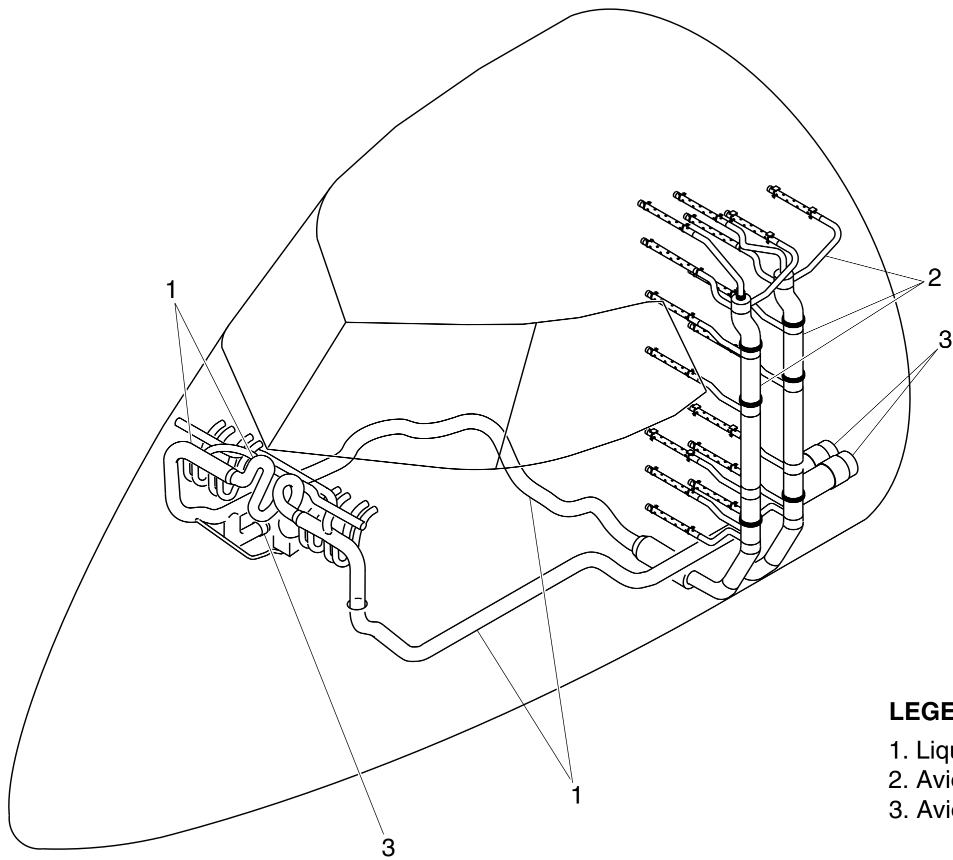
LEGEND

1. Liquid crystal display (LCD) cooling system.
2. Avionics cooling system.
3. Avionics cooling fan.

AVIONICS AND LCD COOLING SYSTEM
Figure 1 (Sheet 1 of 2)



LEGEND

1. Liquid crystal display (LCD) cooling system.
2. Avionics cooling system.
3. Avionics cooling fan.

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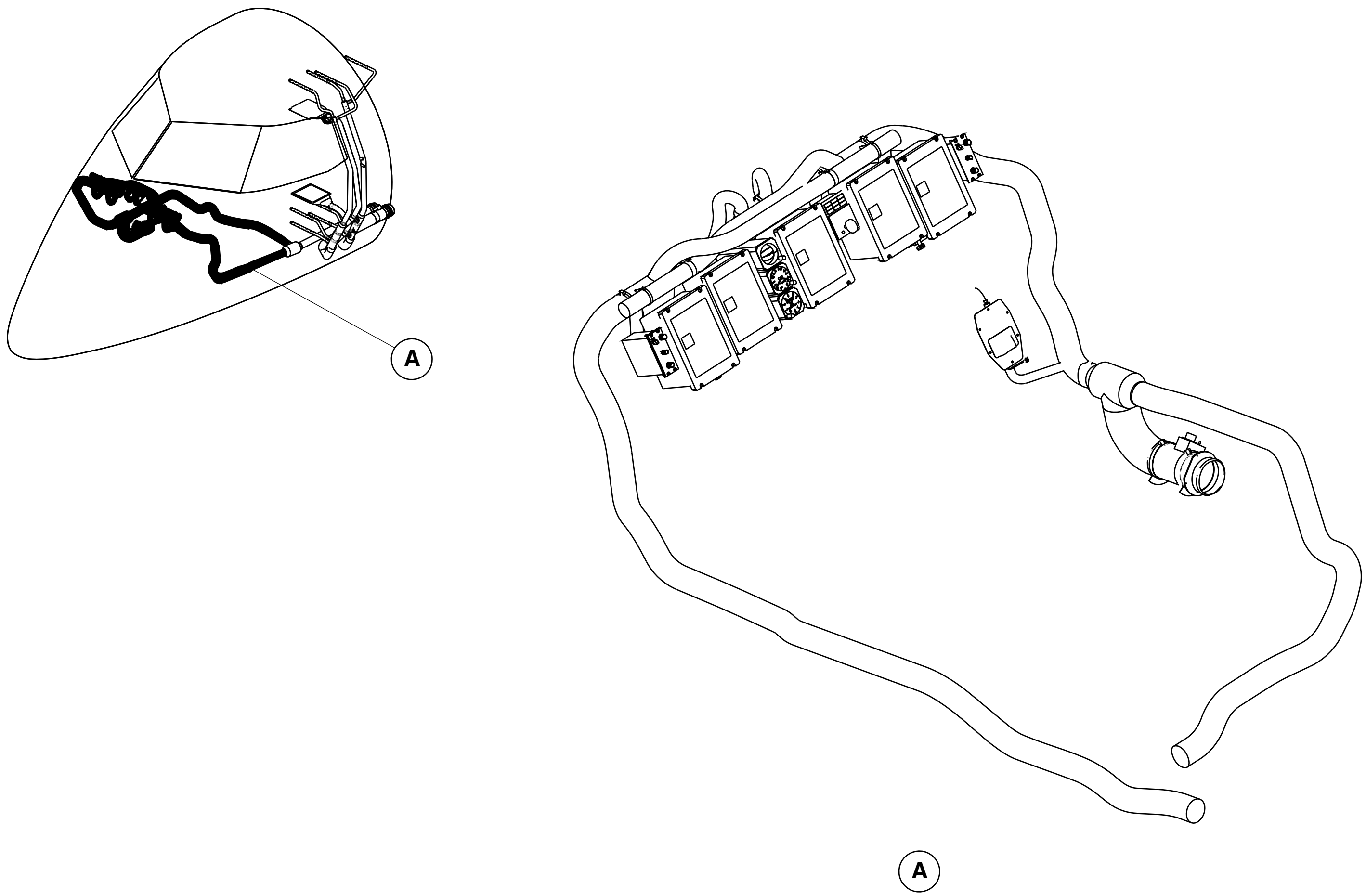
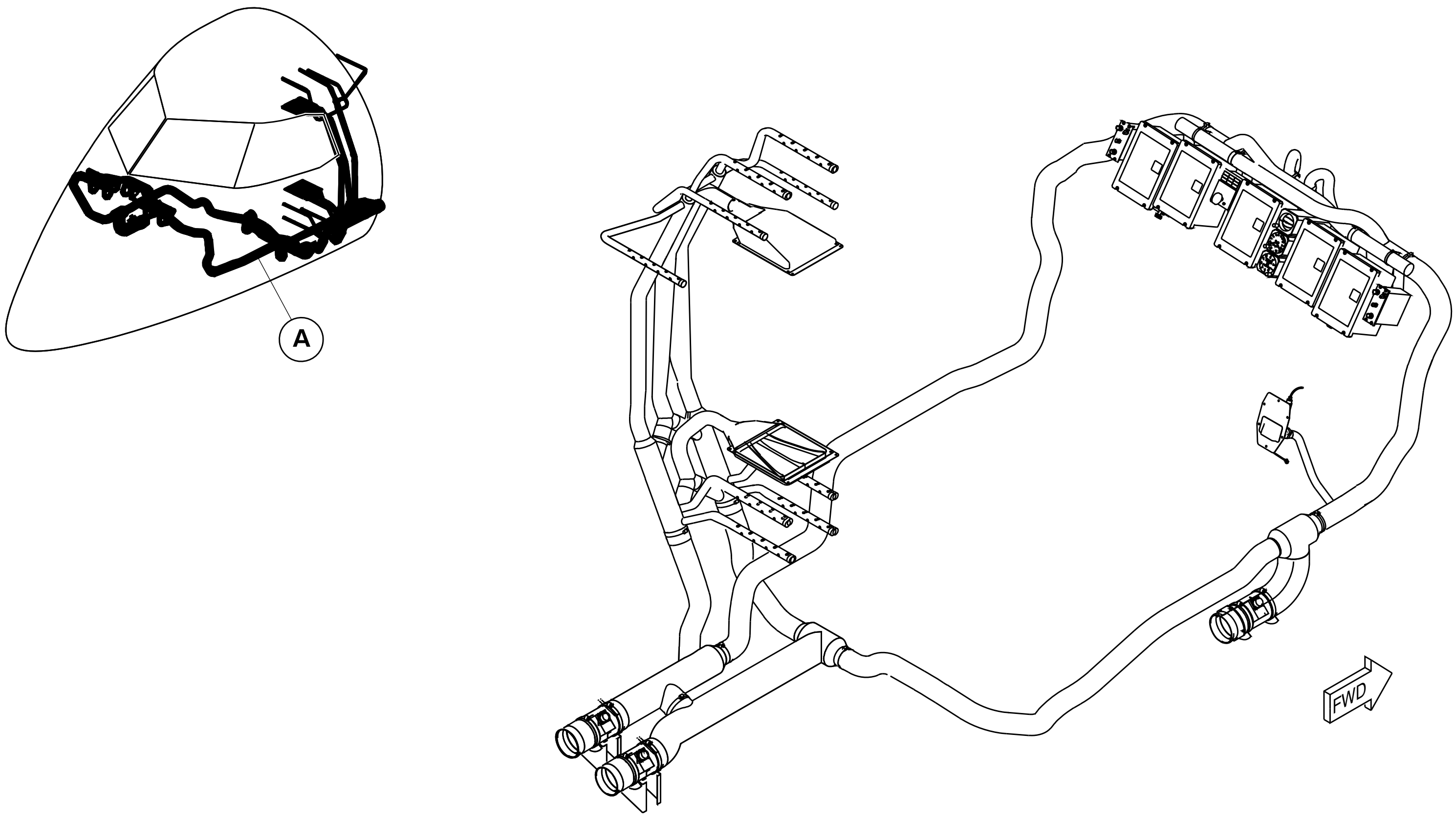
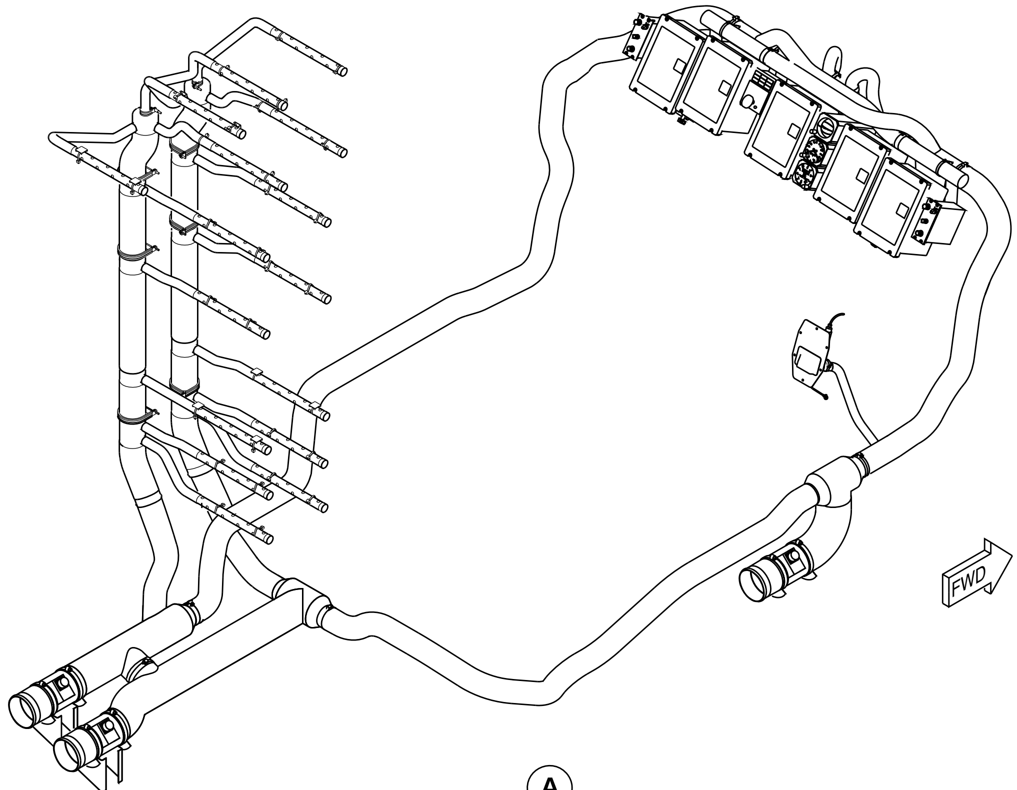
AVIONICS AND LCD COOLING SYSTEM
Figure 1 (Sheet 2 of 2)



LIQUID CRYSTAL DISPLAY (LCD) COOLING SYSTEM
Figure 2



Avionics and LCD Cooling System Ducts Locator
Figure 3 (Sheet 1 of 2)



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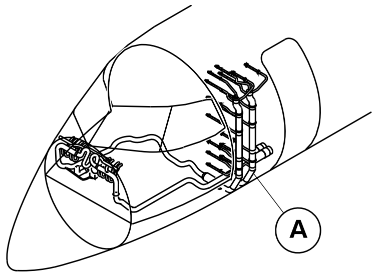
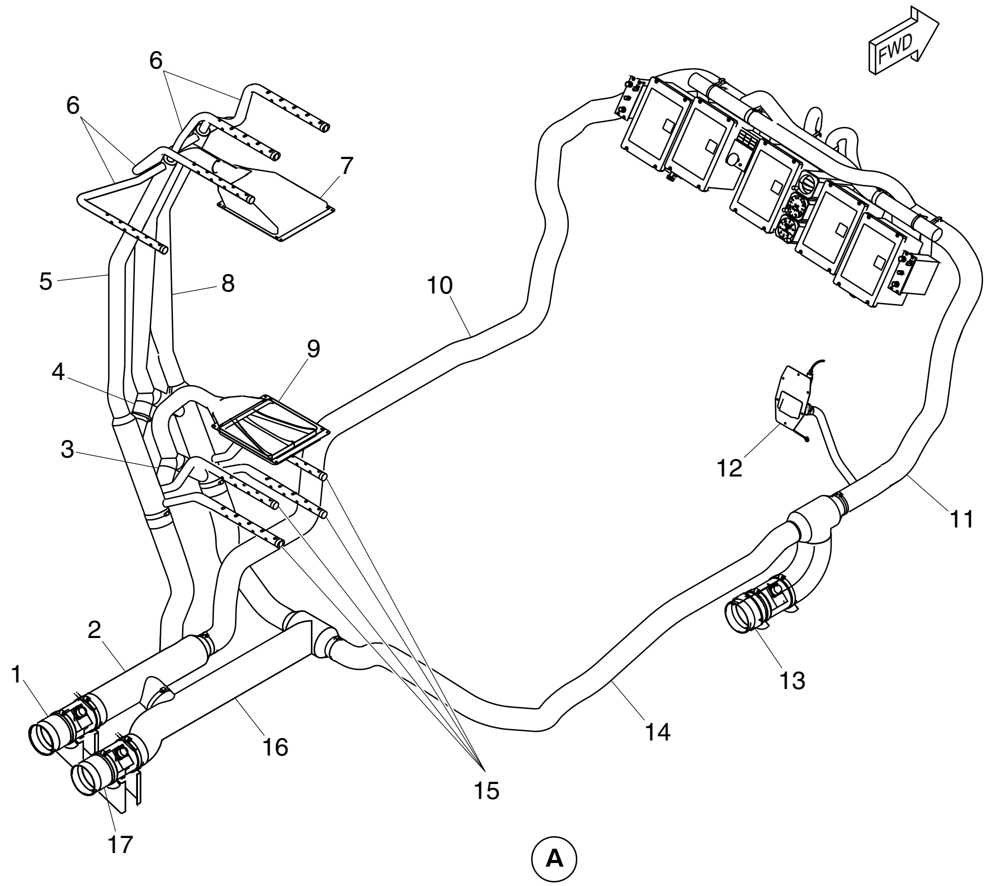
Avionics and LCD Cooling System Ducts Locator
Figure 3 (Sheet 2 of 2)



LEGEND

1. Fan 1 (Pilot Side).
2. Lower Aft Riser.
3. Lower Y–Ducts.
4. Upper Y–Ducts.
5. Upper Aft Riser.
6. Upper Piccolo Tubes.
7. Upper Plenum.
8. Upper Forward Riser.
9. Lower Plenum.
10. Left Underfloor Duct.
11. Right Forward Underfloor Duct.
12. Zone Temperature Sensor and Housing.
13. Fan 3 (Standby).
14. Tee Duct.
15. Lower Piccolo Tubes.
16. Lower Forward Riser.
17. Fan 2 (Copilot Side).



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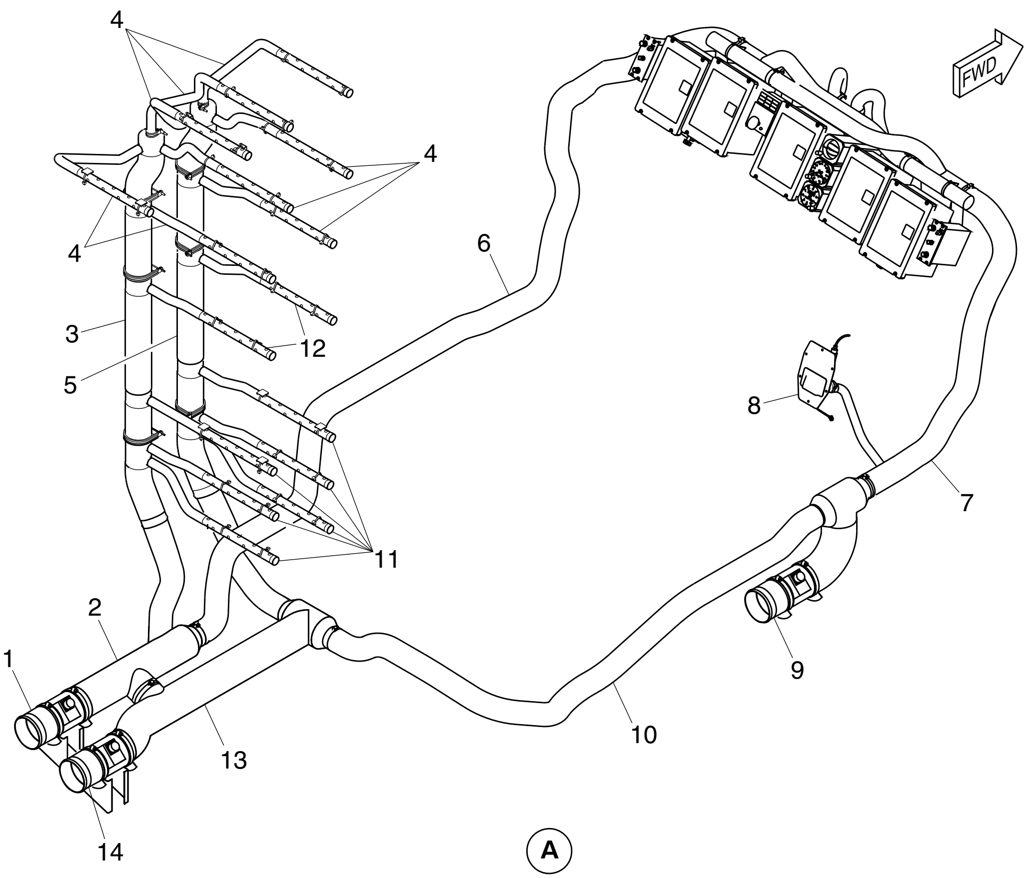
Avionics and LCD Cooling System Ducts Detail Page 1
Figure 4 (Sheet 1 of 2)



LEGEND

1. Fan 1 (pilot side).
2. Lower aft riser.
3. Upper aft riser.
4. Upper piccolo tube.
5. Upper forward riser.
6. Left under floor duct.
7. Right forward under floor duct.
8. Zone temperature sensor and housing.
9. Fan 3 (standby).
10. Tee duct.
11. Lower piccolo tube.
12. Center piccolo tube.
13. Lower forward riser.
14. Fan 2 (co−pilot side).

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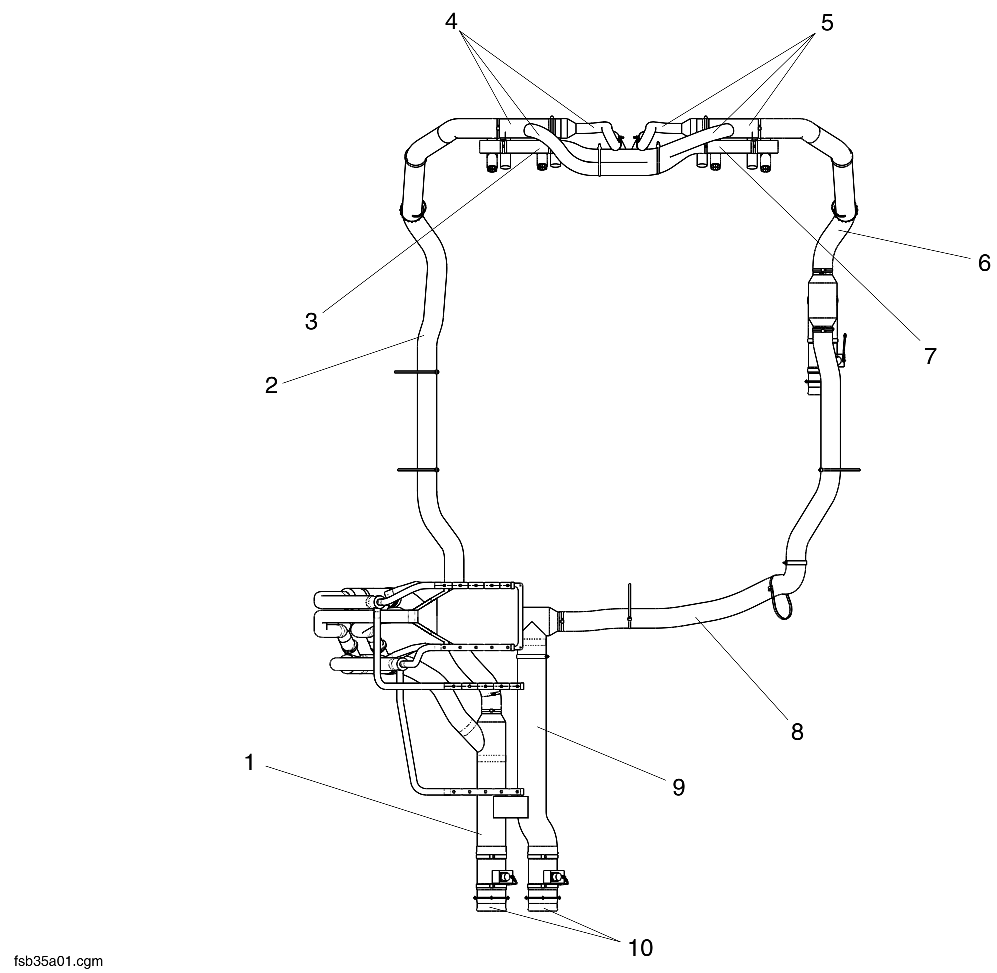
Avionics and LCD Cooling System Ducts Detail Page 1
Figure 4 (Sheet 2 of 2)



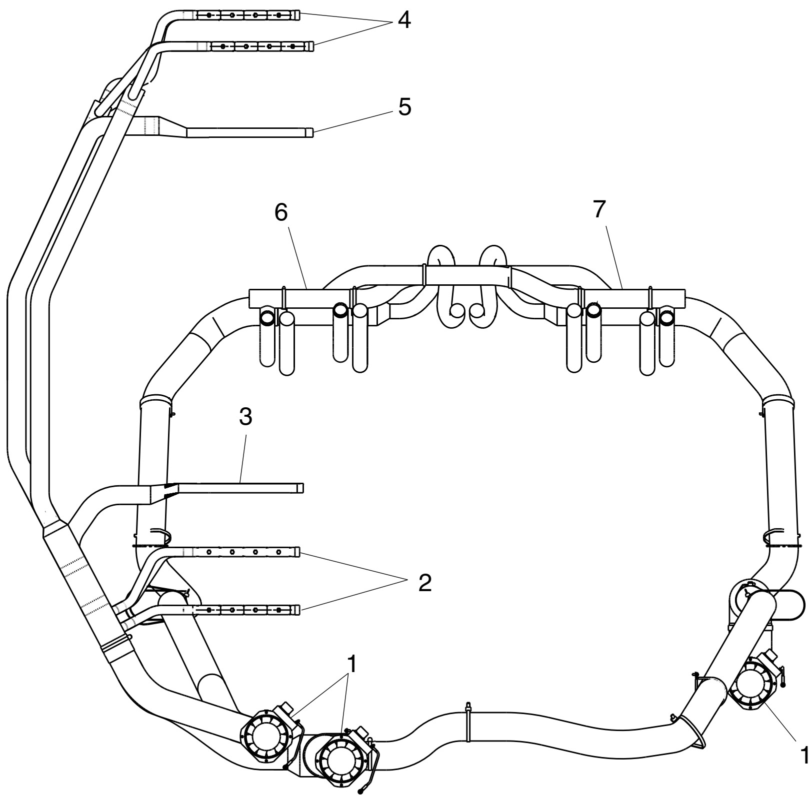
LEGEND

1. Lower Aft Riser.
2. Left Underfloor Duct.
3. Left Double Tee Fitting.
4. Left Manifold Duct.
5. Right Manifold Duct.
6. Right Forward Underfloor Duct.
7. Right Double Tee Fitting.
8. Tee Duct.
9. Lower Forward Riser.
10. Exit Ducts.

NOTE

View looking down.

AVIONICS AND LCD COOLING SYSTEM DUCTS DETAIL PAGE 2
Figure 5



LEGEND

1. Avionics cooling fan.
2. Lower piccolo tube.
3. Lower Plenum.
4. Upper piccolo tube.
5. Upper Plenum.
6. Left double tee fitting.
7. Right double tee fitting.

VIEW LOOKING FORWARD
PRE MODSUM 4–126287

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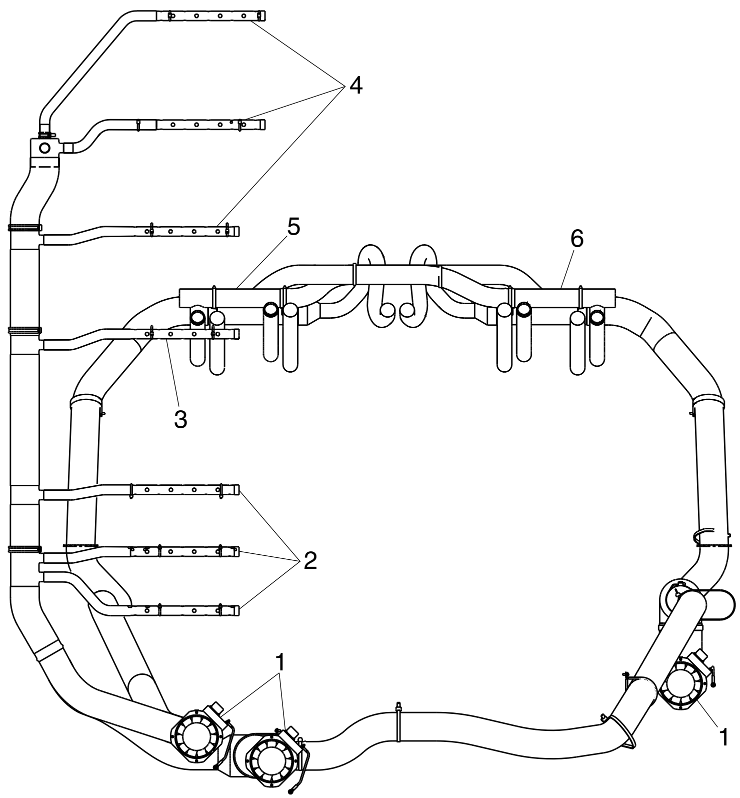
Avionics and LCD Cooling System Ducts Detail Page 3
Figure 6 (Sheet 1 of 2)



LEGEND

1. Avionics cooling fan.
2. Lower piccolo tube.
3. Center piccolo tube.
4. Upper piccolo tube.
5. Left double tee fitting.
6. Right double tee fitting.

VIEW LOOKING FORWARD
POST MODSUM 4–126287

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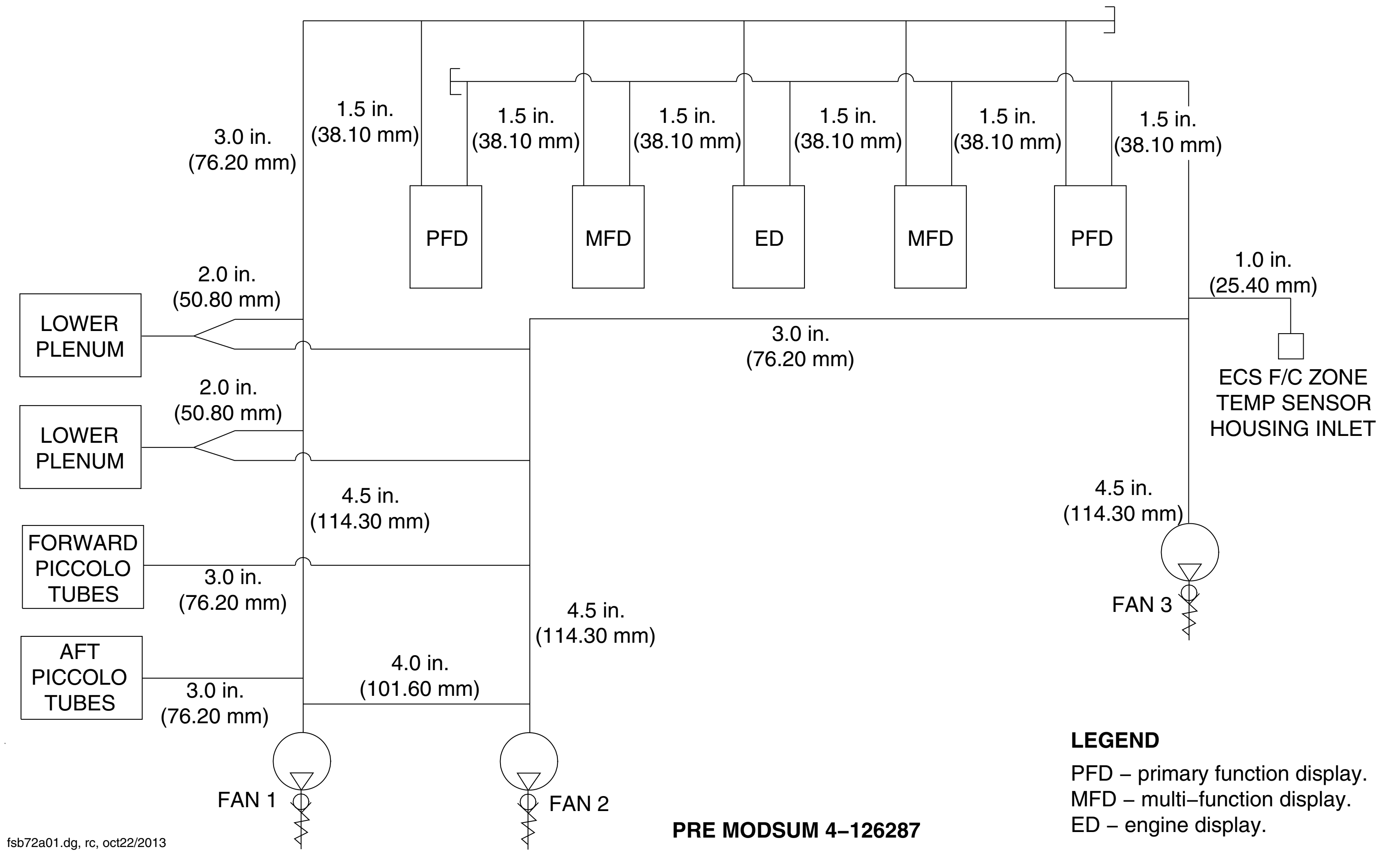
Avionics and LCD Cooling System Ducts Detail Page 3
Figure 6 (Sheet 2 of 2)



LEGEND

PFD – primary function display.
MFD – multi–function display.
ED – engine display.

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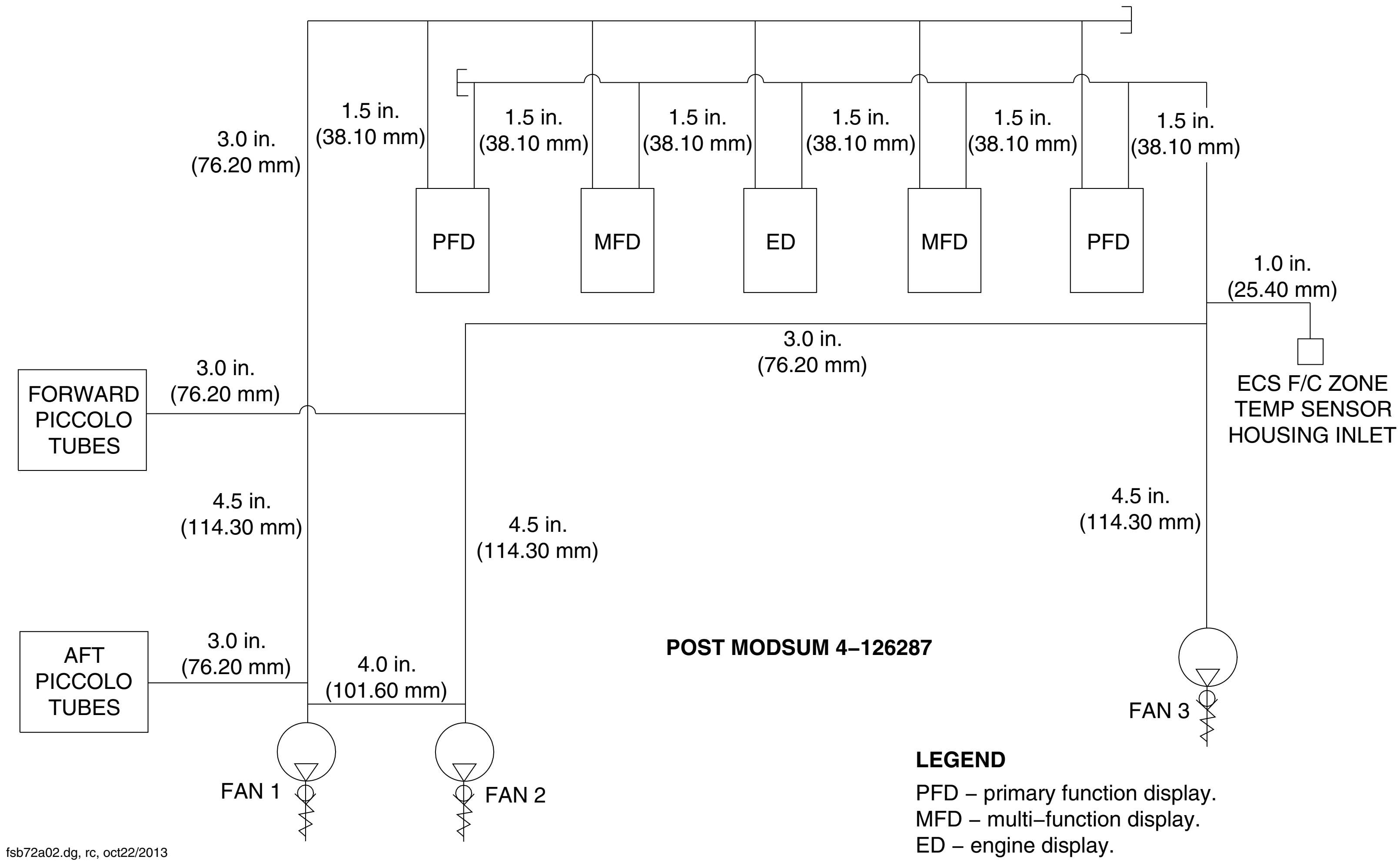
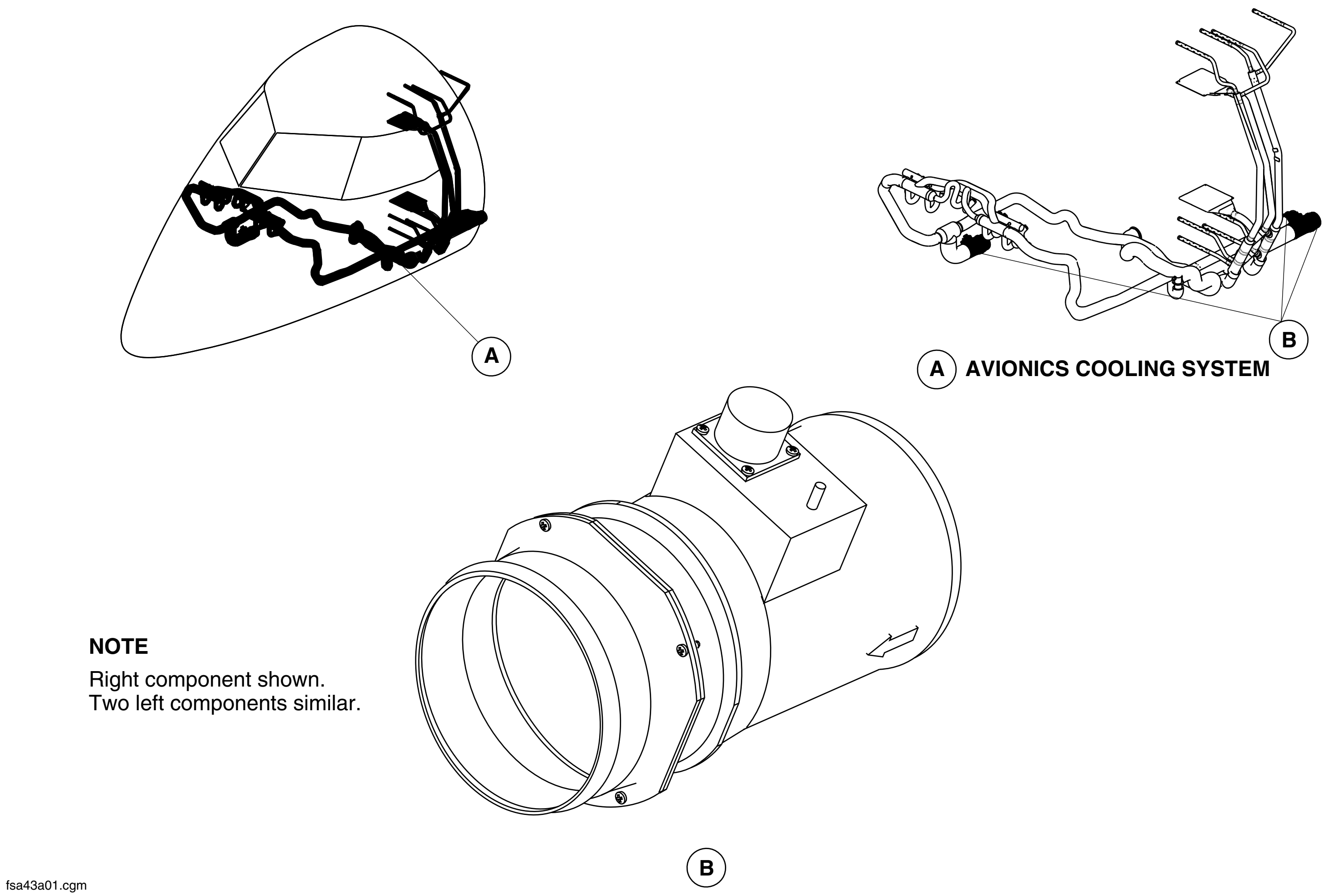
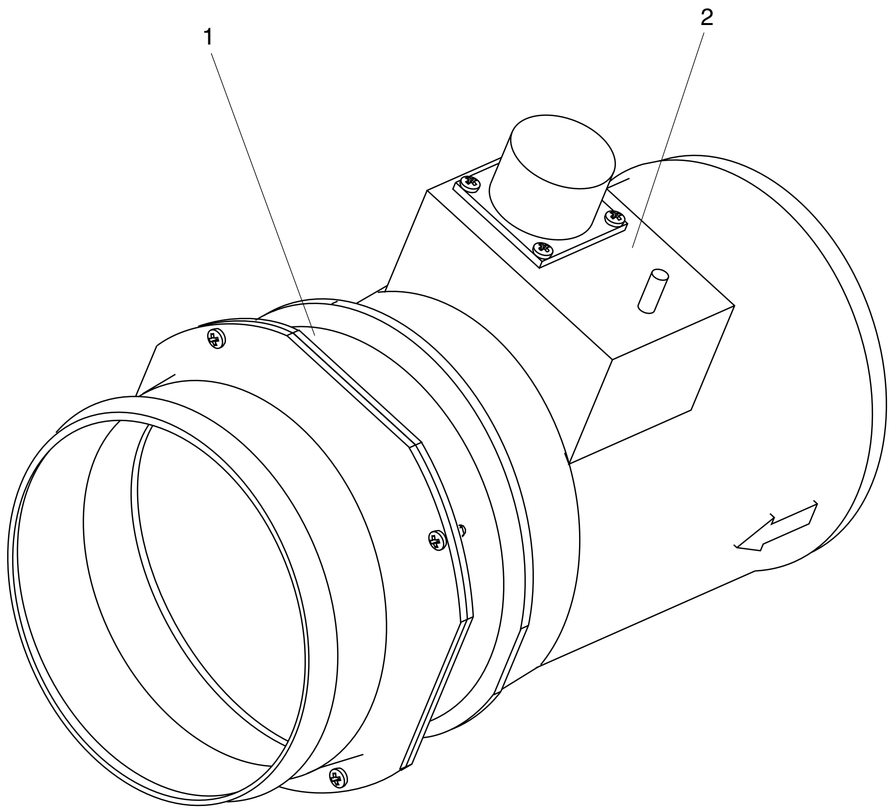
Avionics and LCD Cooling System Ducts Detail Page 3
Figure 7 (Sheet 1 of 2)



Avionics and LCD Cooling System Ducts Detail Page 3
Figure 7 (Sheet 2 of 2)



AVIONICS AND LCD COOLING FANS LOCATOR
Figure 8



LEGEND

1. Fan.
2. Speed Control Unit.

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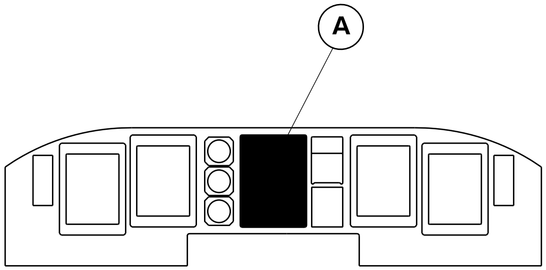
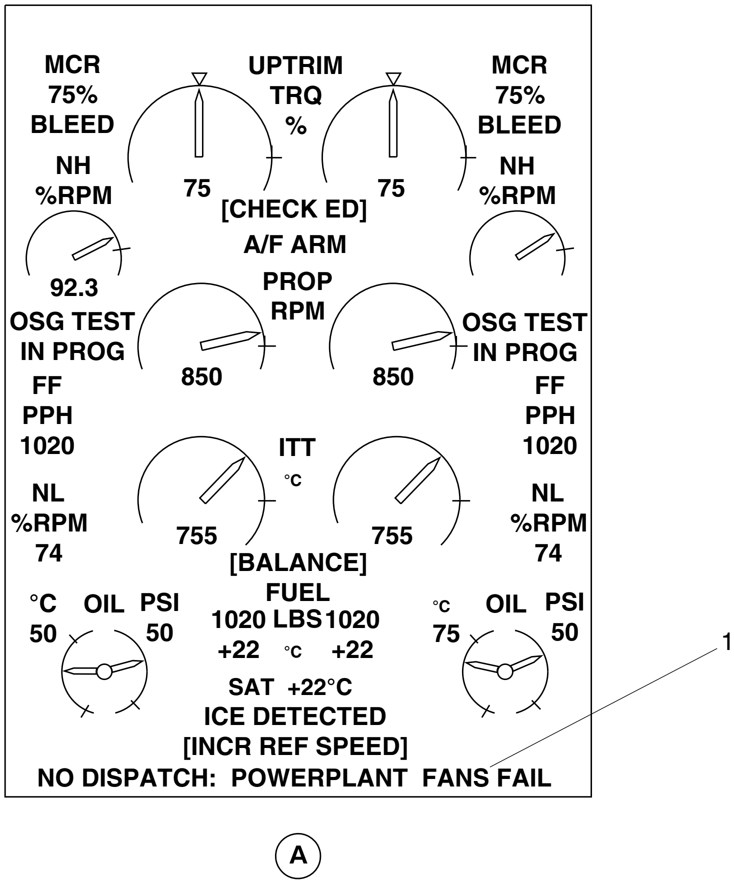
AVIONICS AND LCD COOLING FANS DETAIL
Figure 9



MAIN INSTRUMENT PANEL



LEGEND

1. No Dispatch Advisory Message.

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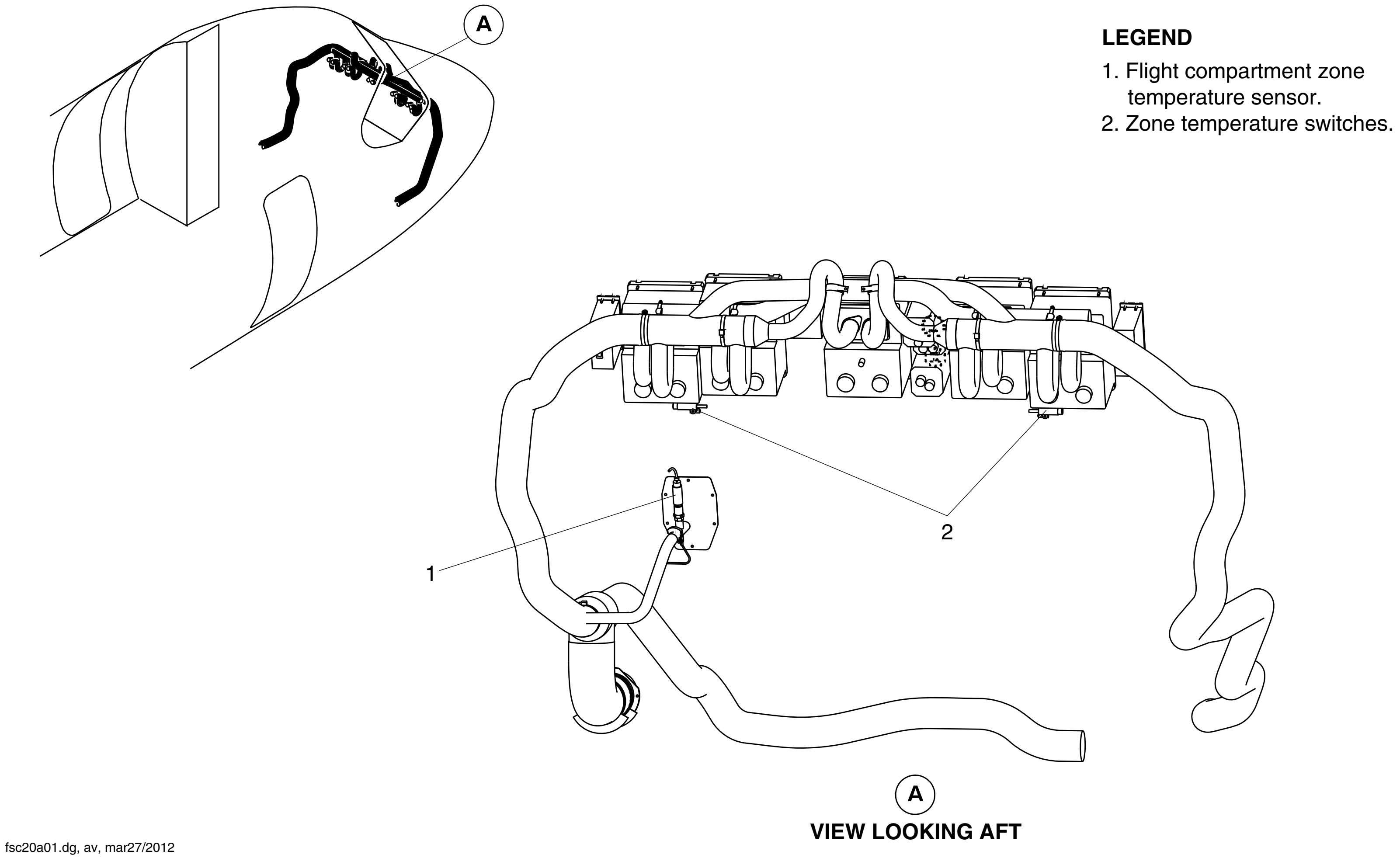
Engine Display / Avionics Advisory Message
Figure 10



Zone Temperature Switches
Figure 11

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21–30–00–001

PRESSURIZATION CONTROL

Introduction

The cabin pressure control system keeps a constant cabin pressure during the ground and flight modes. The system controls the cabin altitude and the cabin rate of change.

General Description

Refer to Figure 1.

The pressurized area of the fuselage extends from the front pressure bulkhead immediately forward of the windshield at aircraft station X −111 to the rear dome at Sta X 856. This area includes the flight compartment, passenger compartment, under floor space and baggage compartment.

The Cabin Pressure Control Panel controls the system in automatic and manual modes. The Cabin Indication Module shows the data to monitor the system.

When cabin altitude is too high, the cabin pressure and the cabin indication module send the warning signal to the aircraft systems.

Electrical power supply for pressurization control is supplied by the 28 VDC left main bus through a CABIN PRESS AUTO CONT circuit breaker.

The pressurization control system has the components that follow:

- Cabin Pressure Control Panel (21–31–01)
- Cabin Pressure Controller (21–31–06)
- Cabin Indication Module (21–31–11)
- Aft Outflow Valve (21–31–16)
- Forward Safety Valve (21–31–21)
- Forward Safety Valve Selector (21–31–26)
- Aft Safety Valve (21–31–31)
- Air Jet Pump (21–31–36).

Detailed Description

The system shows in the flight compartment:

- Cabin altitude
- Cabin altitude rate of change
- Differential pressure
- Cabin altitude warning.

Refer to Figures 2 and 3.

The aft outflow valve is for automatic and manual control of the pressurization. The aft outflow valve can also be used to dump the pressurization. The forward safety valve is for emergency operation and for smoke removal from the flight compartment. The aft safety valve and the forward safety valve keep the positive and negative pressure relief constant.

The system functions in the modes that follow :

- Normal/Automatic
- On ground
- Take–off
- Landing
- Emergency/Smoke Removal
- Pressure dump.

Normal/Automatic: The pressurization control system is electrically operated. The pressurized area of the fuselage is supplied with a constant flow of conditioned air from the engine bleed air systems through the Air Conditioning Unit (ACU). The pressure in the fuselage is controlled by system modulation of the aft outflow valve, to control the amount of air let out of the fuselage. If the external ambient pressure is more than the fuselage pressure, the safety valves will open for negative pressure relief.

Refer to Figures 4 and 5.

When electrical power is first supplied to the system, a full electrical test of the system is done. The FAULT alert light, on the Cabin Pressure Control Panel comes on during the power up test mode. If there is a failure in the system, the light will stay on. The system operation is fully automatic with the data programmed into the controller.

On ground: When the aircraft is on the ground with the weight on the wheels and the No. 2 engine power lever angle is set less than 60 degrees, electrical power is supplied through the open contacts of the energized landing gear relay. The relay is energized through the Proximity Switch Electronic Unit (PSEU). During the ground mode the aft outflow valve is at the fully open position to prevent aircraft pressurization.

Take–off: When the No. 2 engine power lever angle is set to greater than 60 degrees the controller sends a signal to the aft outflow valve to open or close, as necessary, to pressurize the aircraft to 400 ft (121.9 m) less than ambient. The aft outflow valve moves from the fully open position and starts to modulate to control the pressure changes that occur after take–off. When the landing gear relay is de–energized after take–off (through the PSEU), the aft outflow valve modulates to keep the set aircraft pressure.

Landing: The aircraft depressurization is controlled automatically. If the set field altitude is higher than actual field altitude, the aircraft will land unpressurized. If the field altitude is set less than actual field altitude, the aircraft will land pressurized. On landing, cabin altitude will go back to field altitude at the rate programmed into the controller, for one minute before cabin pressure is bled to ambient.

Manual: The manual mode is used if the automatic pressurization mode does not operate. Pressurization can be controlled through the aft outflow valve, when the AUTO–MAN–DUMP switch is set to MAN. The cabin pressure is set with the toggle switch moved and held to the DECR position, to open the aft outflow valve and increase the cabin altitude. When the toggle switch is moved and held to INCR, the aft outflow valve closes and the cabin altitude decreases.

Emergency/Smoke Removal: Pressurization can be controlled through the forward safety valve when the AUTO–MAN–DUMP switch is set to MAN. Cabin pressure can be regulated by turning the FWD OUTFLOW knob, as necessary, to adjust the amount of pressure bleed to get the required pressurization selection. When the control knob is turned clockwise the forward safety valve opens and the cabin pressure decreases. Pressurization can also be reduced by operating the forward safety valve selector on the copilot's side console.

If the AUTO/MAN/DUMP switch is set in automatic mode, and the FWD OUTFLOW knob is turned to open the forward safety valve, the cabin pressure will decrease and the aft outflow valve will start to close. When the FWD OUTFLOW knob is set to a higher cabin altitude (lower cabin pressure) than that set in automatic mode it will override the automatic selection.

During ground operation with the AUTO/MAN/DUMP switch selected to MAN, the FWD OUTFLOW knob turned fully clockwise (forward safety valve set to open) and all doors and hatches closed, the aircraft will begin to pressurize. The forward safety valve will be slow to start to modulate and cause an increase of cabin pressure.

Pressure dump: The fast depressurization function may be done in the automatic and the manual modes. The AUTO/MAN/DUMP switch set to DUMP, fully opens the aft outflow valve. In the manual mode, the aft outflow valve opens when the toggle switch is moved and held in the DECR position. On the ground the AUTO/MAN/DUMP switch is set to DUMP, to make sure that all pressure is bled off, before any doors or hatches are opened.

Cabin Pressure Control Panel

Refer to Figures 4 and 5.

The cabin pressure control panel is located in the flight compartment on the overhead panel. The panel has mechanical and hardware subassemblies.

The mechanical subassembly has:

- FWD OUTFLOW rotary select knob
- AUTO/MAN/DUMP select switch
- Fault alert light
- Landing altitude indicator
- LDG ALT rotary select knob
- Needle mechanically driven by the landing altitude potentiometer
- MAN DIFF INC/DECR toggle switch.

The FWD OUTFLOW knob is connected to the cabin pressure and the reference pressure. The knob controls the pneumatic opening of the forward safety valve. When the knob is turned clockwise, the valve opens and it increases the cabin rate of change.

The AUTO/MAN/DUMP switch is used to set the mode of operation for pressurization control. In the automatic mode, the pressurization control is fully automatic. The manual mode is used if the automatic pressurization mode does not operate. In the DUMP mode, the aft outflow valve is in the open position and the aircraft can be operated unpressurized.

The FAULT alert light gives a visual indication of failures in the cabin pressure control system.

The LDG ALT select knob is set before take–off. A needle is mechanically driven by the LDG ALT knob and gives the set landing altitude information to the flight crew. The LDG ALT knob sends a signal to the Cabin Pressure Controller. If the landing altitude is not set, the controller input becomes the landing altitude. The potentiometer selections are from –2000 to 14,000 ft (–609.6 to 4267 m) .

Cabin Pressure Controller

Refer to Figures 6 and 7.

The cabin pressure controller is located on the avionics rack, and operates the auto channel of the electrical actuator of the aft outflow valve to control the cabin pressure.

A pressure sensor measures the cabin pressure. The sensor gives the a warning, of cabin altitude too high, to the Cabin Indication Module

Cabin Indication Module

Refer to Figures 8 and 9.

The cabin indication module is located in the flight compartment on the overhead panel and has a differential pressure indicator, a cabin altitude indicator, a cabin rate of change indicator, and a pressure sensor. The differential pressure indicator gives a differential pressure from −0.5 to 5.8 ± 0.15 psi (−3.45 to 40 ±1.03 kPa) between the cabin pressure and the outside static pressure. The pressure sensor gives the excessive altitude warning received from the Cabin Pressure Controller to the CABIN PRESS warning light. The CABIN PRESS warning light on the Caution and Warning panel comes on when the cabin altitude reaches 10,000 ft (3048 m), except when take−off or landing altitude is above 8000 ft (2438.4 m). For take−off or landing altitudes above 8000 ft (2438.4 m) the warning light will come on in the conditions that follow:

- The aircraft take−off altitude + 1000 ft (304.8 m)
- Landing altitude + 1000 ft (304.8 m).

Aft Outflow Valve

Refer to Figures 2 and 3.

The aft outflow valve, located on the rear pressure bulkhead, has a subassembly attached to an electrical actuator. The subassembly has a body, a butterfly with a sealing gasket, and a shaft that goes through the butterfly and is supported by two bearings. The electrical actuator has a dc motor, a gearbox and a control box.

When the cabin pressure controller is set to pressurize the cabin, the aft outflow valve is closed. Air from the cooling packs is supplied to the cabin. Since no air can be released, the cabin pressure increases. Once the cabin has reached the set pressure, the cabin sensor automatically sends a signal to the pressure controller to modulate the outflow valve through the electrical actuator. The dc motor of the electrical actuator moves the shaft to open the butterfly chamber of the aft outflow valve. The air is released from the cabin to regulate the pressure. The chamber pressure is kept below the cabin pressure.

Forward Safety Valve

Refer to Figures 10 and 11.

The forward safety valve body is located in the nose compartment and is installed on the bulkhead flange. The valve has a cover that includes an overpressure relief box and a filter box, a poppet/diaphragm assembly, a deflector, a spring, and a pneumatic relay that includes a manometric capsule and a check valve.

The forward safety valve automatically releases the cabin pressure when preset pressure limits are approached. The cabin pressure is applied to the inside of the overpressure relief box and the pneumatic relay. The poppet/diaphragm assembly releases the cabin pressure through the filter box. When the cabin pressure is more than the combined force of the outside static pressure and the spring, the check valve of the manometric capsule is opened. When the check valve opens, the cabin pressure of the poppet/diaphragm assembly starts to flow out to outside static pressure through the hollow center of the forward safety valve. The reduced pressure in the chamber of the poppet/diaphragm assembly lets the spring open the poppet/diaphragm assembly. The cabin air flows out.

To prevent high cabin pressure, the safety valves have an automatic positive pressure–relief control system operation. The force of the cabin pressure applied on one side of the poppet/diaphragm assembly is more than the force of the outside static pressure and the spring on the other side of the poppet/diaphragm assembly. The cabin pressure in the chamber is released and the cabin pressure decreases.

The safety valves also have a negative pressure relief function. During an emergency descent or without cabin airflow, the outside static pressure becomes more than the cabin pressure. The force of the cabin pressure applied on one side of the poppet/diaphragm assembly is less than the force of the outside static pressure and the spring on the other side of the poppet/diaphragm assembly. The differential pressure between the outside and the cabin moves the deflector and the poppet/diaphragm assembly opens. The air in the chamber is released into the cabin through the filter box. The outside air flows into the cabin and the cabin pressure increases.

The forward safety valve has a pneumatic emergency RAM air and smoke evacuation function. The forward safety valve can be opened with the forward safety valve selector or with the FWD OUTFLOW knob of the cabin pressure control panel. The pneumatic relay opens and the outside static pressure creates a negative pressure in the chamber and the poppet/diaphragm assembly opens. The air in the chamber is released.

Forward Safety Valve Selector

Refer to Figures 12 and 13.

The forward safety valve selector is located on the copilot side console and completely opens the forward safety valve to dump the cabin pressure. The selector is a shut–off valve. NORMAL or OPEN positions can be selected with the rotary switch.

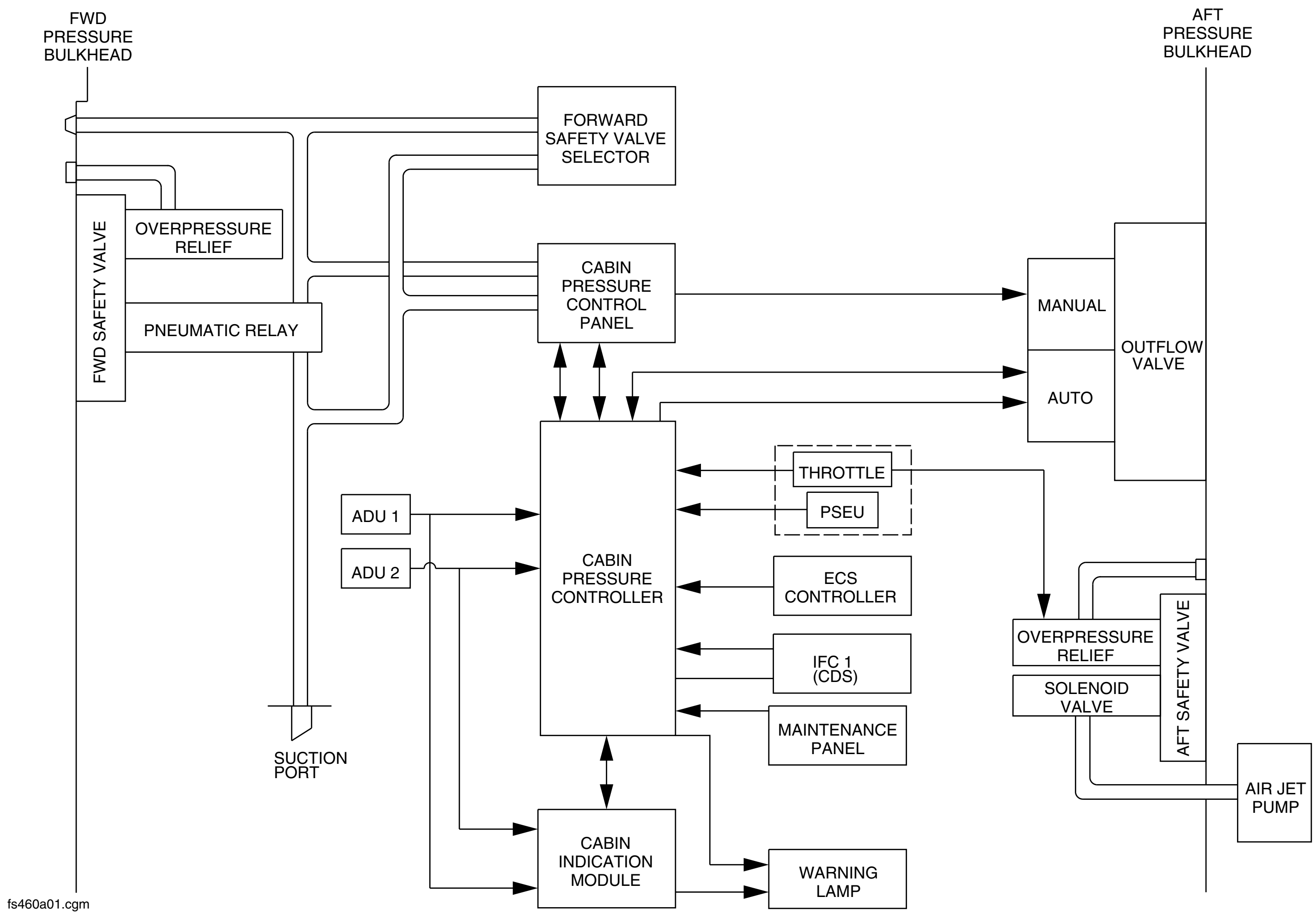
Aft Safety Valve

Refer to Figures 14 and 15.

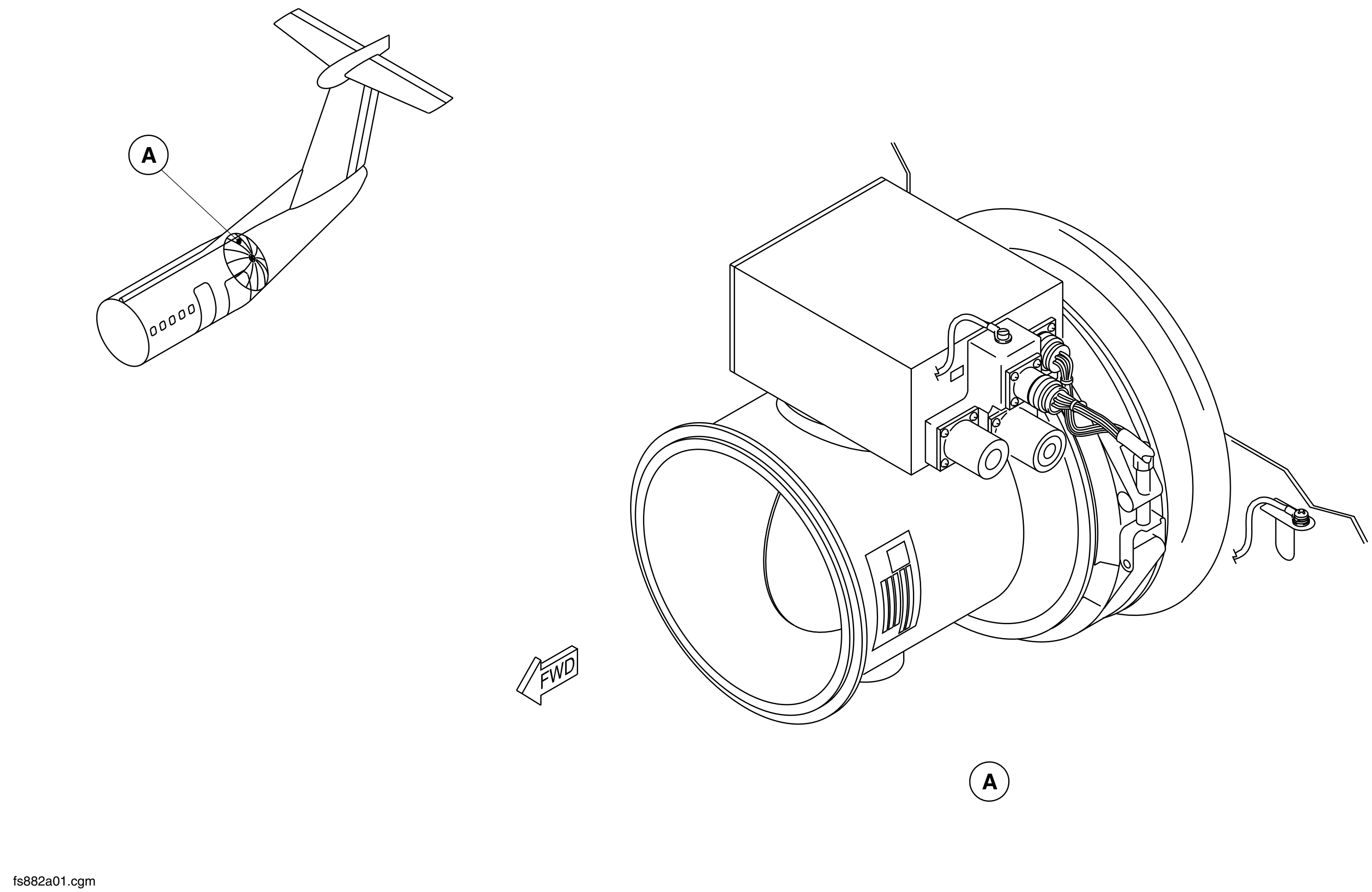
The aft safety valve is located on the aft pressure bulkhead and opens on the ground when the engines are in the idle condition. The aft safety valve is the same as the forward safety valve, but has a solenoid valve instead of a pneumatic relay.

Air Jet Pump

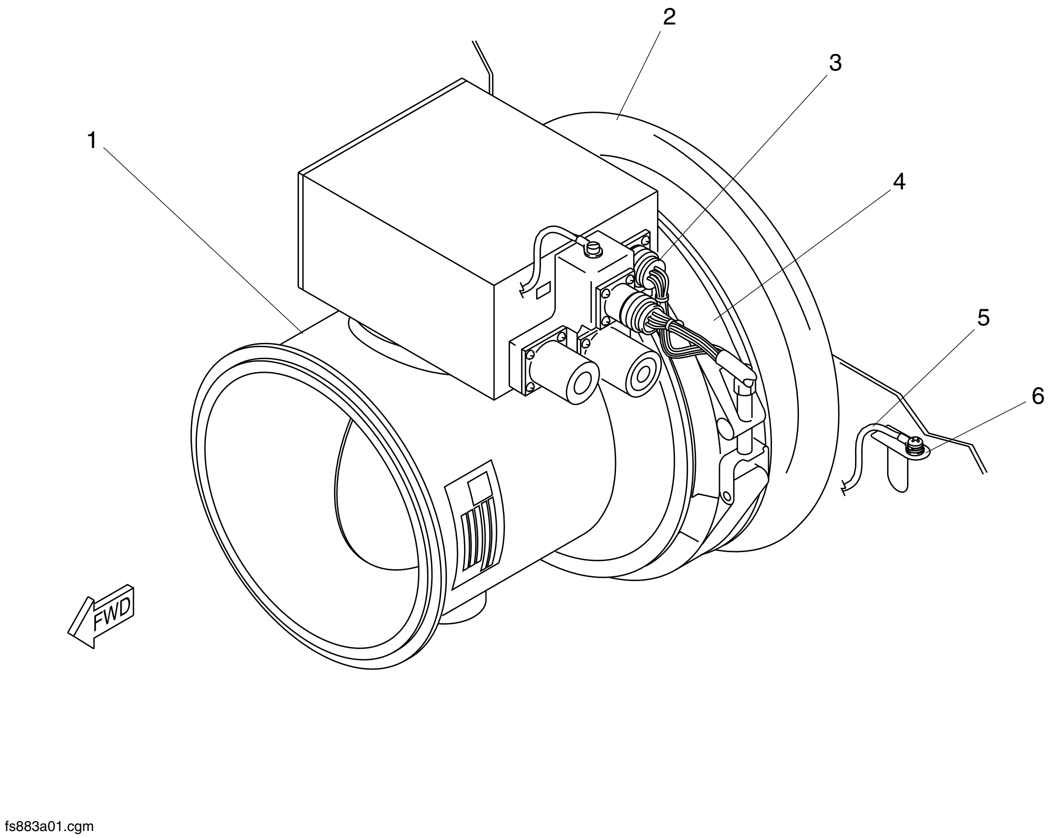
The air jet pump located on the rear pressure bulkhead, has a body and three orifices. The pump uses the bleed air to cause a negative pressure on the solenoid valve of the aft safety valve. The pump fully opens the aft safety valve on the ground. The pump is supplied with air from the de–icing line or the APU.



Pressurization Control Synoptic.
Figure 1



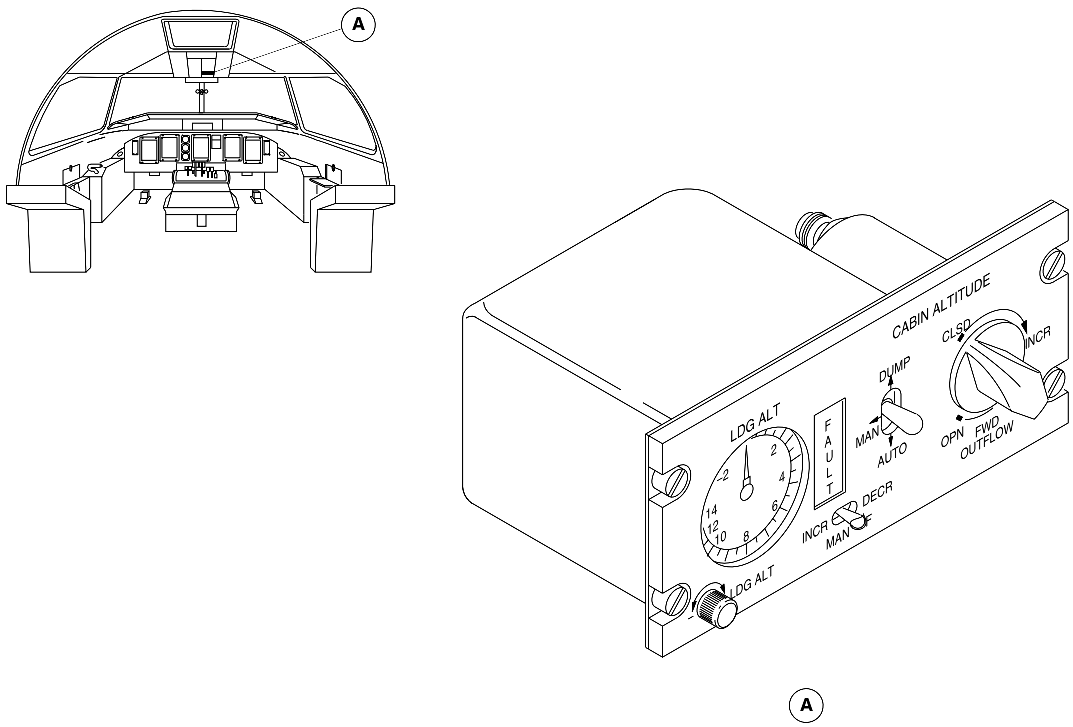
Outflow Valve Locator
Figure 2



LEGEND

1. Aft Outflow Valve.
2. Adapter.
3. Electrical Connector.
4. Coupling.
5. Bonding Jumper.
6. Bracket.

Outflow Valve Detail
Figure 3



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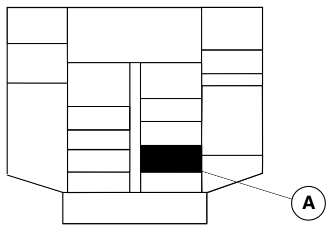
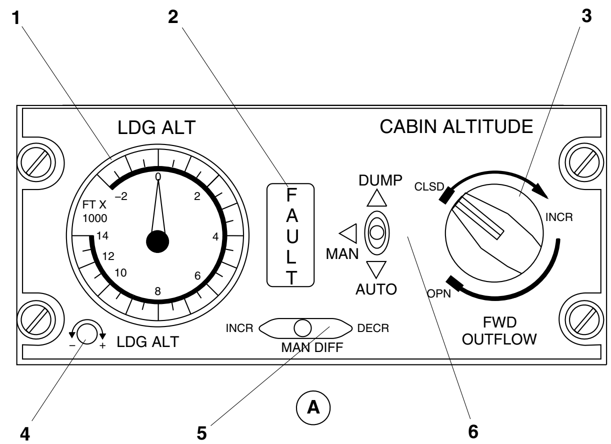
Cabin Pressure Control Panel Locator
Figure 4



OVERHEAD CONSOLE



LEGEND

1. Landing Altitude Indicator.
2. Fault Warning Lamp.
3. Pressure Rate of Change Rotary Knob.
4. Landing Altitude Potentiometer.
5. Manual Differential Pressure Toggle Switch.
6. Selection Switch.

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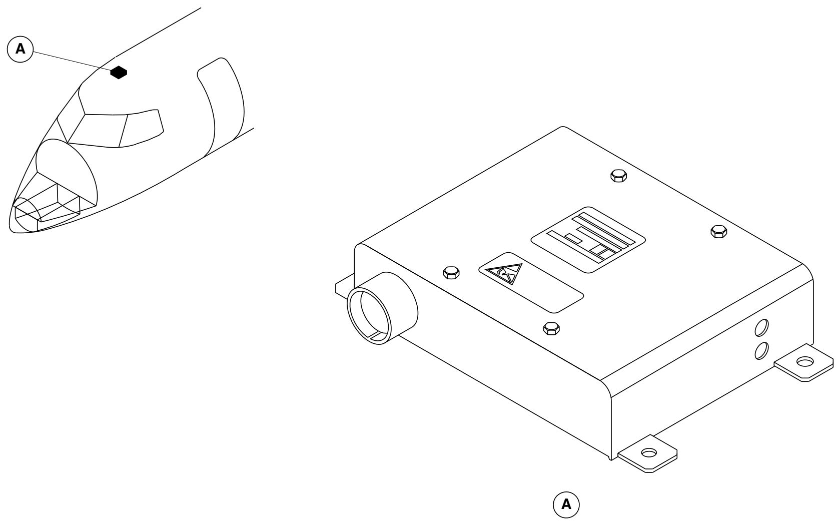
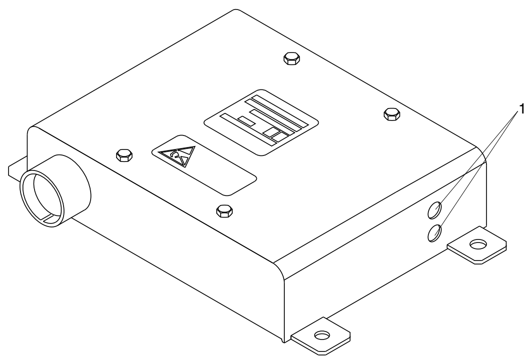
Cabin Pressure Control Panel Detail
Figure 5



Cabin Pressure Controller Locator
Figure 6

LEGEND

1. Pressure Sensor.



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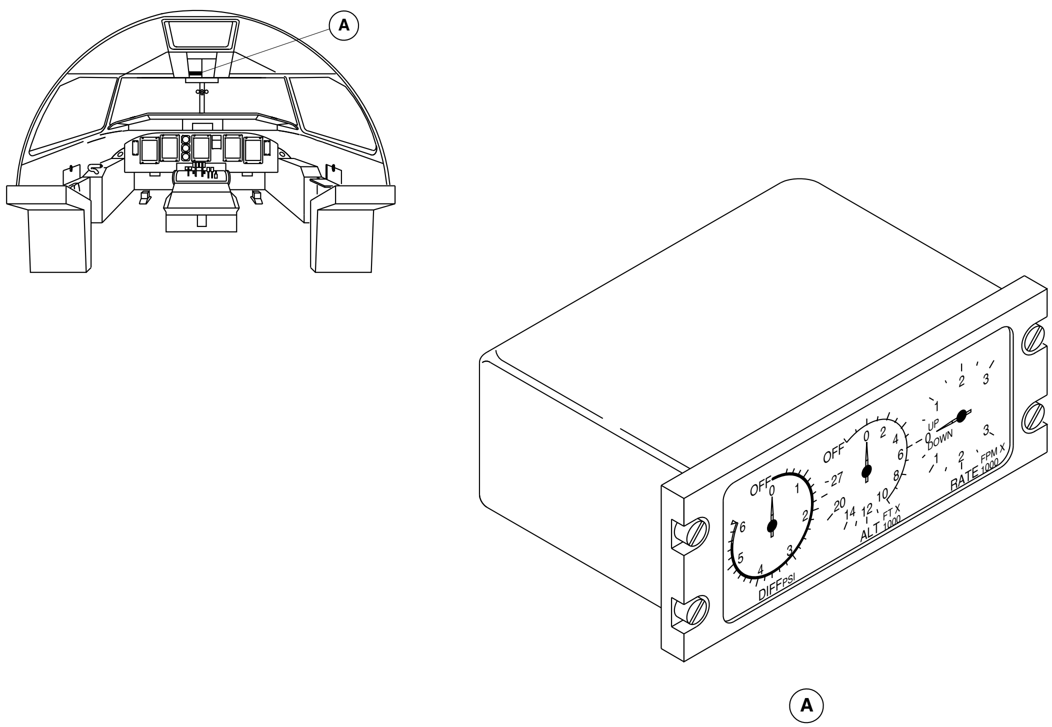
Cabin Pressure Controller Detail
Figure 7



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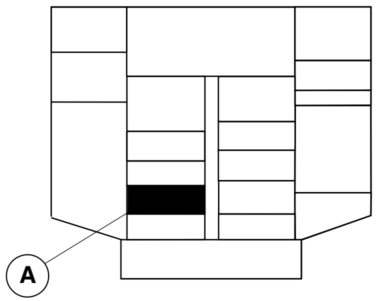
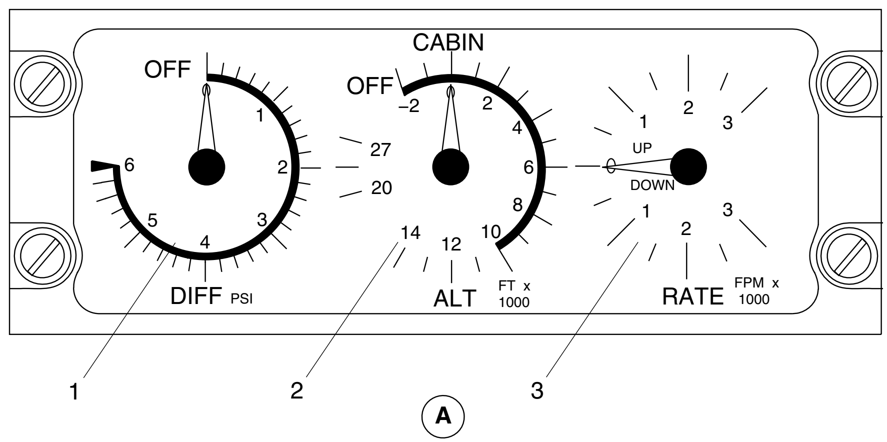
Cabin Indication Module Locator
Figure 8



OVERHEAD CONSOLE



LEGEND

1. Differential Pressure Quadrant.
2. Cabin Altitude Quadrant.
3. Cabin Rate of Change Quadrant.

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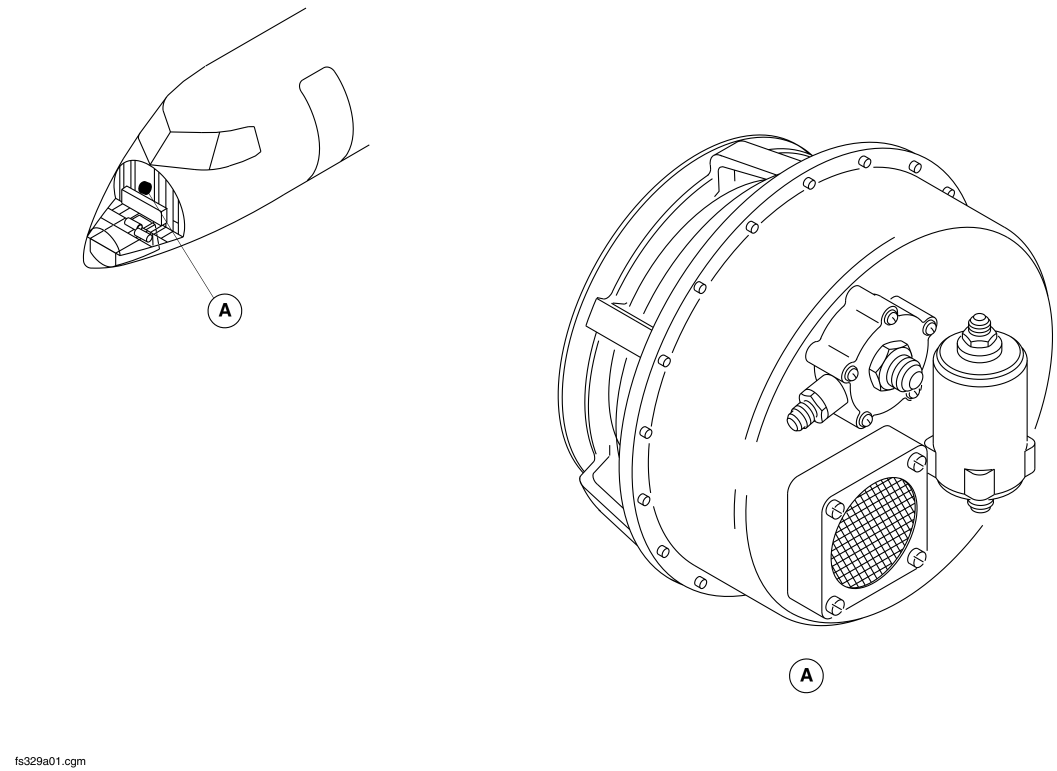
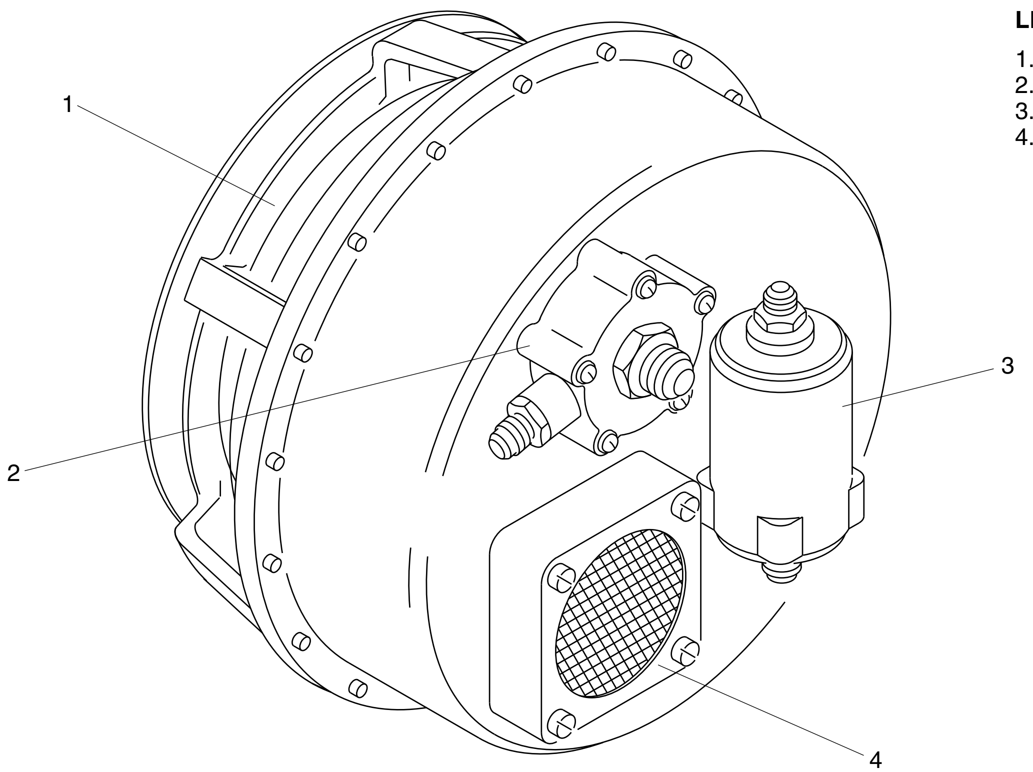
Cabin Indication Module Detail
Figure 9



Forward Safety Valve Locator
Figure 10



LEGEND

1. Poppet Assembly.
2. Overpressure Relief Box.
3. Pneumatic Relay.
4. Filter.

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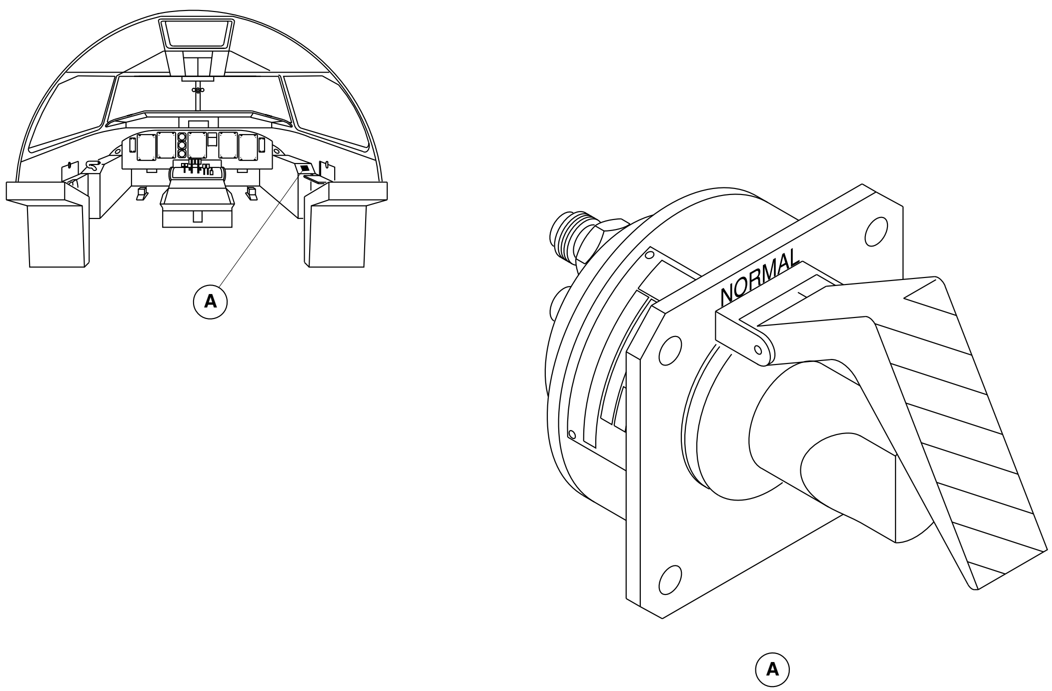
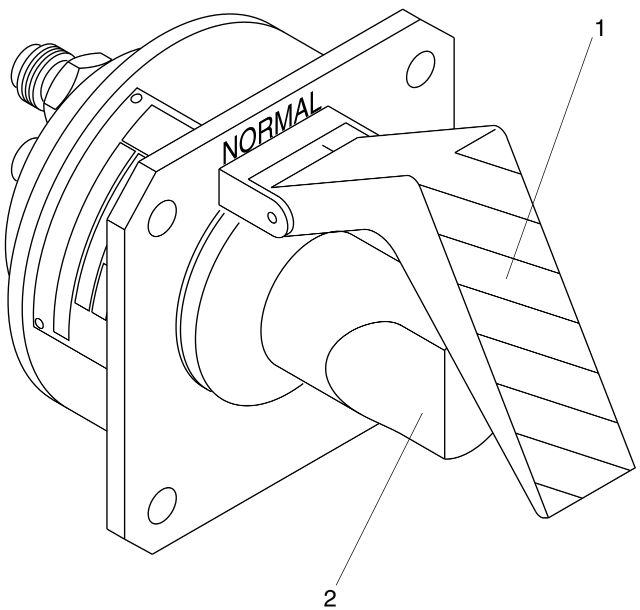
Forward Safety Valve Detail
Figure 11



Forward Safety Valve Selector Locator
Figure 12

LEGEND

1. Switch Guard.
2. Rotary Switch.



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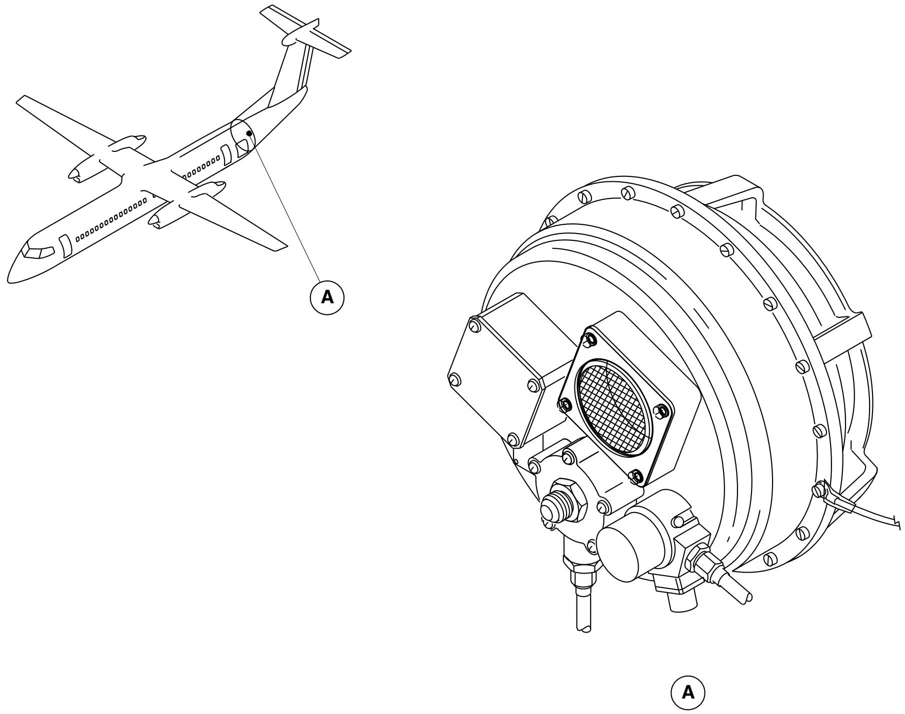
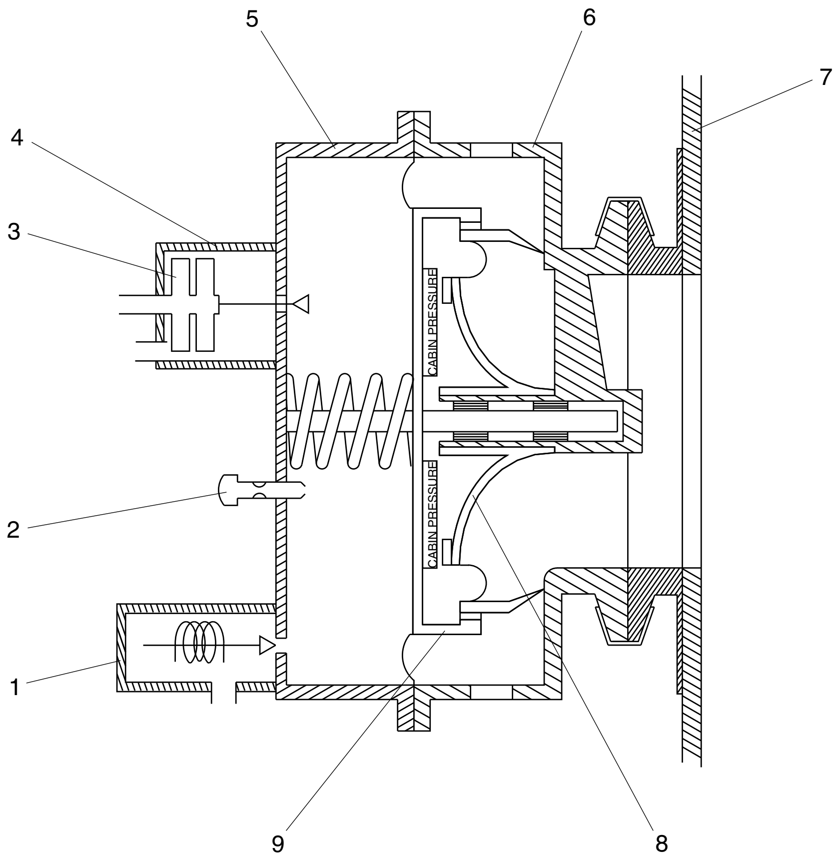
Forward Safety Valve Selector Detail
Figure 13



Aft Safety Valve Locator
Figure 14



LEGEND

1. Solenoid Valve.
2. Filter Box.
3. Manometric Capsule.
4. Overpressure Relief Box.
5. Cover.
6. Body.
7. A/C Pressure Bulkhead.
8. Deflector.
9. Poppet/Diaphragm Assembly.

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Aft Safety Valve Detail
Figure 15

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21–50–00–001

COOLING

Introduction

The air conditioning system supplies the desired flow of conditioned air to the aircraft air distribution system. The air is conditioned to the proper temperature and humidity required to maintain a comfortable environment for the crew and passengers.

General Description

The air conditioning pack is part of the Environmental Control System (ECS). It uses bleed air from the engines or Auxiliary Power Unit (APU) to supply conditioned air to the cabin and flight compartment. The system responds to flight crew commands and compartment temperature controls to supply a comfortable temperature environment for passengers and crew.

The flow control system uses pressure and temperature data to control the flow of the bleed air that enters the air conditioning pack. The Electronic Control Unit (ECU) opens and closes the pack flow control and shutoff valve to regulate the flow of bleed air.

Detailed Description

The cooling system has the sub–systems that follow:

- Air Conditioning Pack (21–51–00)
- Flow Control System (21–52–00)

Air Conditioning Pack (21–51–00)

The main functions of the air conditioning packs are:

- Condition the bleed air from the engines (or APU) and supply it to the air distribution system
- Protect the Air Cycle Machines (ACMs) from excessively high temperatures
- Recirculate cool air to the pack inlet to counteract high bleed air temperatures
- Remove condensed water from the conditioned air.

Refer to Figures 1 , 2 and 3.

The ECS Electronic Control Unit (ECU) monitors and controls the operation of the air conditioning packs. The ECU has input from the selector switches on the AIR CONDITIONING control panel and from the sensors and switches located throughout the packs. The ECU uses these inputs to monitor and control the temperature and flow rate of the conditioned air leaving the air conditioning packs.

Two air cycle machines (ACMs), with one dual heat exchanger, cool the bleed air coming from the two engines. This configuration has the redundancy of two ACMs while allowing access to a much larger dual heat exchanger during single ACM operation.

The dual heat exchanger receives warm air from the pack flow control and shutoff valve. The primary section of the dual heat exchanger cools the air which then goes through two compressor inlet check valves. These valves prevent any back flow from the downstream ACM compressor into the primary section of the dual heat exchanger during single ACM operation.

The ACM compresses the air and sends the warm compressed air to the secondary circuit of the dual heat exchanger for cooling. The air then goes through the condenser section of the condenser/reheater/mixer, where water in the air is condensed into droplets. The cooled air then goes through the water collector where the water droplets are collected to flow out through a drain. The drain water then goes to a nozzle where it is sprayed against the ram inlet face of the dual heat exchanger for evaporative cooling.

The air continues on through the reheater, where it is warmed. This relatively warm, compressed air is then expanded in the turbine section of the ACM, where it is cooled. The temperature at the pack outlet is modulated by the pack bypass valves, that introduce warm bleed air into the turbine outlet. The signals that control the positions of these valves come from the Electronic Control Unit (ECU).

When necessary, the temperature of the bleed air at the pack inlet is lowered by opening the temperature reduction valve. This valve allows cooler air downstream of the secondary heat exchanger to flow into the pack inlet. The temperature switch of the temperature reduction valve sends a signal to the ECU that is compared with the signal from the pack inlet temperature sensor. The ECU then determines if the temperature reduction valve is operating correctly.

The compressor outlet overtemperature sensor supplies a signal to reduce the pack flow if the temperature in the compressor is too high. The compressor outlet overtemperature switch turns the ACM off if the compressor temperature is too high. This switch supplies backup protection in case the compressor outlet overtemperature sensor fails.

The secondary heat exchanger outlet temperature sensor senses the temperature at the secondary heat exchanger outlet. This signal is used only for Built In Test (BIT) purposes to monitor the efficiency of the heat exchanger.

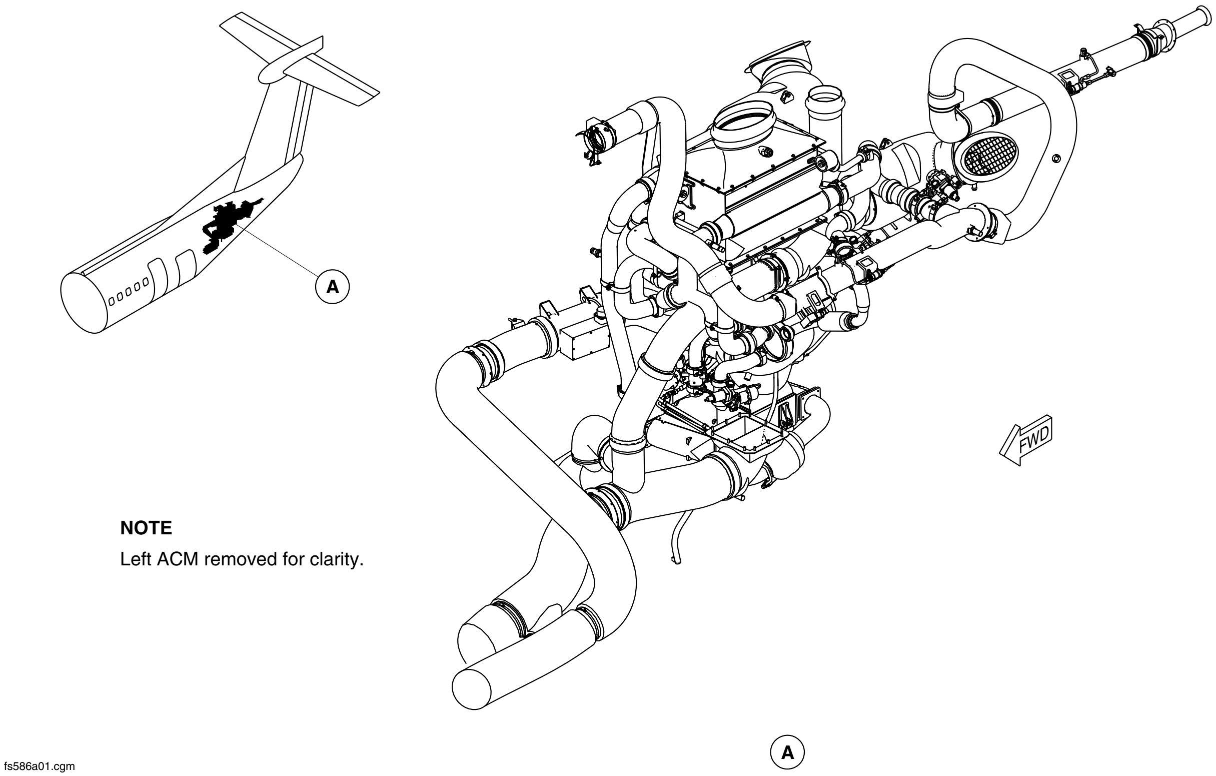
Flow Control System (21–52–00)

Refer to Figure 4.

The flow control system uses pressure and temperature data to control the flow rate of the bleed air that enters the air conditioning pack. The flow control system has three components:

- Valve, Pack Flow Control and Shutoff (21–52–01)
- Sensor, Absolute Pressure (21–52–06)
- Sensor, Pack Inlet Temperature (21–52–11)

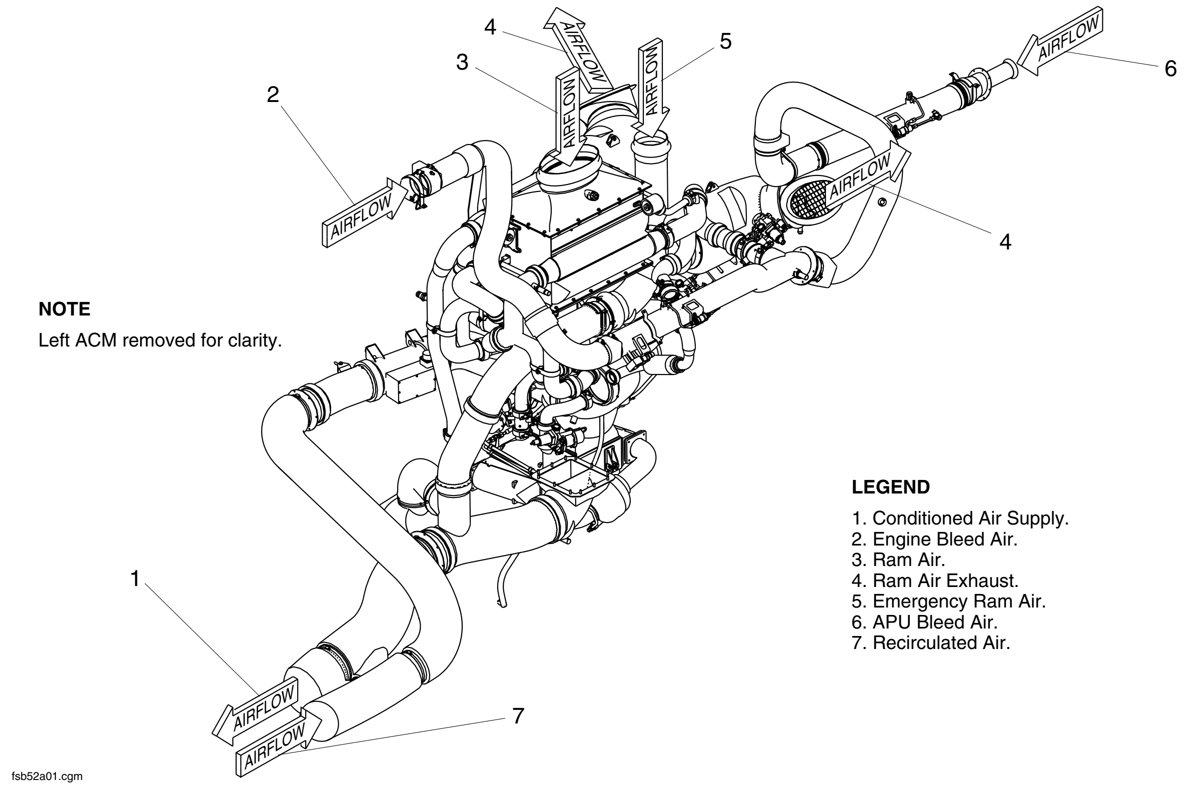
These components are installed in the ducting that supplies bleed air to the air conditioning pack. The two sensors supply the pressure and temperature data of the bleed air to the Electronic Control Unit (ECU). The ECU uses this data to open and close the pack flow control and shutoff valve to regulate the flow of bleed air to the air conditioning pack.



NOTE

Left ACM removed for clarity.

AIR CONDITIONING PACK LOCATOR
Figure 1



NOTE

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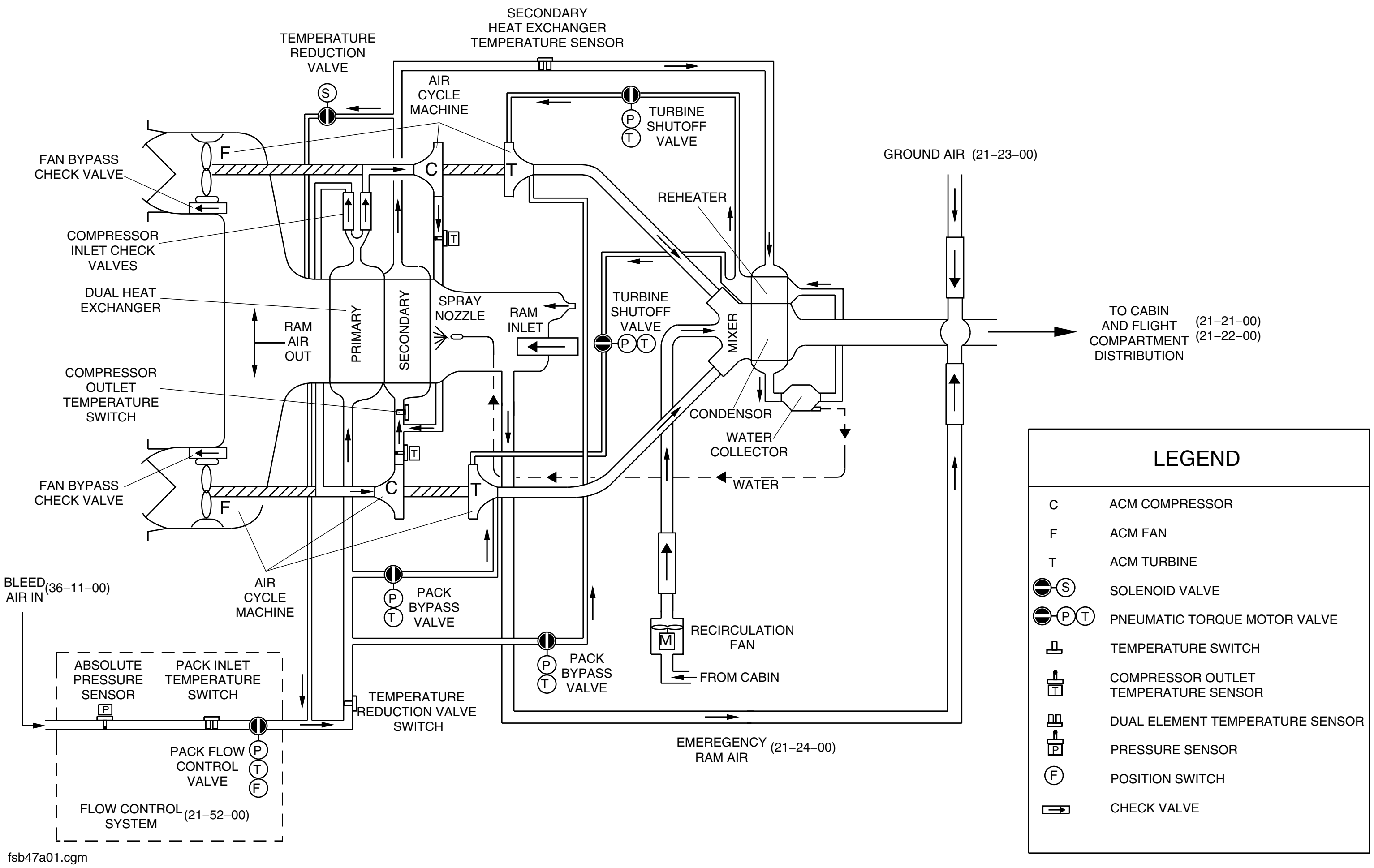
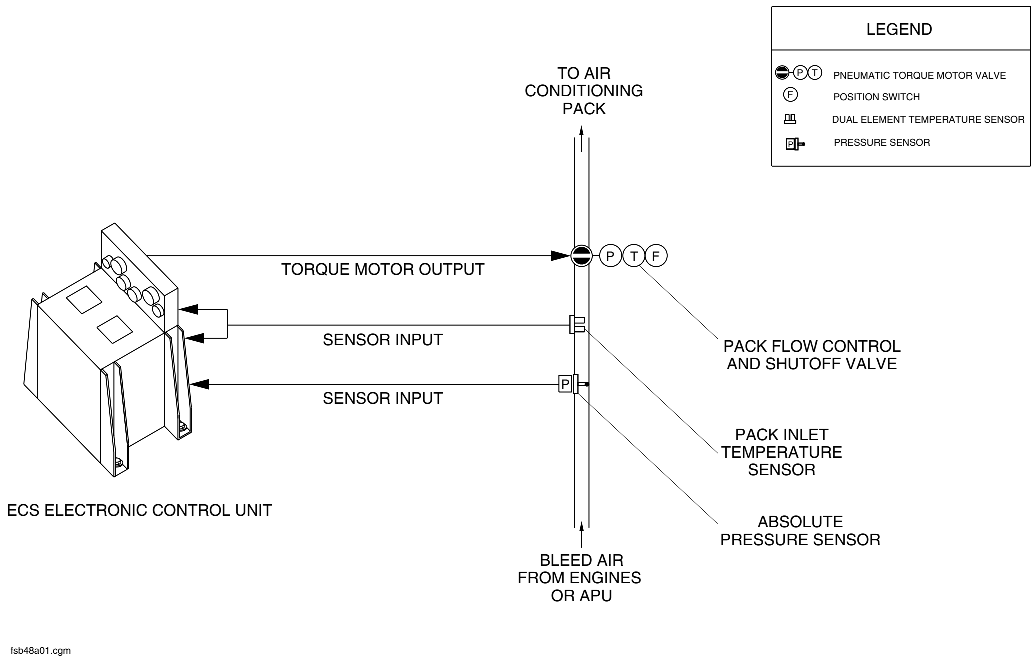
LEGEND

1. Conditioned Air Supply.
2. Engine Bleed Air.
3. Ram Air.
4. Ram Air Exhaust.
5. Emergency Ram Air.
6. APU Bleed Air.
7. Recirculated Air.

AIR CONDITIONING PACK AIRFLOW
Figure 2



AIR CONDITIONING PACK SCHEMATIC
Figure 3



FLOW CONTROL SYSTEM
Figure 4

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21–51–00–001

AIR CONDITIONING PACK

Introduction

The air conditioning pack conditions the bleed air to the proper temperature and humidity and delivers it to the air distribution system for environmental control of the cabin and flight compartment.

General Description

The air conditioning pack is part of the Environmental Control System (ECS). It uses bleed air from the engines or Auxiliary Power Unit (APU) to supply conditioned air to the cabin and flight compartment. The system responds to flight crew commands and compartment temperature controls to supply a comfortable temperature environment for passengers and crew.

Two air cycle machines (ACM), with one dual heat exchanger located in the aft fuselage, cool the bleed air coming from the two engines. This configuration provides the redundancy of two packs while allowing access to a much larger dual heat exchanger during operation with a single ACM.

The air conditioning pack has the components that follow:

- Exchanger, Heat–Dual (21–51–01)
- Condenser/Reheater/Mixer (21–51–06)
- Nozzles, Spray (21–51–11)
- Collector, Water (21–51–16)
- Machine, Air Cycle (21–51–21)
- Housing, Fan Inlet Diffuser (21–51–26)
- Valve, Pack Bypass (21–51–31)
- Check Valve, Compressor Inlet (21–51–36)
- Valve, Turbine Shutoff (21–51–41)
- Check Valve, Fan Bypass (21–51–46)
- Check Valve, Ram Outlet (21–51–51)
- Check Valve, Ram Inlet (21–51–56)
- Valve, Temperature Reduction Shutoff (21–51–61)
- Ducts, Air Conditioning Pack (21–51–66)
- Struts, Air Conditioning Pack (21–51–71)
- Mounts, Air Conditioning Pack (21–51–76)
- Switch, Valve–Temperature Reduction (21–51–81)
- Switch, Compressor Overtemperature (21–51–86)
- Sensor, Compressor Outlet Temperature (21–51–91)
- Sensor, Outlet Temperature–Secondary Heat Exchanger (21–51–96)

Detailed Description

Operation

The air conditioning system is started by selecting BLEED 1 and/or BLEED 2 switches on the AIR CONDITIONING control panel, or the

BL AIR switchlight on the APU CONTROL panel. These switch settings determine the selection of the bleed air source, manual or automatic ECS operation and air flow rates for the flight and passenger compartments.

With the air conditioning PACKS switch set to AUTO, the bleed air comes from the left and right engines. With pilot and crew switch selections made, the ECS Electronic Control Unit (ECU) measures temperatures, pressures, and flow rates and uses this information to adjust valve positions in the ECS to get the selected conditioned air temperatures and flow rates (refer to SDS 21–61–00).

The ECU controls the two nacelle shutoff valves to regulate the air flow to the air conditioning packs. The ECU sets these two valves to get an equal bleed air flow from each engine (refer to SDS 36–11–00). Bleed air passes through the fully open pack flow control and shutoff valve to the air conditioning pack. The ECU receives the pressure and temperature data of the bleed air from the pack inlet absolute pressure sensor and the pack inlet temperature sensor. The ECU uses this data to control the flow of bleed air through the pack flow control and shutoff valve during single engine operation. The ECU also uses this data to control bleed air flow rate when APU bleed air is selected (refer to SDS 21–52–00).

Refer to Figures 1 , 2 and 3.

As soon as the bleed air flows into the air conditioning pack, it passes through the primary part of the dual heat exchanger. The primary heat exchanger uses ram air to remove some of the heat from the bleed air. The primary–cooled bleed air then divides into two air flow paths which pass through two compressor inlet check valves (one check valve for each air flow path) to the compressors of both Air Cycle Machines (ACMs). Whenever the air conditioning system is on, both air cycle machines are operating.

Refer to Figures 4 , 5 , 6 and 7.

The ACM compressors increase the temperature and pressure of the bleed air. The two compressor outlet flows combine into one flow to pass through the secondary part of the dual heat exchanger. Two compressor outlet temperature sensors (one for each compressor outlet) measure the compressor outlet temperatures. The secondary heat exchanger also uses cooler ram air to remove heat from the compressed bleed air. A temperature sensor measures the cooled bleed air as it leaves the secondary heat exchanger.

The cooled bleed air from the secondary heat exchanger passes through the reheater and then to the condenser. The condenser, mixer, and reheater are all part of the Condenser/Mixer/Reheater unit.

The condenser is a heat exchanger which uses the cooler air from the ACM turbines to remove enough heat from the bleed air. This decrease in temperature causes water vapor in the bleed air to condense. The bleed air with condensed water vapor then passes to the water collector. The water collector uses centrifugal action (no moving parts) to remove the condensed water from the bleed air flow. The water in the collector then passes through a water line to a spray nozzle in the secondary heat exchanger. The high pressure of the bleed air forces the water out of the nozzle as a spray. The spray nozzle has two functions:

- Evaporates the water sprayed into the air flow to increase the efficiency of the secondary heat exchanger
- Discards the water into the ram air flow which gets exhausted out of the aircraft.

Dried bleed air from the water collector then passes through the reheater (another heat exchanger). The reheating of the bleed air vaporizes any condensation that gets past the water collector.

Bleed air from the reheater divides into two streams. Each air stream passes through a fully open turbine shutoff valve to an ACM expansion turbine. The bleed air expands as it passes through the turbine. The expanding air spins the turbine to supply the mechanical power to rotate the ACM compressor and fan. The temperature of the expanding bleed air greatly decreases to become the conditioned (cooled) air for the flight and cabin compartments. The volume of the expanding air increases and the pressure decreases to approximately 1 atmosphere (14.7 psi (101.35 kPa)).

The expanded air from both turbines enters the mixer which combines it with recirculated air from the cabin. This mixed air enters the condenser to cool the bleed air from the secondary heat exchanger and condense water vapor. The mixed air then passes to the cabin and flight compartment distribution system.

The ECU controls two pack bypass valves to regulate the temperature of the conditioned air sent to the cabin and flight compartment. These valves pass a portion of the warmer and higher–pressure bleed air (taken before the dual heat exchanger) to the outputs of each ACM expansion turbine. Opening the pack bypass valve increases the flow of warm bleed air into the turbine outlet. This raises the temperature of the air supplied to the flight and cabin compartment. Closing the pack bypass valve decreases the flow of warm bleed air into the turbine outlet. This decreases the temperature of the air supplied to the flight and cabin compartments. Bleed air, bypassed by the pack bypass valves, raises the humidity of the very dry conditioned air sent to the flight and cabin compartments.

Each ACM turbine drives a fan which keeps the ram air flow through the dual heat exchanger at acceptable levels. This becomes necessary when the aircraft's speed cannot generate enough ram air flow through the dual heat exchanger (such as when the aircraft is on the ground). When the aircraft is airborne, the ACM fans can limit the flow of ram air. As the flow of ram air increases, the ram air pressure on the fans increases. This increasing pressure opens two bypass valves in the ram air outlet ducts to permit the maximum flow of ram air.

A recirculation fan takes air from the cabin to mix with the conditioned air from the ACM turbines. A RECIRC switch turns the recirculation fan on or off. Operating conditions determine the automatic control of the recirculation fan speed.

<u>Exchanger, Heat–Dual (21–51–01)</u>

Refer to Figures 8 and 9.

The dual heat exchanger is an aluminum plate–and–fin, primary crossflow/secondary counterflow heat exchanger. It provides primary cooling of the pack inlet air and secondary cooling of the air from the ACM compressors.

The dual heat exchanger is mounted at the top of the air conditioning pack between the ram air inlet duct and fan inlet diffuser housing.

<u>Condenser/Reheater/Mixer (21–51–06)</u>

Refer to Figures 10 and 11.

The condenser/reheater/mixer is a crossflow, plate and fin, aluminum heat exchanger. It has three air circuits. The condenser circuit receives warm, moist (ACM compressor) air from the secondary heat exchanger. The mixer circuit contains the cool air from the ACM turbines and the recirculated air from the cabin and flight compartment. The air in this circuit is cool enough to condense the water vapour from the surrounding air. The condensed water then goes to the water collector. After passing through the condenser, this air in the mixer circuit is delivered to the cabin and flight compartment. The reheater circuit contains the air that is downstream of the water collector. This air is reheated before it is delivered to the ACM turbine inlet.

The condenser/reheater/mixer is located at the bottom of the air conditioning pack.

Nozzles, Spray (21–51–11)

Refer to Figure 12.

The spray nozzle is installed on the top left side of the ram air header duct. The nozzles spray water against the ram air inlet face of the dual heat exchanger. This water evaporates and improves the efficiency of the heat exchanger.

Removal of the water from the system is important for extending the life of the air cycle machine. If the water cannot be removed from the air before it enters the ACM turbines, it causes turbine nozzle erosion, which degrades ACM performance and decreases its reliability.

Collector, Water (21–51–16)

Refer to Figure 13.

The water collector is installed on the left side of the condenser and removes water from the bleed air. It has swirl vanes which use centrifugal action to push the condensed water to its inside walls. The water goes out through a tube to the spray nozzle, for spray against the ram inlet face of the dual heat exchanger. A small overflow provides an outlet for excess water.

Machine, Air Cycle (21–51–21)

Refer to Figure 14.

The air cycle machine is a 3–wheeled, air bearing device that provides cooling of the warm bleed air. There are three rotors in each ACM: a compressor, a turbine and a fan. The rotors are held together with interference fits on shaft segments so the three rotors turn together. The ACM compresses the air (compressor section) and then expands it (turbine section). The expansion of the air causes a decrease in temperature. The fan section draws cooling air through the dual heat exchanger during ground operations.

There are two air cycle machines which are located below the fan inlet diffuser housing.

Housing, Fan Inlet Diffuser (21–51–26)

Refer to Figures 15 and 16.

The fan inlet diffuser housing is located in the air conditioning pack between the dual heat exchanger outlet and the ACM fans. The housing has a common duct for cooling airflow from the dual heat exchanger to the ACM fans and ram air exhaust outlets. It collects cooling air from the dual heat exchanger outlet and delivers it to the fans. It also collects the air from the fan outlets and delivers it to the ram air exhaust outlets which pass the air overboard.

Valve, Pack Bypass (21–51–31)

Refer to Figure 17.

The pack bypass valve is an electrically–controlled, pneumatic valve, which controls the pack discharge temperature for all operating conditions. The valve opens to let warm bleed air go to the ACM turbine outlet when warmer air is needed. The valve consists of:

- Torque motor
- Pneumatic actuator
- Butterfly valve

The valve is fitted with a metal wire mesh filter in the servo air line, to prevent contamination from entering the torque motor and pneumatic actuator. The valve also has a locking pin that, when in the locked position, bleeds servo air to ambient and mechanically locks the valve in the closed position.

The pack (ACM) bypass valves are located between the bleed air inlet and the ACM turbine outlets. The valve is used to regulate the temperature of conditioned air sent to the cabin and flight compartments.

Check Valve, Compressor Inlet (21–51–36

Refer to Figure 18.

The compressor inlet check valve is a spring–loaded, dual–flapper device that prevents back flow of air into the primary dual heat exchanger.

Valve, Turbine Shutoff (21–51–41)

Refer to Figure 19.

The turbine shutoff valve is an electrically–controlled, pneumatic valve that opens to allow air into the ACM turbine. The valve consists of:

- Torque motor
- Pneumatic actuator
- Position switch
- Butterfly valve.

The valve has a metal wire mesh filter in the servo air line to prevent contamination from entering the torque motor and pneumatic actuator. The valve also has a locking pin that, when in the locked position, bleeds servo air to ambient and mechanically locks the valve in the closed position.

The turbine shutoff valves are located between the condenser/reheater/mixer and the ACM turbine inlets.

Check Valve, Fan Bypass (21–51–46)

Refer to Figure 20.

The fan bypass check valve is a spring–loaded, dual–flapper device that allows ram air to flow overboard in parallel with the ACM fan to supply the necessary ram flow during flight.

Check Valve, Ram Outlet (21–51–51)

Refer to Figure 21.

The ram outlet check valve is a spring–loaded, dual–flapper device that keeps debris from entering fan exhaust ducts during pack shutdown operation.

Check Valve, Ram Inlet (21–51–56)

Refer to Figure 22.

The ram inlet check valve is a rectangular, aluminum, bolted flange, spring loaded, dual flapper valve. It is visible and accessible from inside the ECS compartment in the tail of the aircraft. The valve allows additional ram air to flow through the dual heat exchanger during ground operation.

Valve, Temperature Reduction Shutoff (21–51–61)

Refer to Figure 23.

The temperature reduction shutoff valve is a pneumatically actuated, solenoid–controlled gate type device that opens to recirculate secondary heat exchanger air, back into the primary heat exchanger inlet. This is done when the temperature at the primary heat exchanger inlet is too hot.

The temperature reduction shutoff valve is installed on the aft side of the heat exchanger.

Ducts, Air Conditioning Packs (21–51–66)

Refer to Figures 24 and 25.

After the bleed air flows through the primary heat exchanger of the air conditioning pack, it divides into two separate airflows. One airflow is ducted through a check valve to the right ACM compressor inlet. Similarly, the other airflow is ducted through a check valve to the left ACM compressor inlet.

The two compressor outlet flows combine into one flow to pass through the secondary part of the dual heat exchanger. Two compressor outlet temperature sensors (one for each compressor outlet) measure the compressor outlet temperatures.

After the cooled air leaves the secondary heat exchanger, it flows down a long vertical duct to the reheater part of the condenser/reheater/mixer unit. A sensor monitors the temperature of the air leaving the secondary heat exchanger.

Mounts, Air Conditioning Pack (21–51–76)

Refer to Figure 26.

There is a mounting strap attached to either side of the fan inlet diffuser housing. The straps are also attached to the airframe and support the weight of the air conditioning pack.

Switch, Valve—Temperature Reduction (21–51–81)

Refer to Figure 27.

The temperature reduction valve switch is a thermal switch which trips if the pack inlet temperature exceeds 560 °F (293 °C). This signal is used only for Built In Test (BIT).

The ECU compares the signal from the temperature reduction valve switch with that from the pack inlet temperature sensor, to determine if the temperature reduction shutoff valve is operating correctly.

The temperature reduction valve switch is installed on the bleed air inlet ducting in the aft fuselage.

Switch, Compressor Overtemperature (21–51–86)

Refer to Figure 28.

The compressor overtemperature switch is mounted in the secondary heat exchanger inlet duct and monitors air flow temperature entering the secondary heat exchanger. It is a thermal switch which trips if the compressor outlet temperature exceeds 415 °F (212 °C). The ECU uses the signal from the switch to determine if the air conditioning pack should be turned off.

Sensor, Compressor Outlet Temperature (21–51–91)

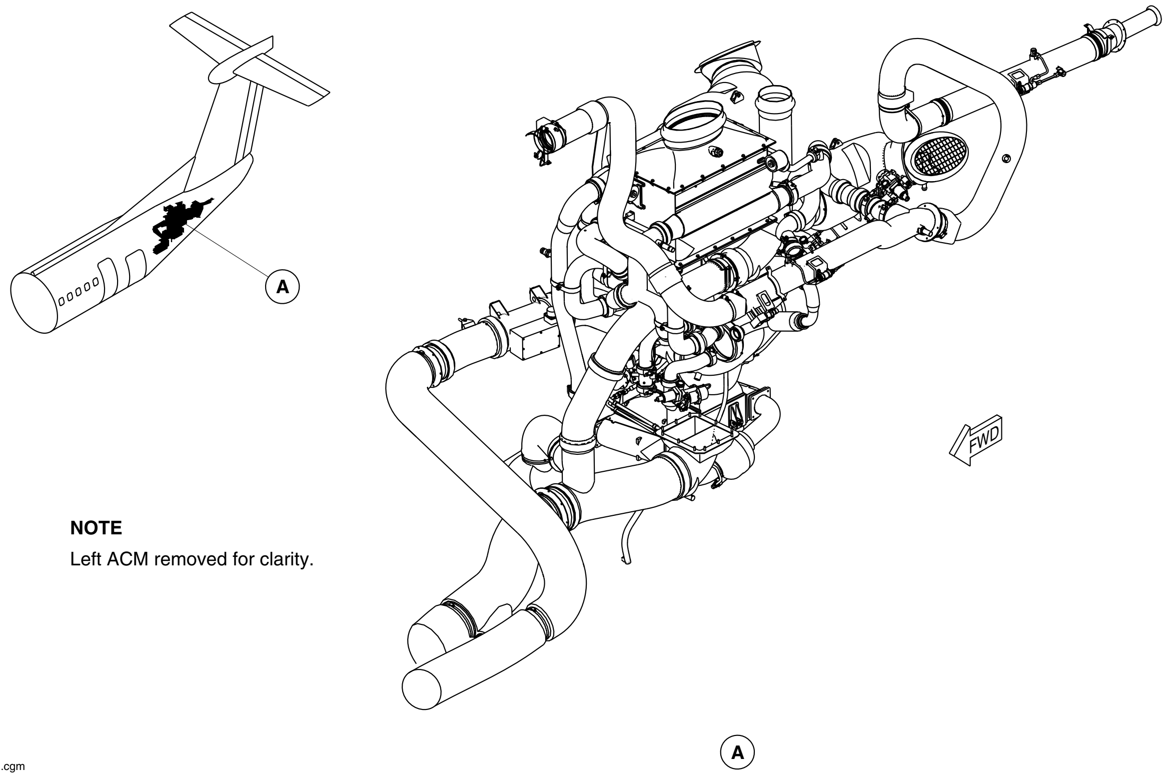
Refer to Figure 29.

The compressor outlet temperature sensors are located in ACM compressor outlet ducting. They monitor the temperature of the air leaving the ACM compressors. The temperature signal is used to reduce air flow through the air conditioning pack at 380 °F (193.33 °C) and to close the pack flow control and shutoff valve at 415 °F (212.78 °C).

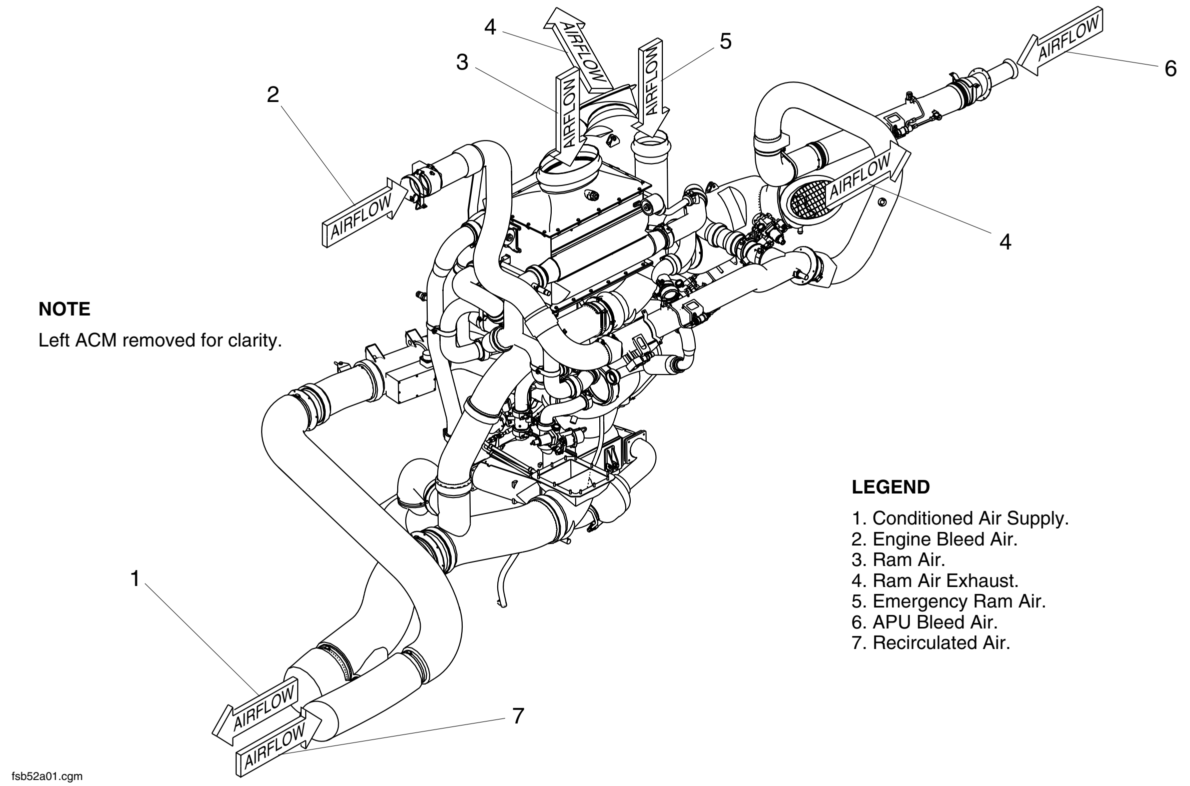
Sensor, Outlet Temperature–Secondary Heat Exchanger (21–51–96)

Refer to Figure 30.

The secondary heat exchanger outlet temperature sensor is installed in the secondary heat exchanger outlet duct and senses the temperature of the air leaving the heat exchanger. This signal is used only for Built In Test (BIT). The ECU compares the signal from this switch, to the signal from the pack inlet temperature sensor, to determine whether the dual heat exchanger is operating efficiently.



AIR CONDITIONING PACK LOCATOR
Figure 1



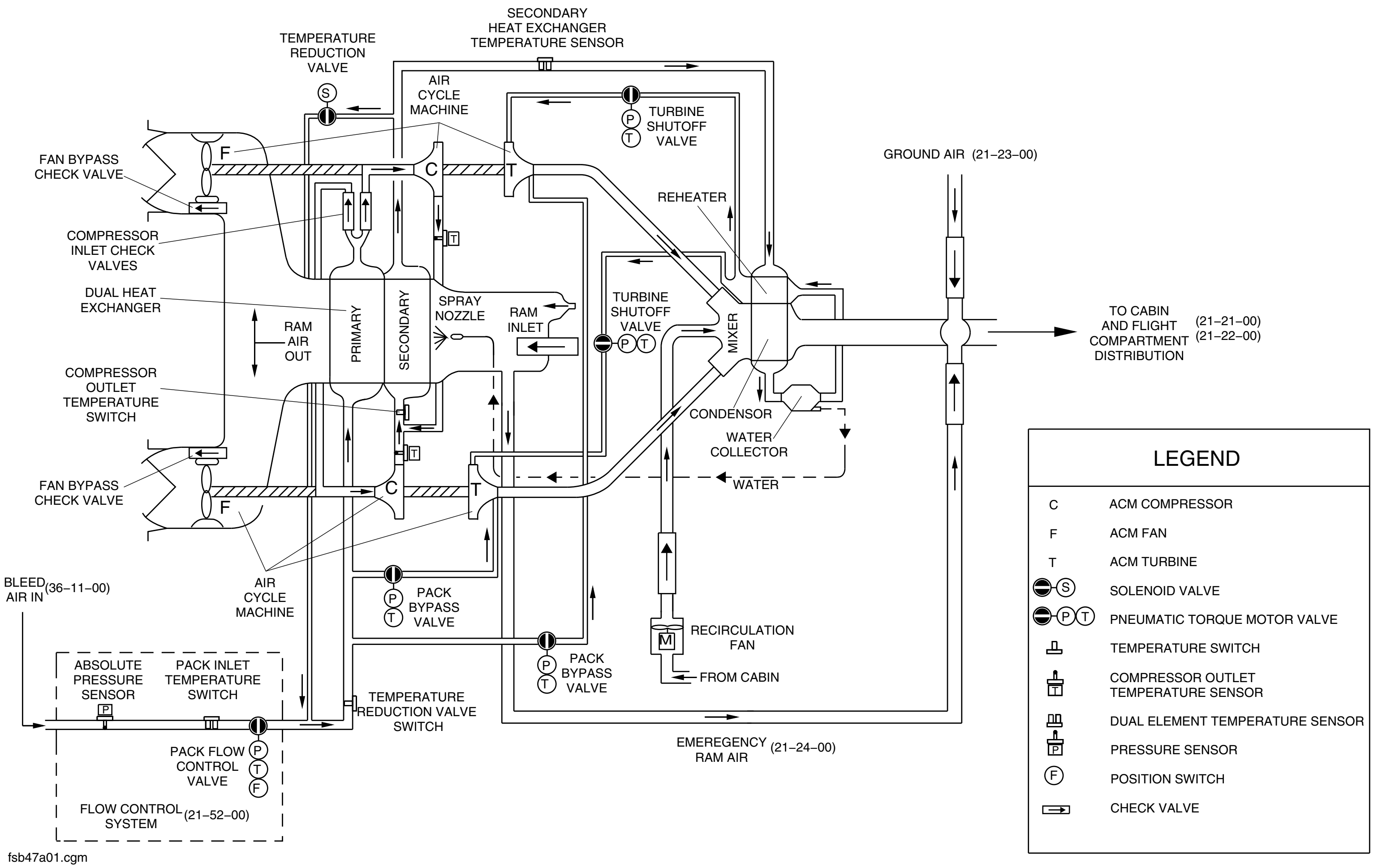
NOTE

Left ACM removed for clarity.

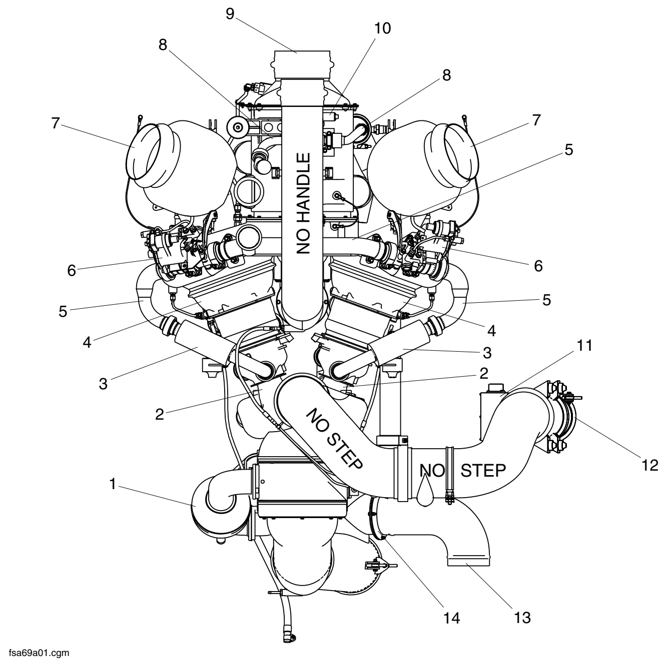
LEGEND

1. Conditioned Air Supply.
2. Engine Bleed Air.
3. Ram Air.
4. Ram Air Exhaust.
5. Emergency Ram Air.
6. APU Bleed Air.
7. Recirculated Air.

AIR CONDITIONING PACK AIRFLOW
Figure 2



AIR CONDITIONING PACK SCHEMATIC
Figure 3



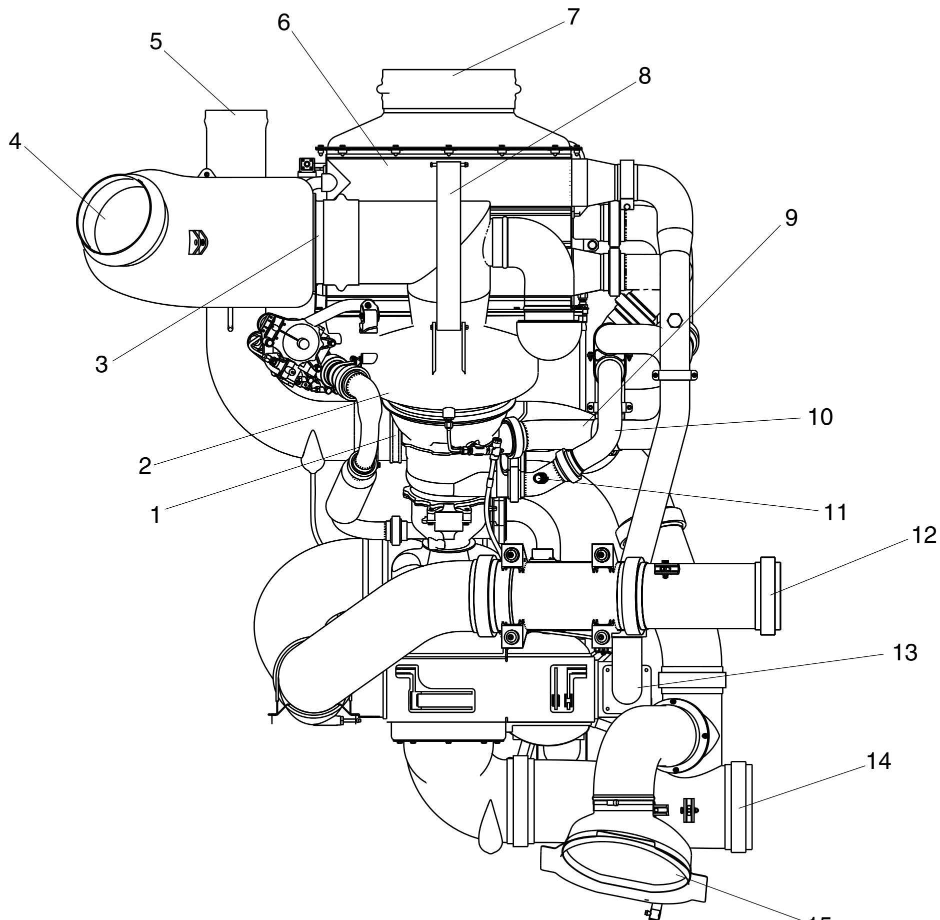
LEGEND

1. Water Collector.
2. Turbine Outlet Duct.
3. Bypass Duct (Muffler).
4. Air Cycle Machine.
5. Bypass Duct.
6. Pack Bypass Valve.
7. Ram Air Exhaust.
8. Temperature Reduction Duct.
9. Ram Air Inlet.
10. Temperature Reduction Shutoff Valve.
11. Recirculation Fan.
12. Recirculated Air Inlet.
13. Ground Air Connection.
14. Ground Air Check Valve.

NOTE

View looking forward.

AIR CONDITIONING PACK DETAIL PAGE 1
Figure 4

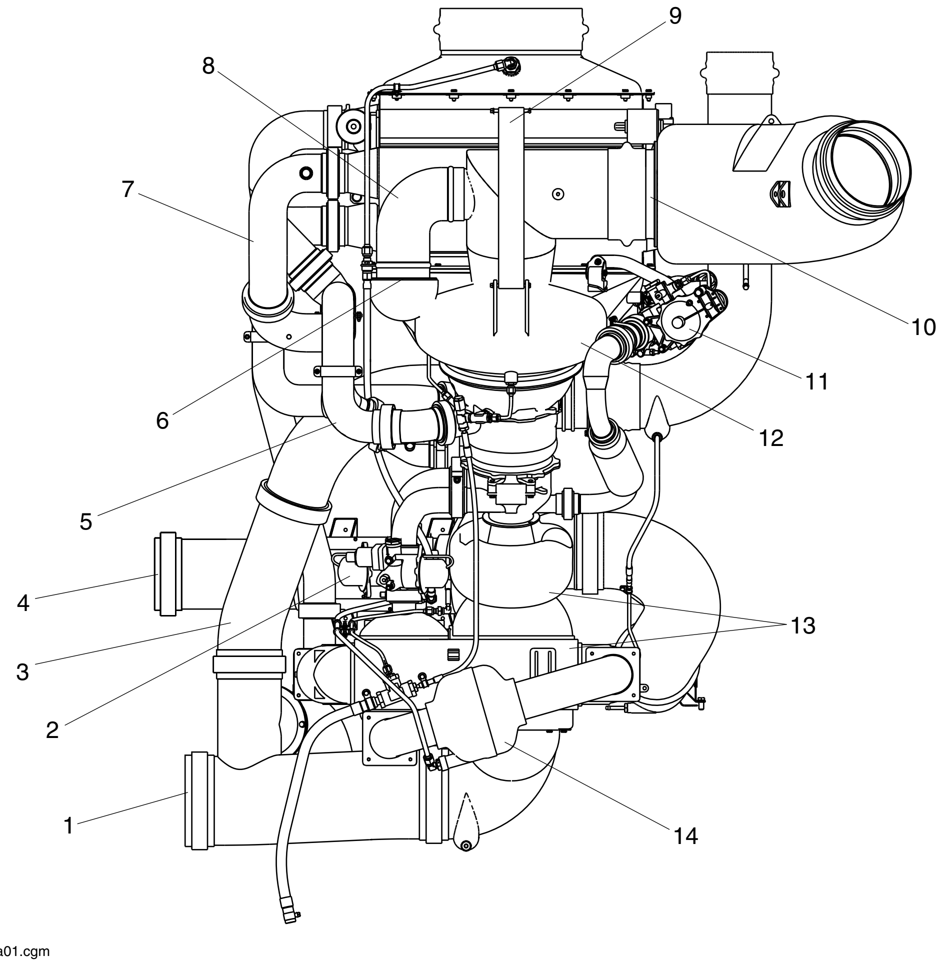


LEGEND

1. Emergency Ram Air Check Valve.
2. Fan Inlet Diffuser Housing.
3. Ram Outlet Check Valve.
4. Ram Air Outlet.
5. Emergency Ram Air Inlet.
6. Dual Heat Exchanger.
7. Ram Air Inlet.
8. Mount Straps.
9. Compressor Inlet Duct.
10. Compressor Outlet Duct.
11. Compressor Outlet Temperature Sensor.
12. Recirculated Air.
13. Reheater Inlet Duct.
14. Conditioned Air Outlet.
15. Ground Air Connection.

NOTE
View looking port.

AIR CONDITIONING PACK DETAIL PAGE 2
Figure 5



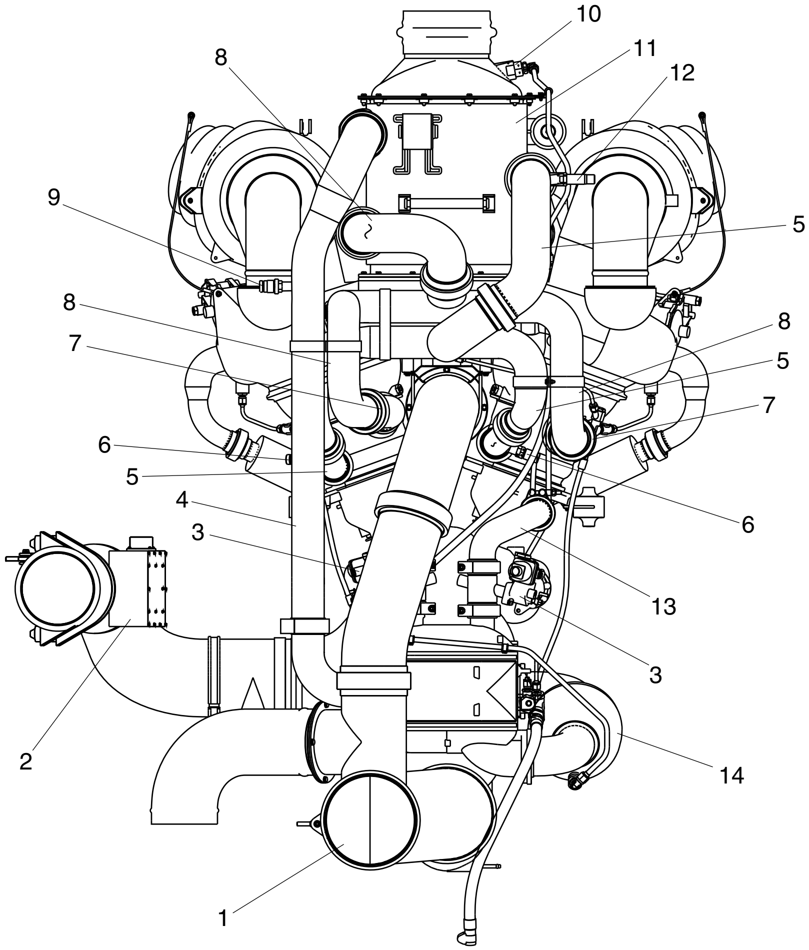
LEGEND

1. Conditioned Air Outlet.
2. Turbine Shutoff Valve.
3. Emergency Ram Air Duct.
4. Recirculated Air Inlet.
5. Compressor Inlet Duct.
6. Fan Bypass Check Valve.
7. Compressor Outlet Duct.
8. Fan Bypass Duct.
9. Mount Strap.
10. Ram Outlet Check Valve.
11. Pack Bypass Valve.
12. Fan Inlet Diffuser Housing.
13. Condenser Reheater Mixer.
14. Water Collector.

NOTE

View looking starboard.

AIR CONDITIONING PACK DETAIL PAGE 3
Figure 6



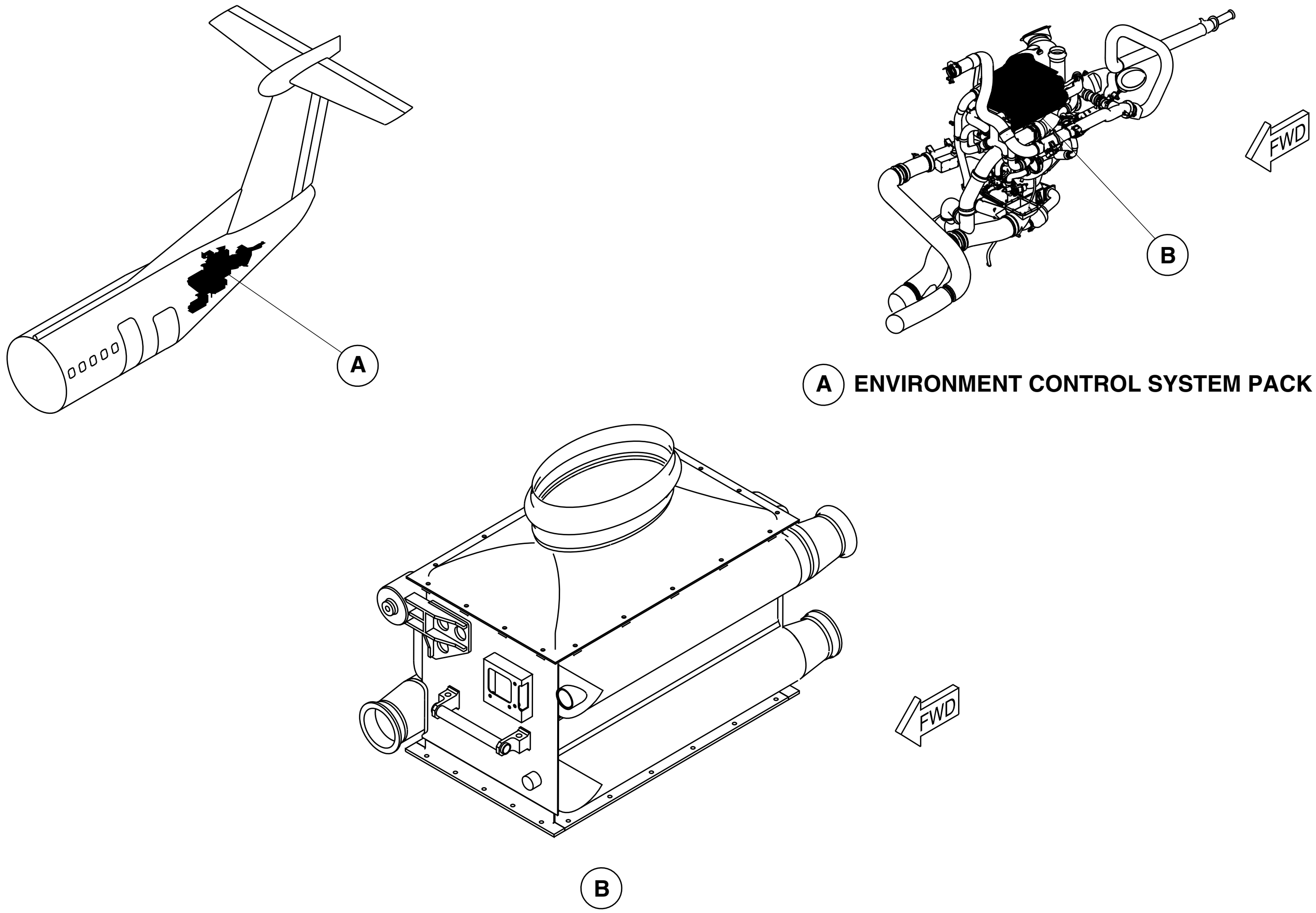
LEGEND

1. Conditioned Air Supply Duct.
2. Recirculation Fan.
3. Turbine Shutoff Valve.
4. Reheater Inlet Duct.
5. Compressor Outlet Duct.
6. Compressor Outlet Temperature Sensor.
7. Compressor Inlet Check Valve.
8. Compressor Inlet Duct.
9. Secondary Heat Exchanger Outlet Temperature Sensor
10. Spray Nozzle.
11. Dual Heat Exchanger.
12. Compressor Overtemperature Switch.
13. Turbine Inlet Duct.
14. Water Collector.

NOTE

View looking aft.

AIR CONDITIONING PACK DETAIL PAGE 4
Figure 7



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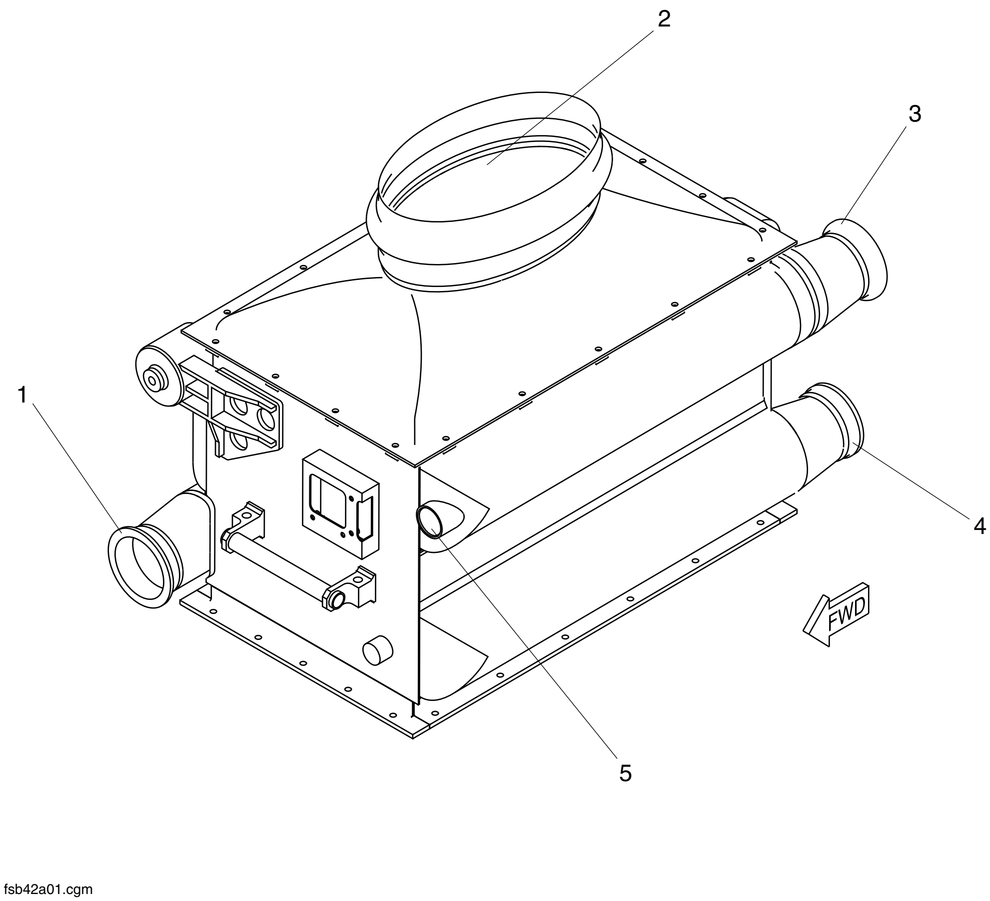
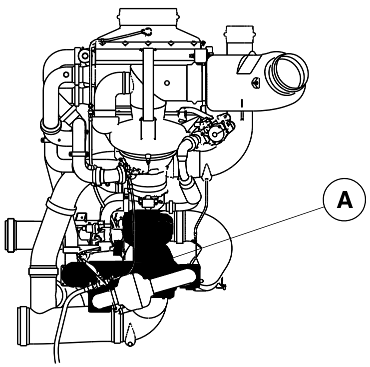
DUAL HEAT EXCHANGER LOCATOR
Figure 8



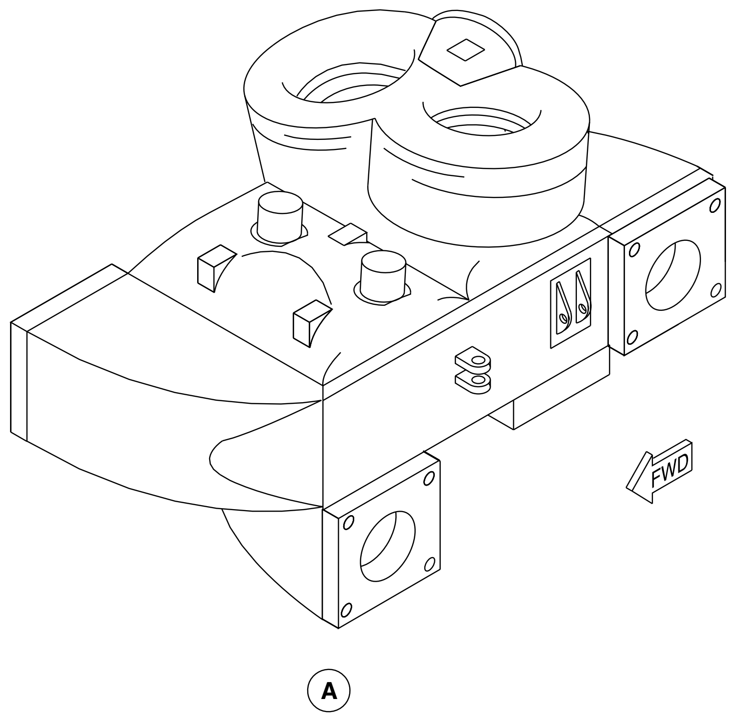
LEGEND

1. Bleed Air Inlet.
2. Ram Air Inlet.
3. Reheater Air Outlet.
4. Compressor Inlet.
5. Temperature Reduction Outlet.

DUAL HEAT EXCHANGER DETAIL
Figure 9



VIEW LOOKING STARBOARD



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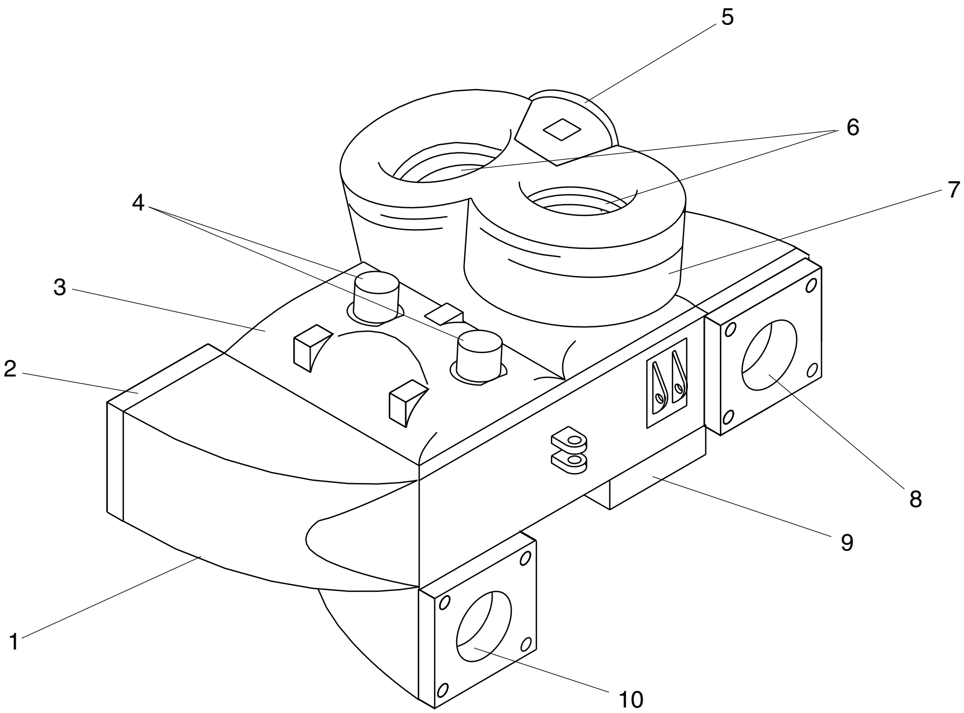
CONDENSER/REHEATER/MIXER LOCATOR
Figure 10



LEGEND

1. Condenser.
2. Condenser Inlet.
3. Reheater.
4. Reheater Outlets.
5. Mixer Inlet from Recirc Fan.
6. Mixer Inlets from ACM.
7. Mixer.
8. Condenser Outlet.
9. Mixer Outlet.
10. Reheater Inlet.

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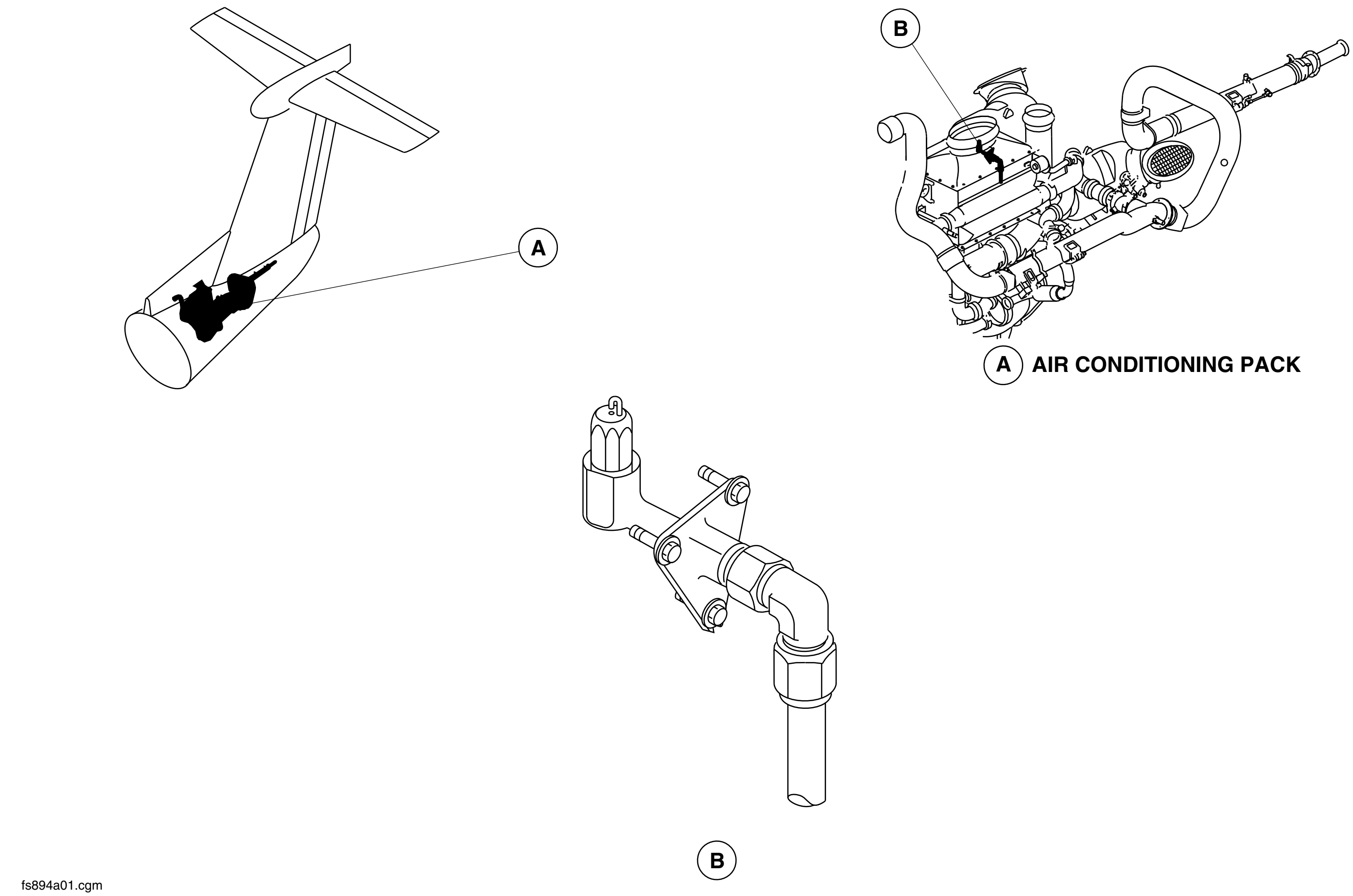
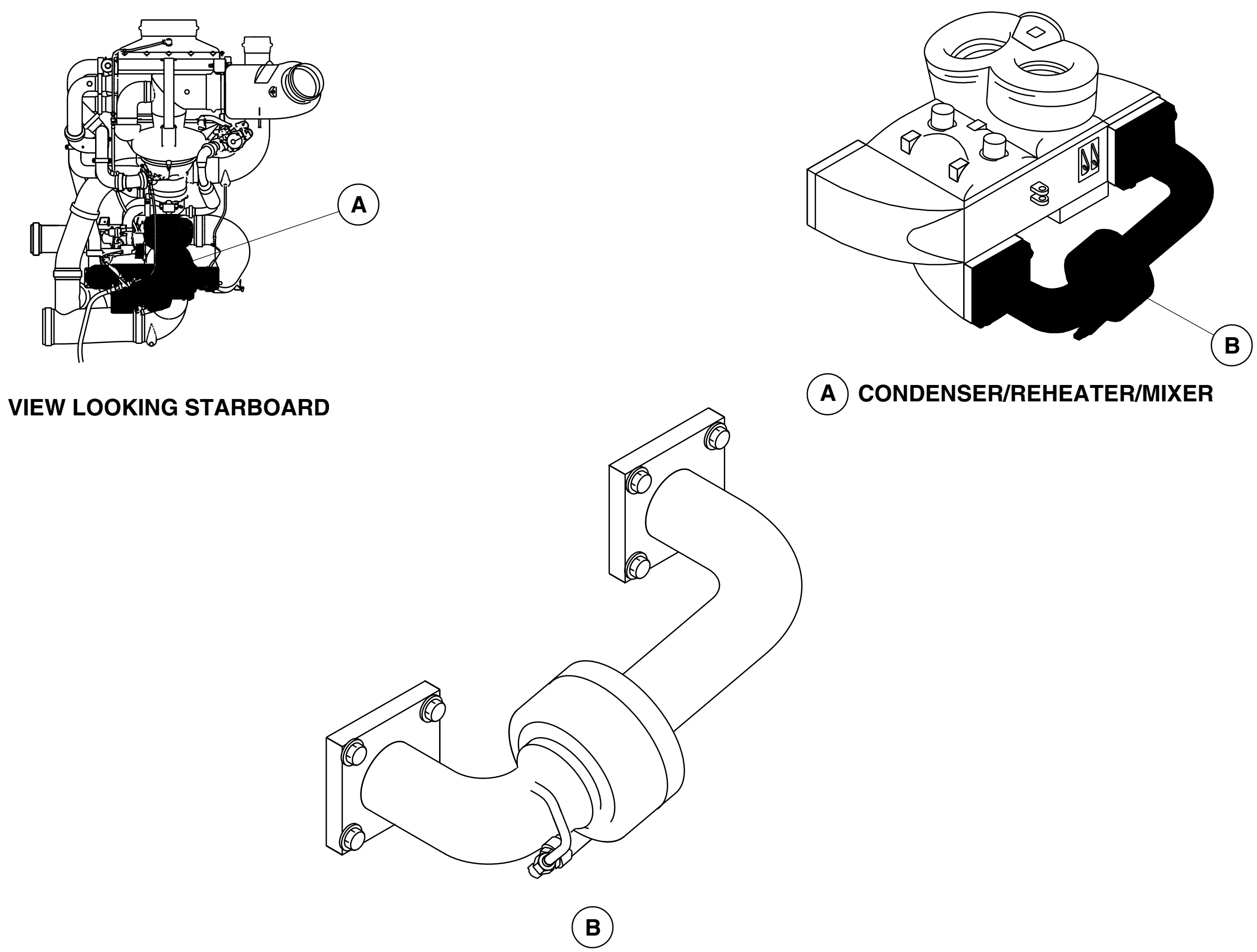
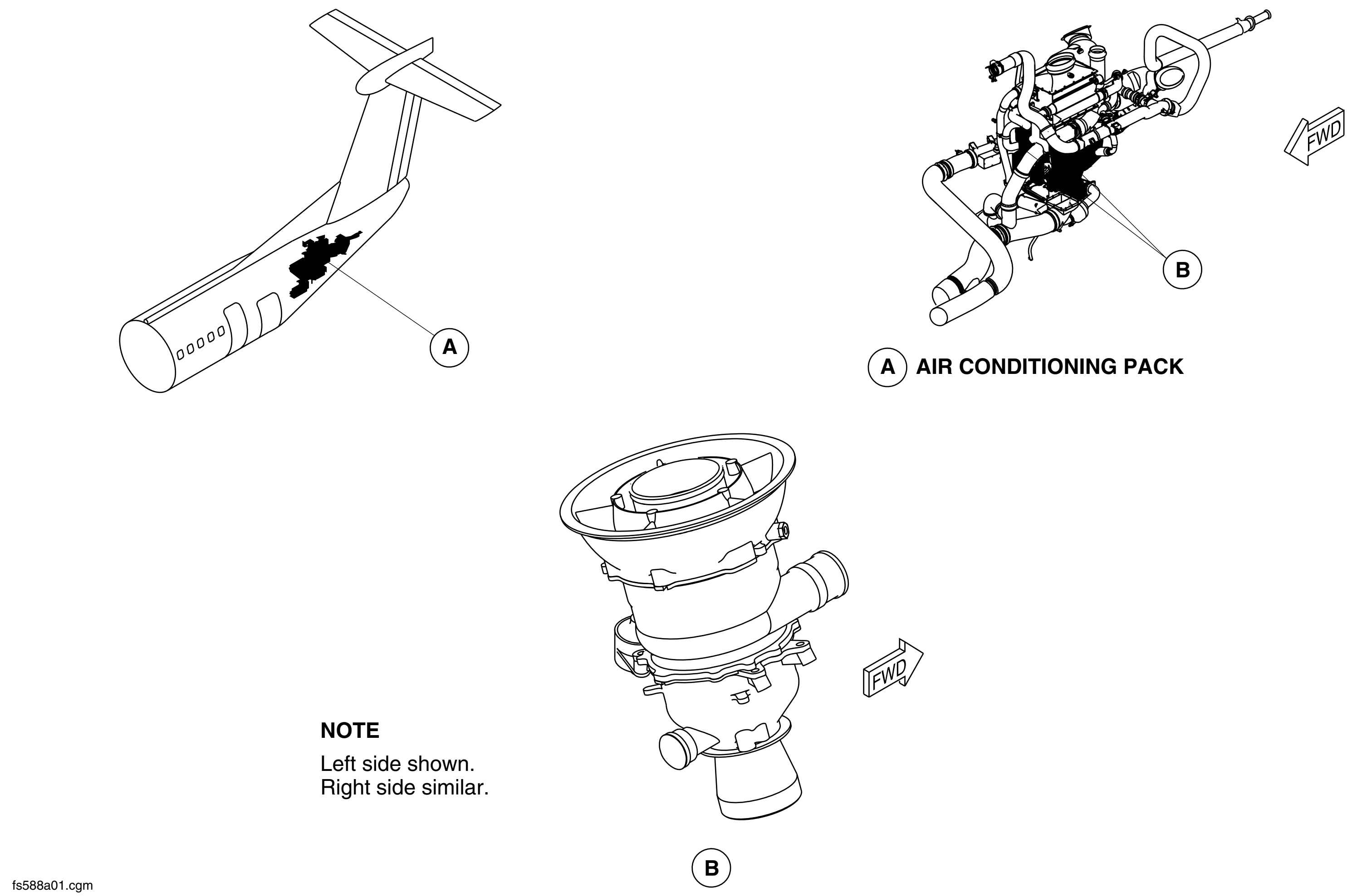
CONDENSER/REHEATER/MIXER DETAIL
Figure 11



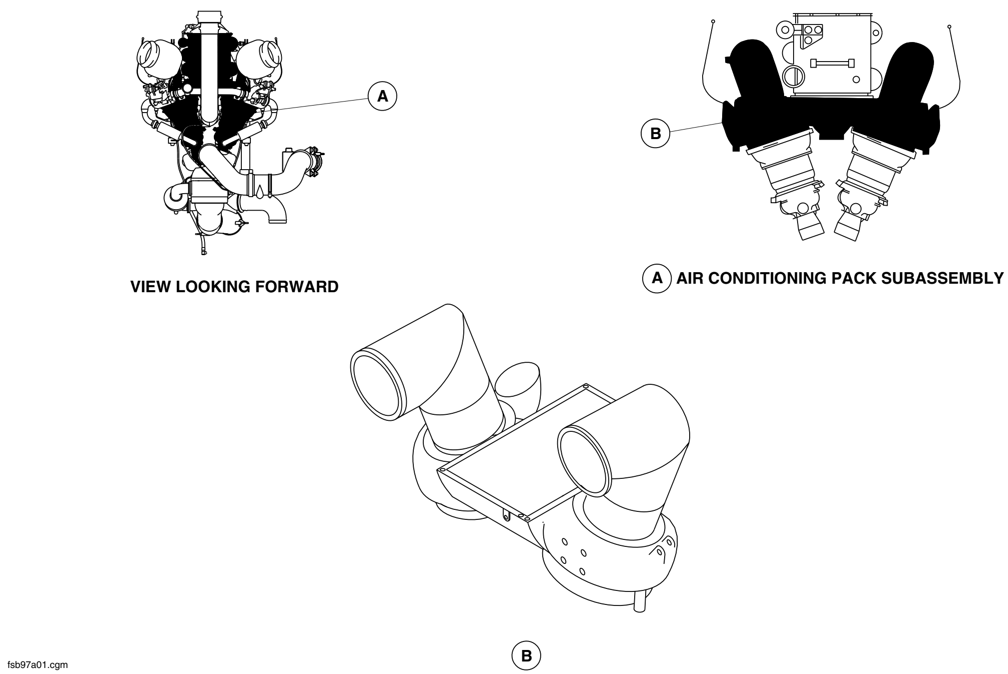
SPRAY NOZZLE
Figure 12



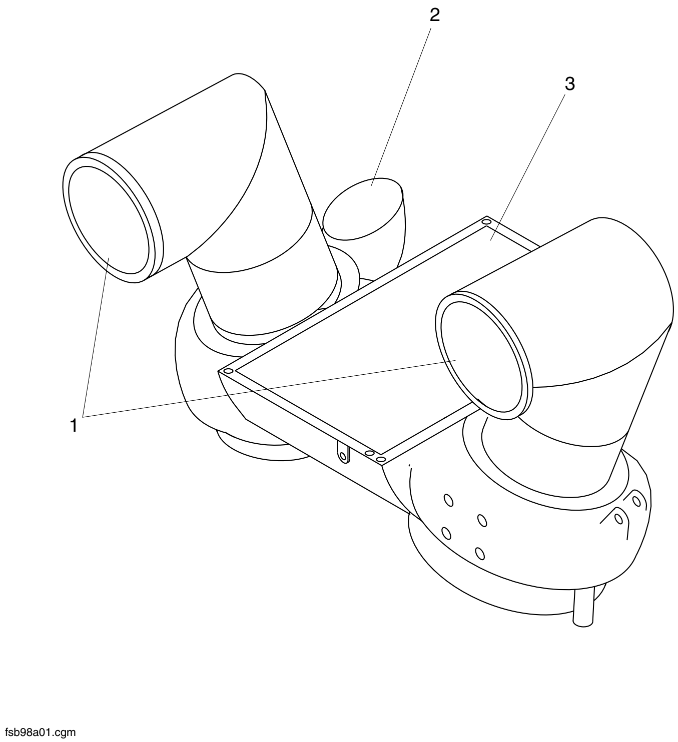
WATER COLLECTOR
Figure 13



AIR CYCLE MACHINE
Figure 14



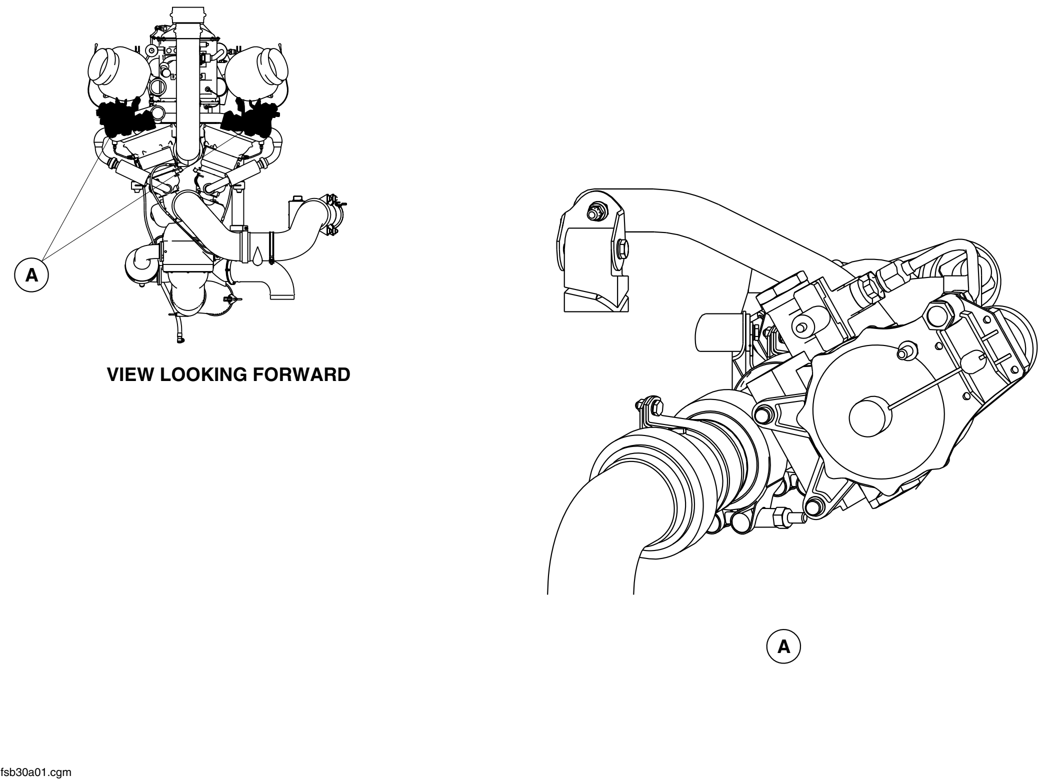
FAN INLET DIFFUSER HOUSING LOCATOR
Figure 15



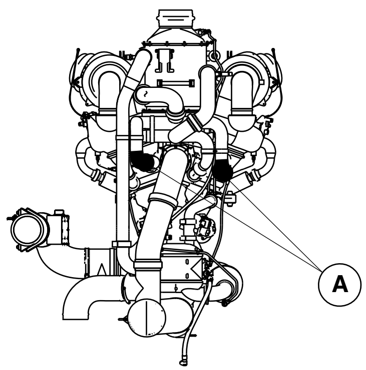
LEGEND

1. Ram Air Exhaust Outlet.
2. Fan Bypass Outlet.
3. Ram Air Inlet from Dual Heat Exchanger.

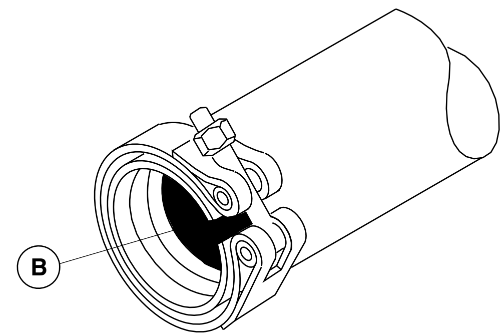
FAN INLET DIFFUSER HOUSING DETAIL
Figure 16



PACK BYPASS VALVE
Figure 17



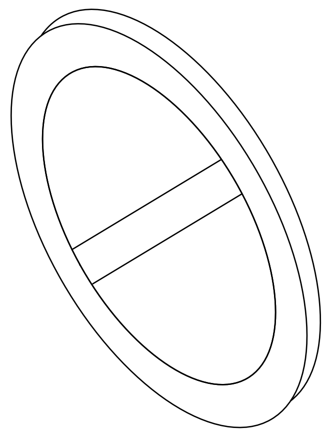
VIEW LOOKING AFT



A COMPRESSOR INLET DUCT ASSEMBLY

NOTE

Left component shown.
Right component similar.



B

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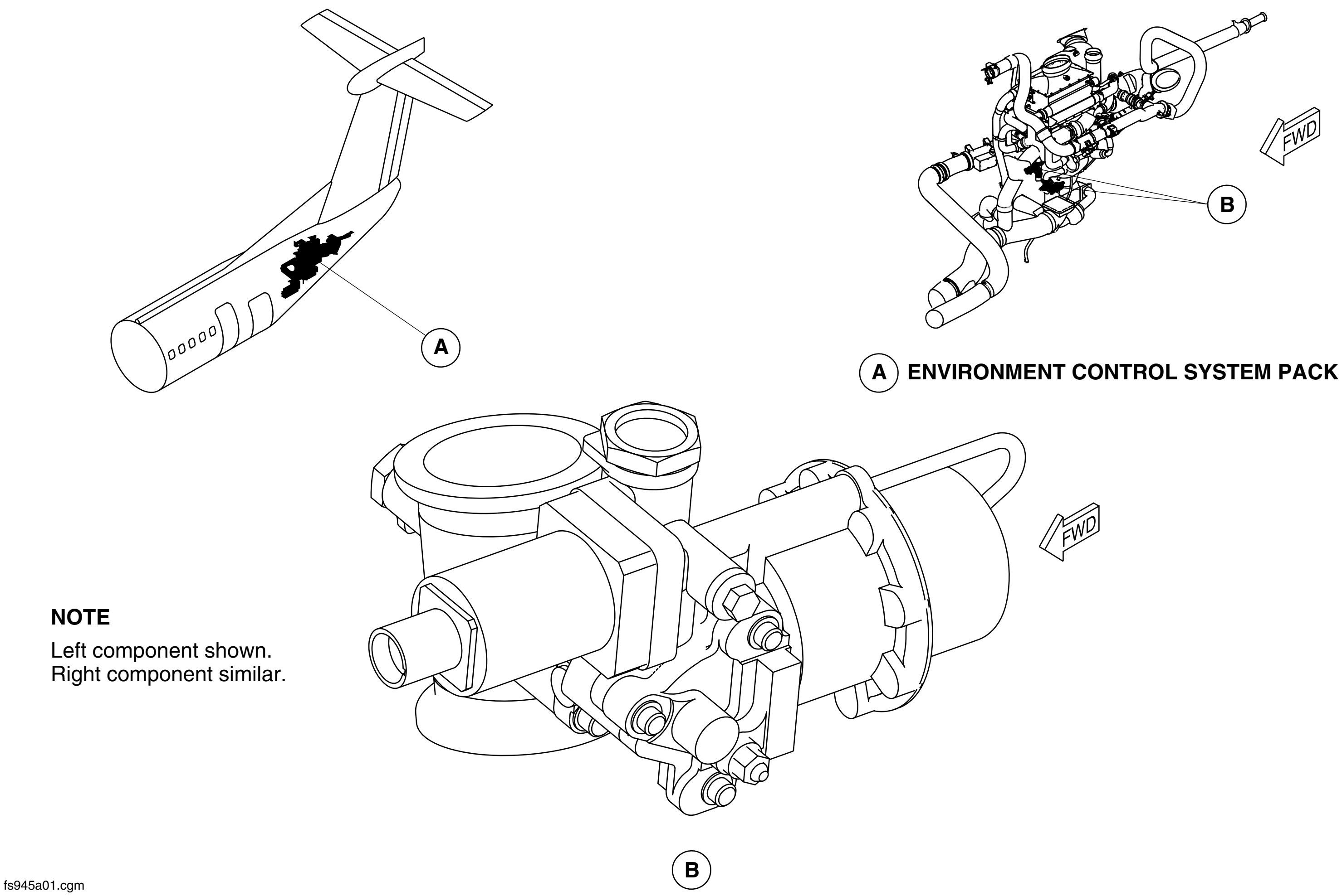
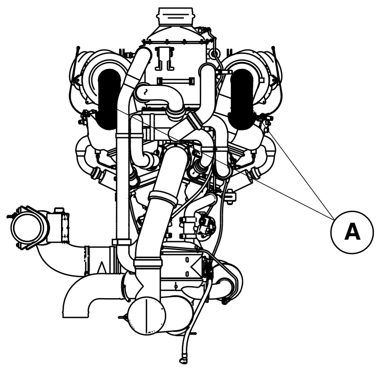
COMPRESSOR INLET CHECK VALVE
Figure 18



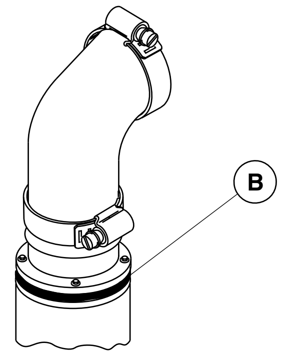
NOTE

Left component shown.
Right component similar.

TURBINE SHUTOFF VALVE
Figure 19



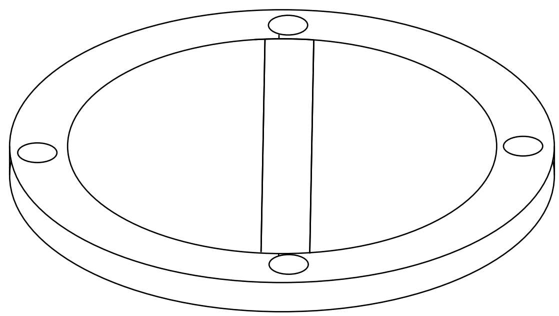
VIEW LOOKING AFT



(A) FAN BYPASS DUCT ASSEMBLY

NOTE

Left component shown.
Right component similar.



(B)

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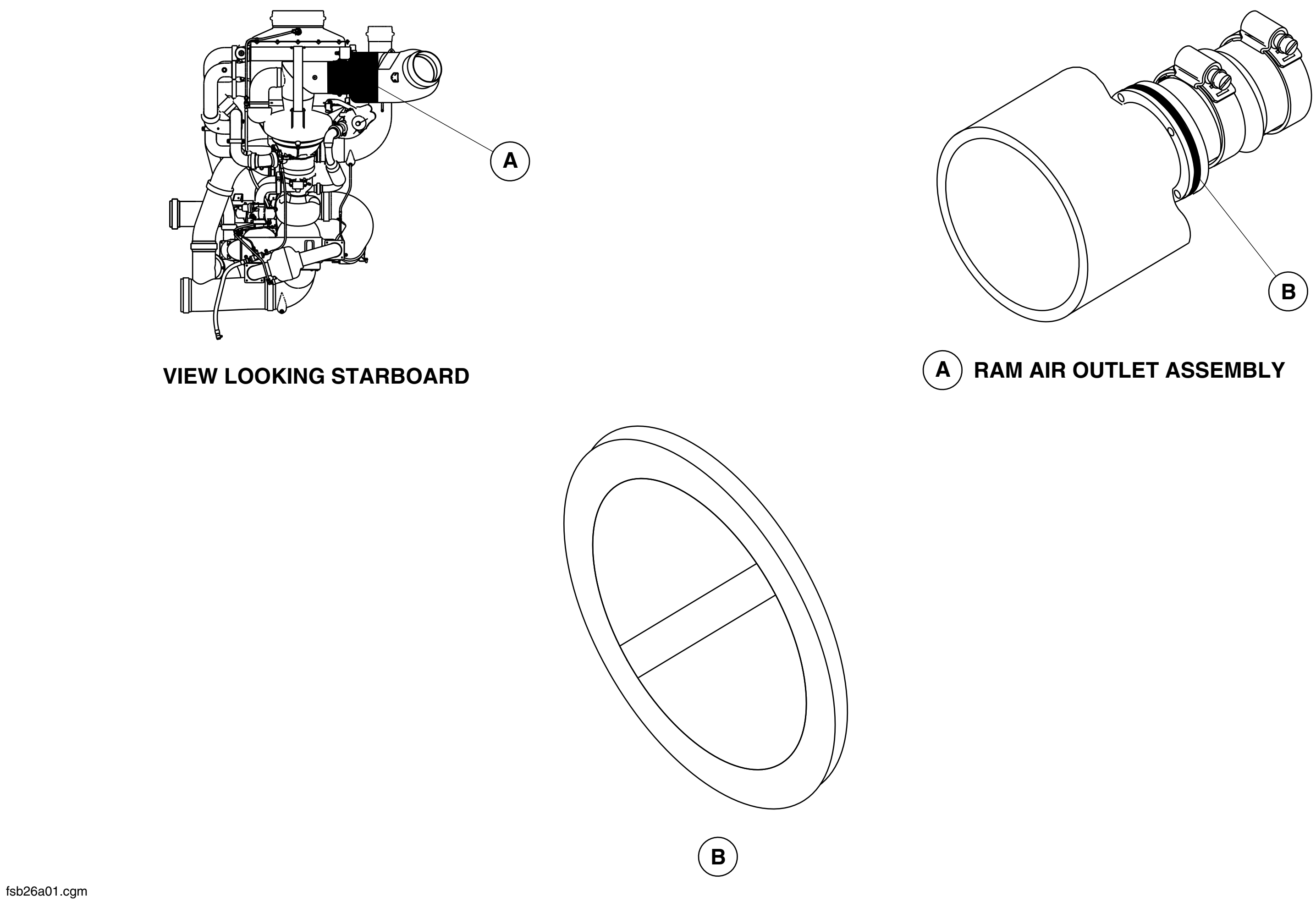
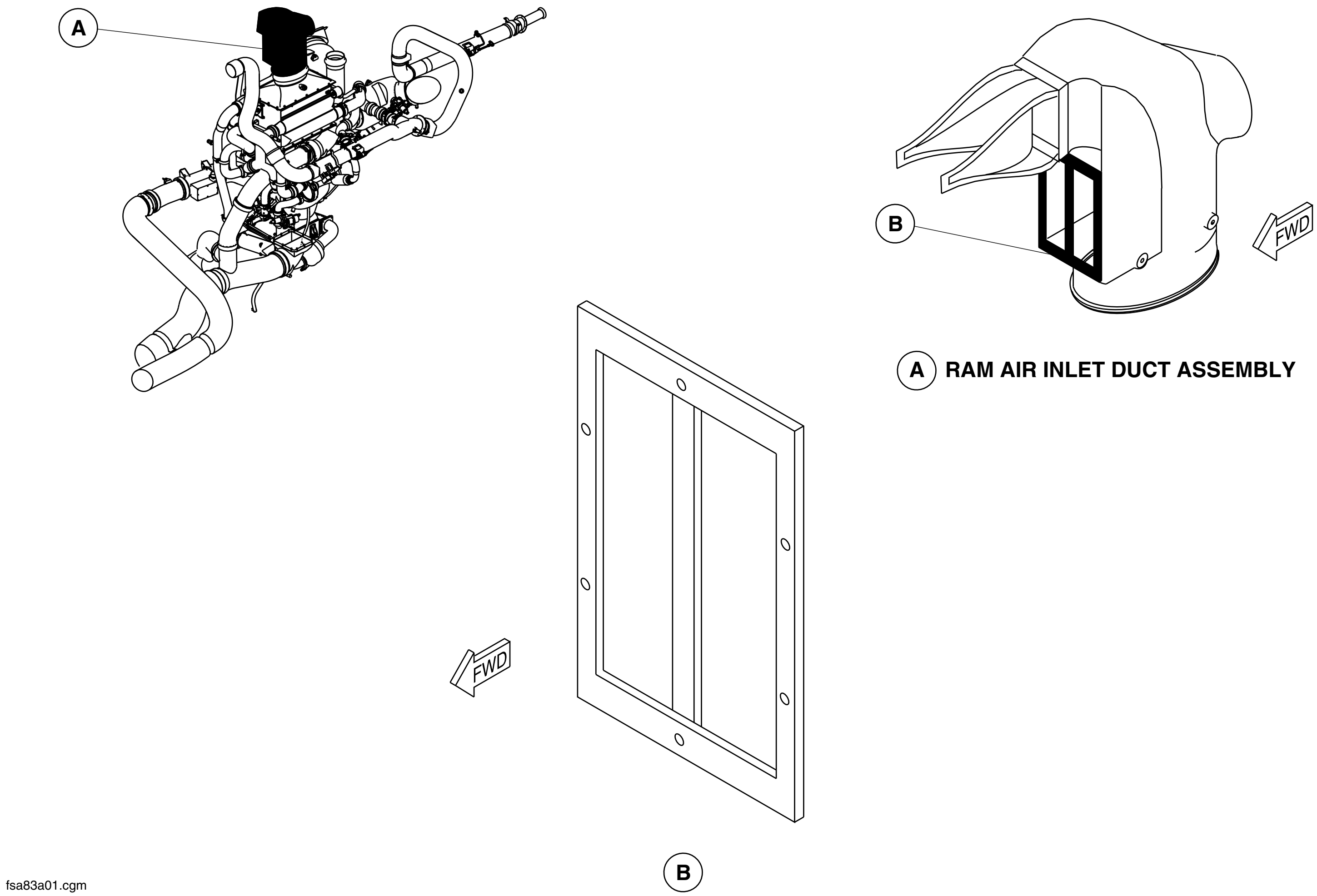
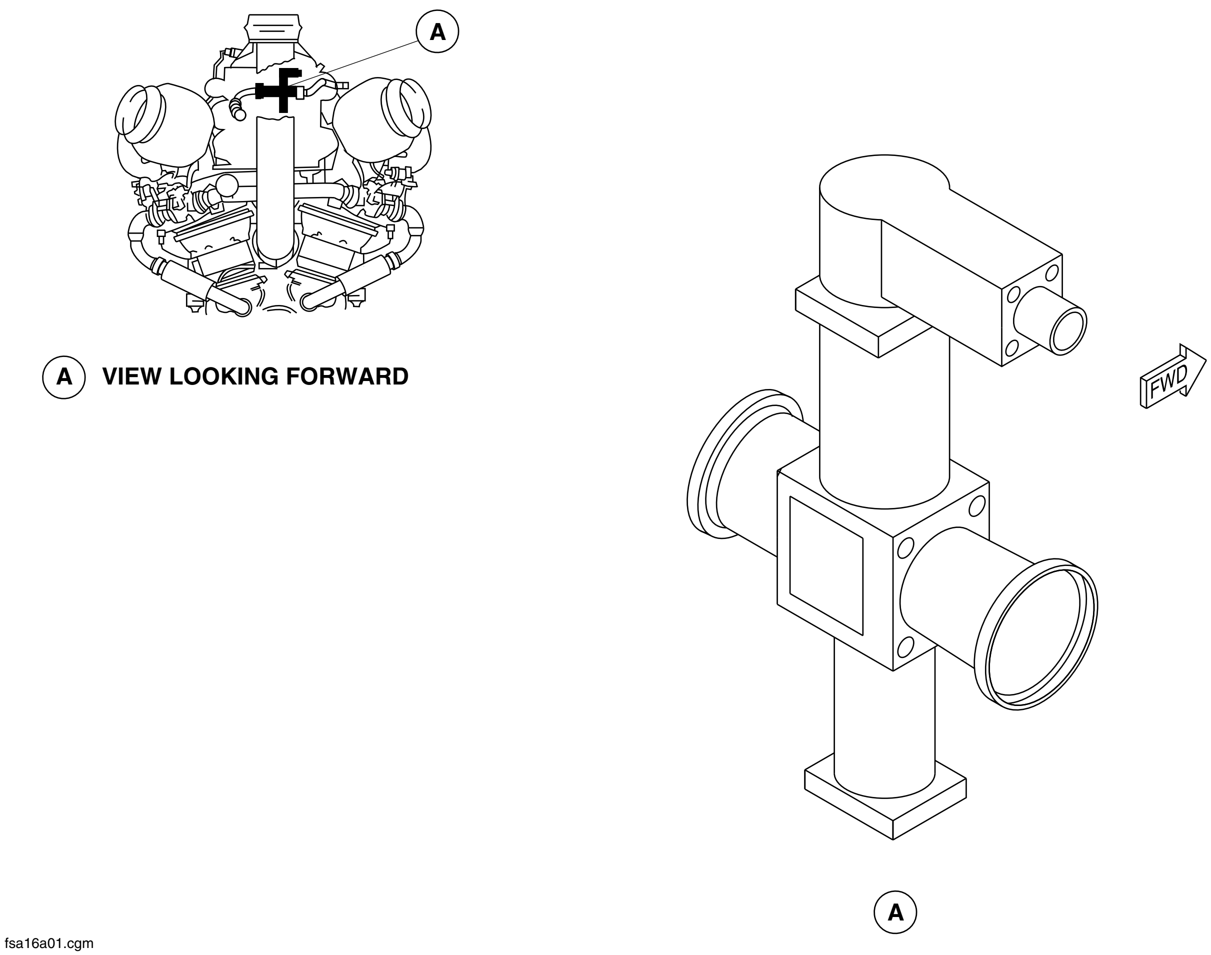
FAN BYPASS CHECK VALVE
Figure 20



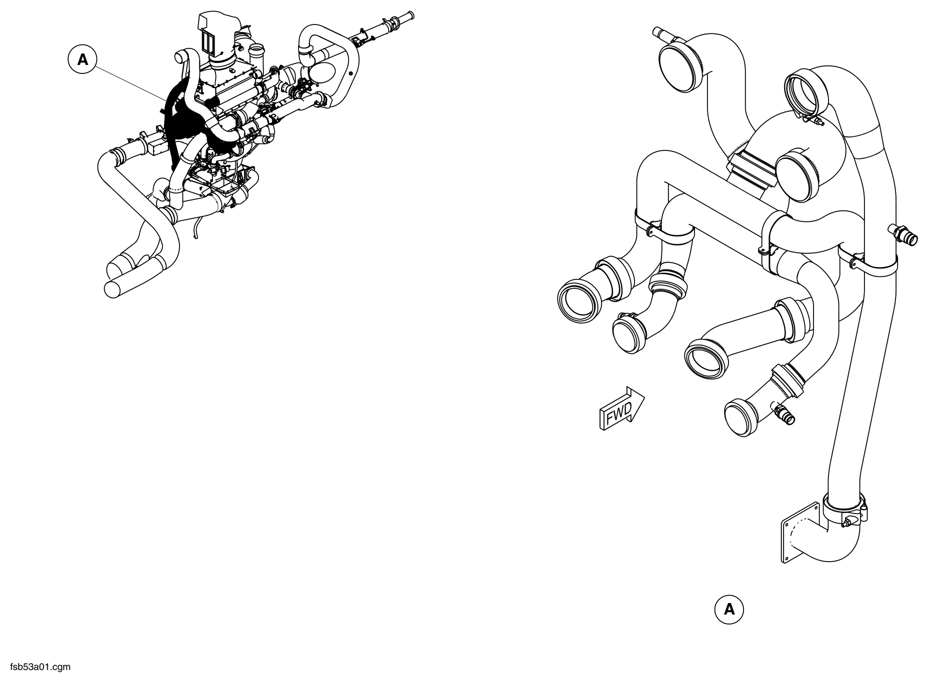
RAM OUTLET CHECK VALVE
Figure 21



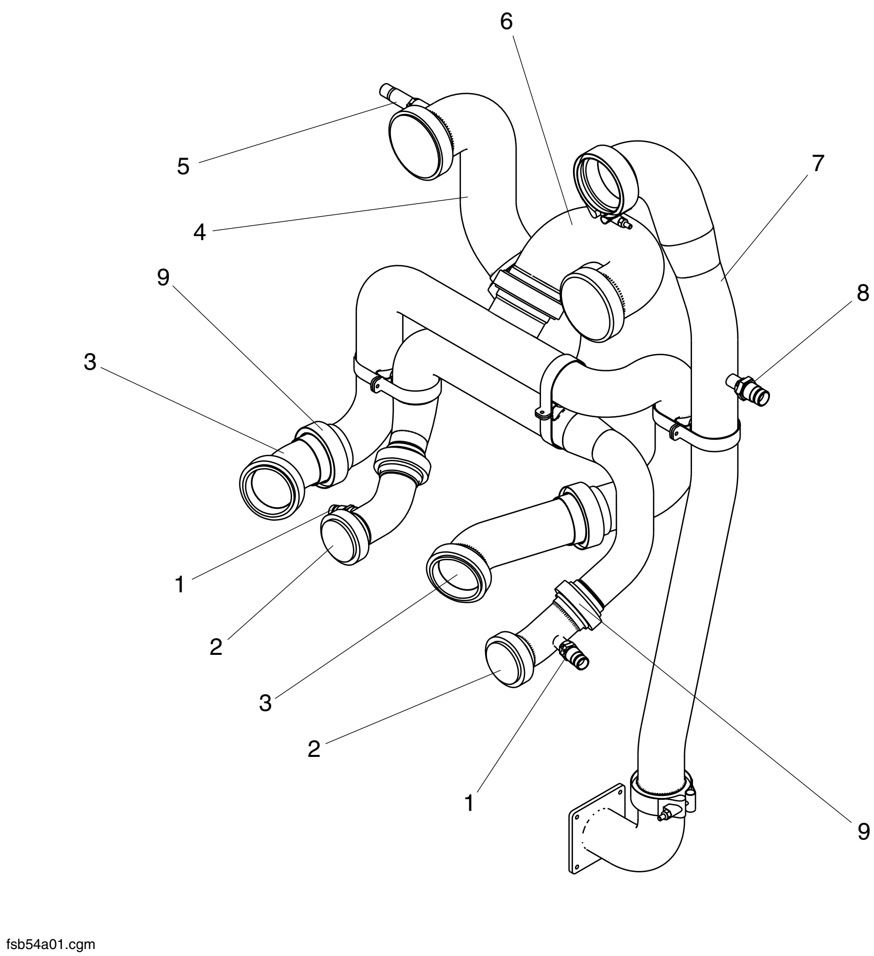
RAM INLET CHECK VALVE
Figure 22



TEMPERATURE REDUCTION SHUTOFF VALVE
Figure 23



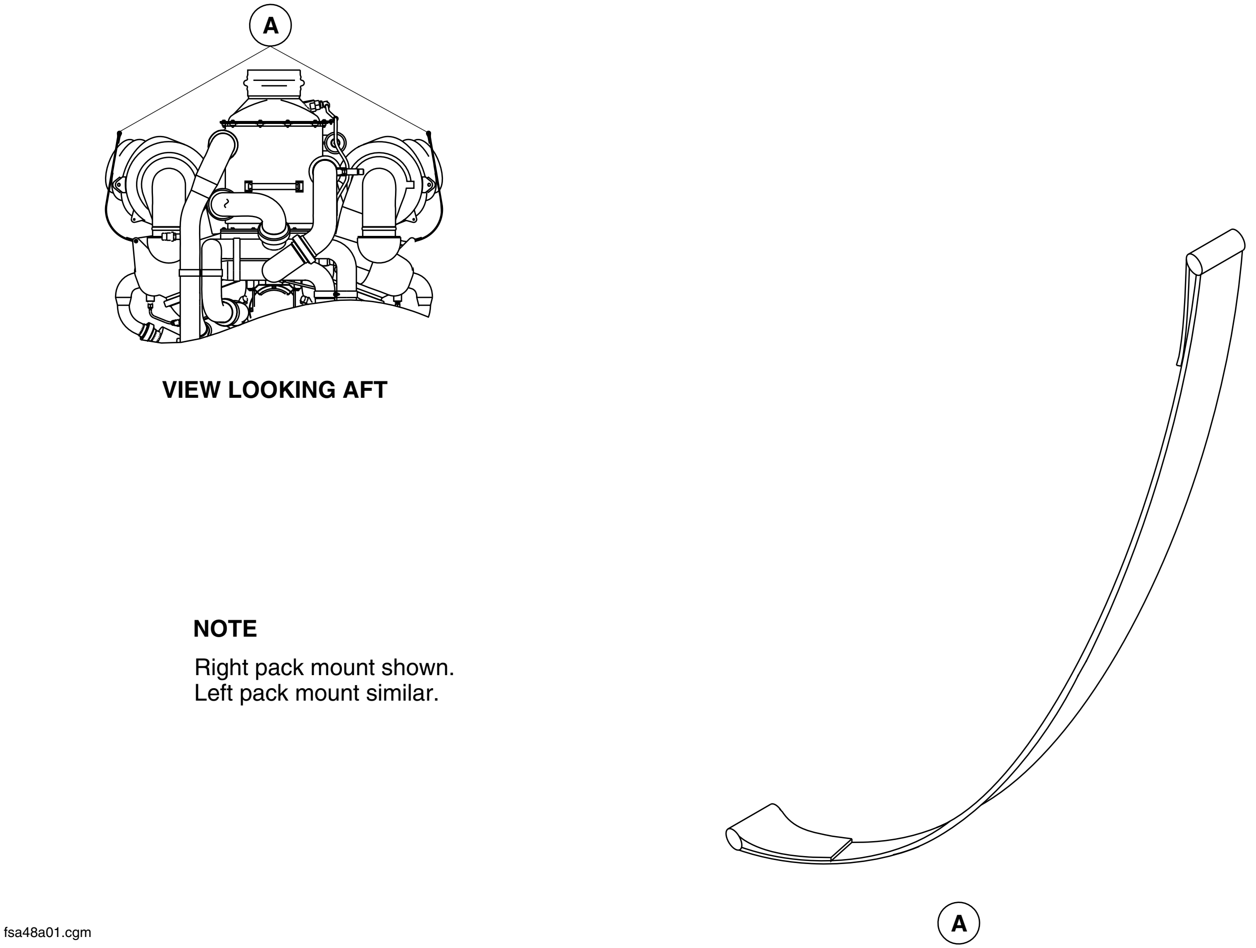
AIR CONDITIONING PACK DUCTS LOCATOR
Figure 24



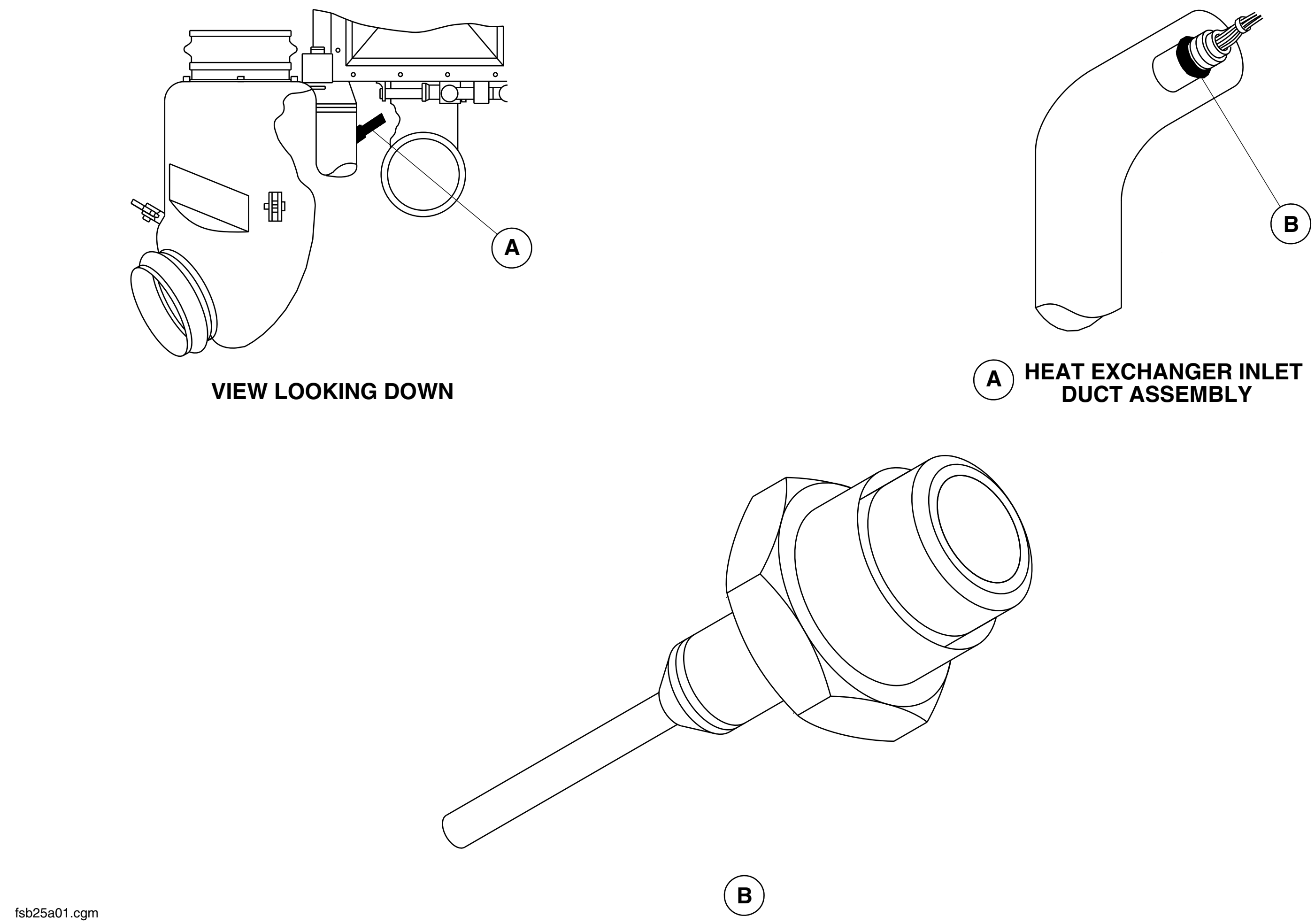
LEGEND

1. Compressor Outlet Temperature Sensors.
2. Compressor Outlet Ducts.
3. Compressor Inlet Ducts.
4. Secondary Heat Exchanger Inlet Duct.
5. Compressor Overtemperature Switch.
6. Primary Heat Exchanger Outlet Duct.
7. Reheater Duct.
8. Secondary Heat Exchanger Outlet Temperature Sensor.
9. Compressor Outlet Check Valves.

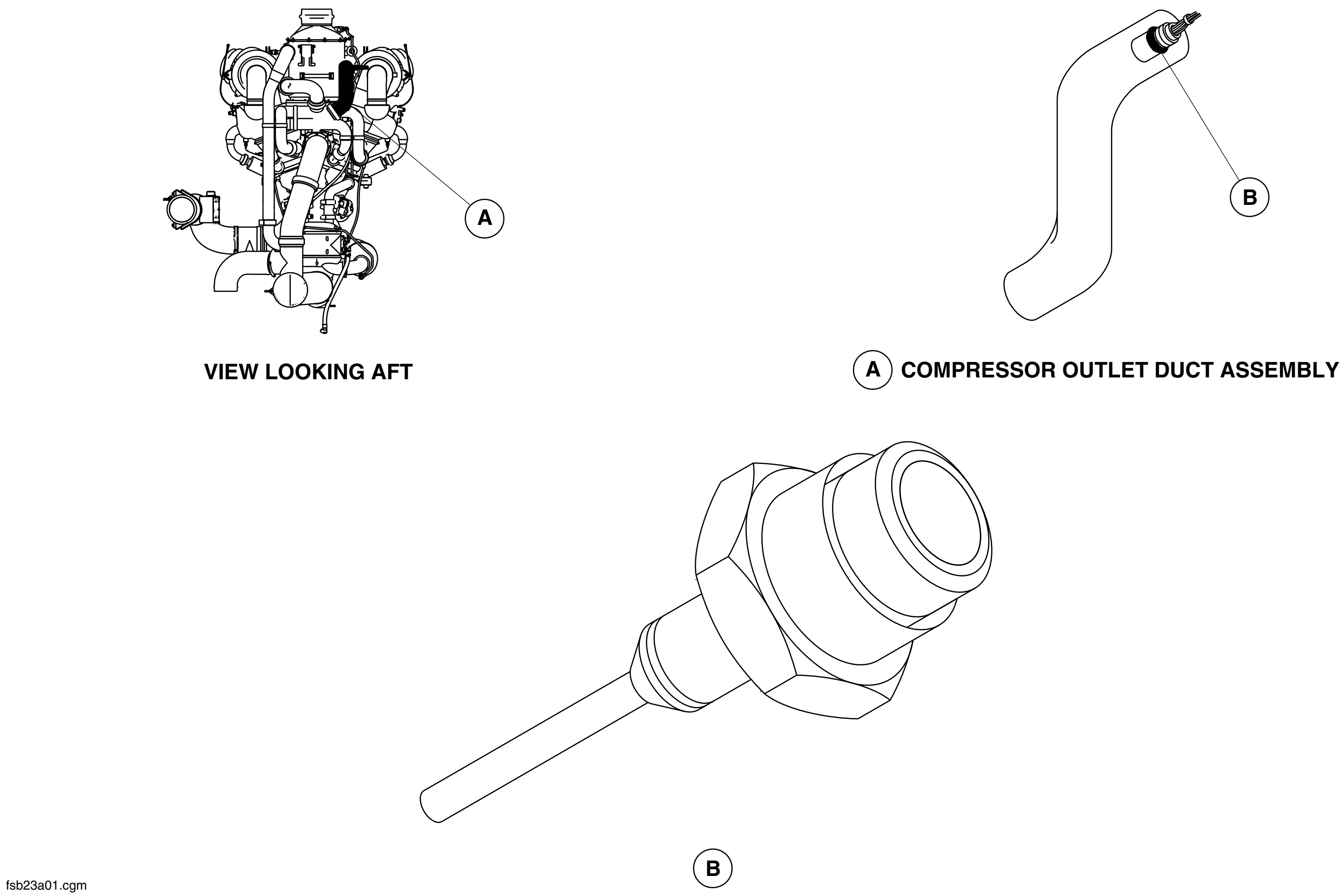
AIR CONDITIONING PACK DUCTS DETAIL
Figure 25



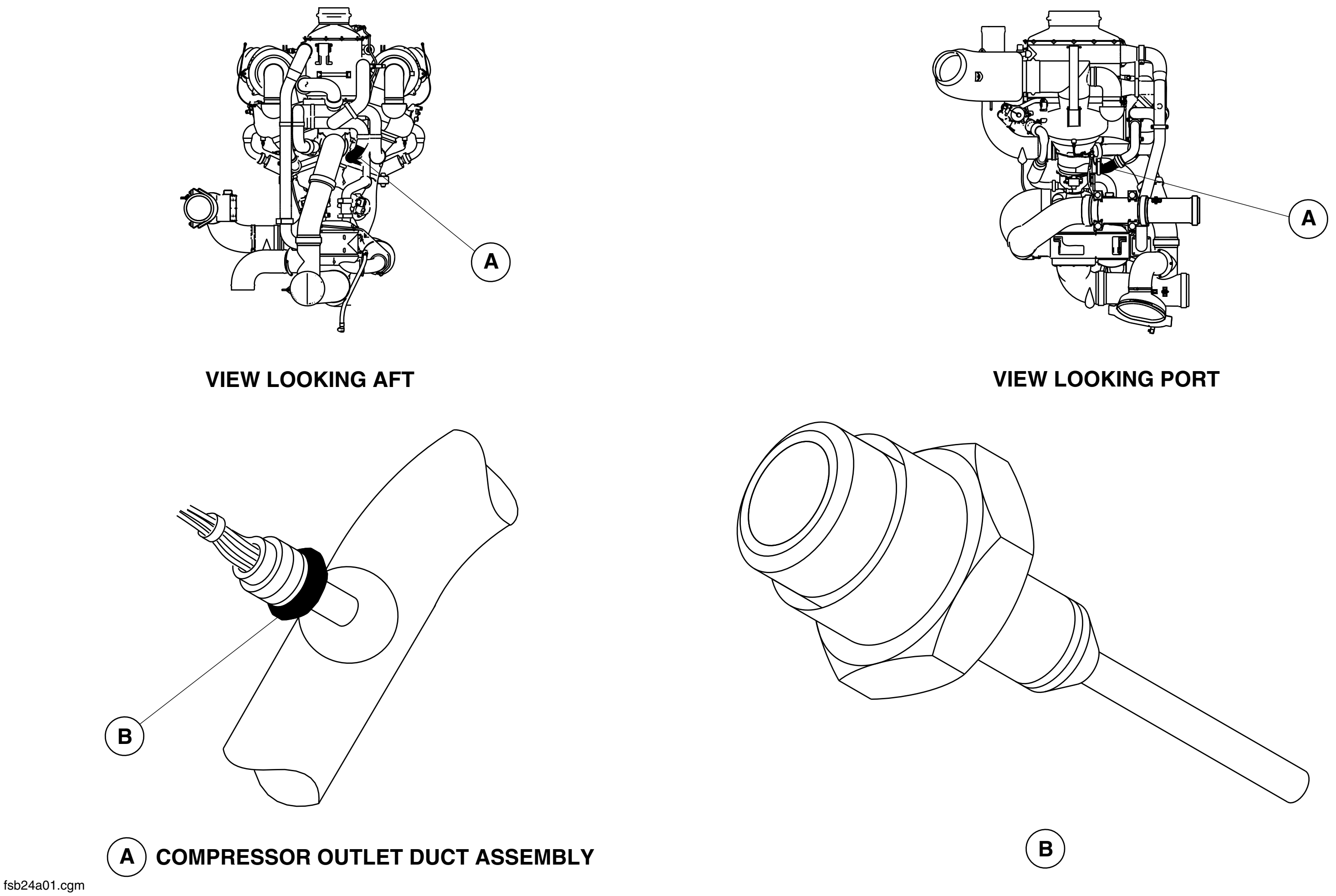
AIR CONDITIONING PACK MOUNTS
Figure 26



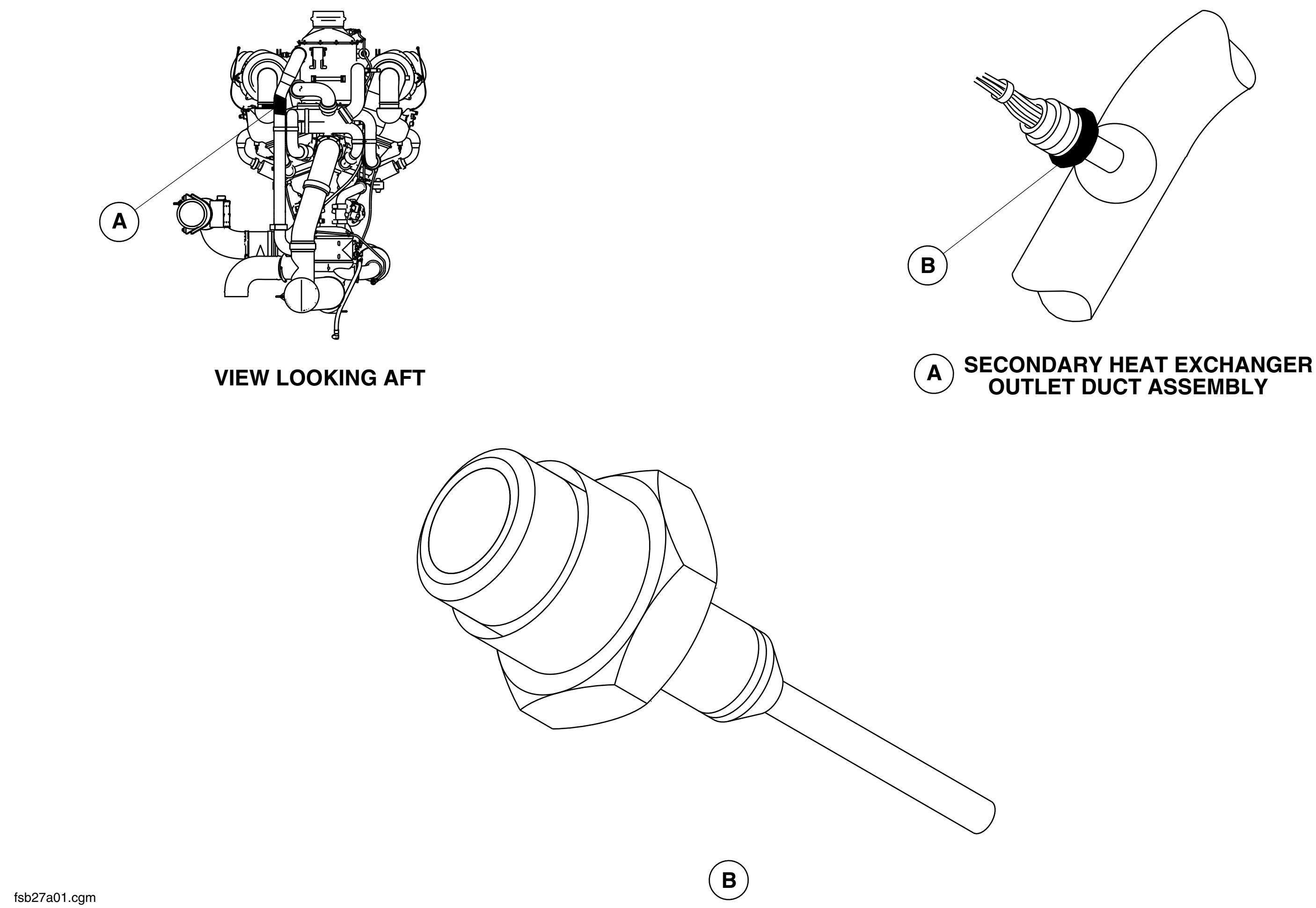
TEMPERATURE REDUCTION VALVE SWITCH
Figure 27



COMPRESSOR OVERTEMPERATURE SWITCH
Figure 28



COMPRESSOR OUTLET TEMPERATURE SENSOR
Figure 29



SECONDARY HEAT EXCHANGER OUTLET TEMPERATURE SENSOR
Figure 30

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21–52–00–001

FLOW CONTROL SYSTEM

Introduction

The flow control system supplies temperature and pressure signals to the Electronic Control Unit (ECU) which regulates the airflow from the bleed source to the air conditioning pack. The flow control system is the part of the Environmental Control System (ECS) that supplies conditioned air to the flight and cabin compartments of the aircraft.

General Description

The flow control system uses pressure and temperature data to control the flow rate of the bleed air that enters the air conditioning pack. The flow control system has three components:

– Valve, Pack Flow Control and Shutoff (21–52–01)

– Sensor, Absolute Pressure (21–52–06)

– Sensor, Pack Inlet Temperature (21–52–11)

These components are installed in the ducting that supplies bleed air to the air conditioning pack. The two sensors supply the pressure and temperature data of the bleed air to the Electronic Control Unit (ECU). The ECU uses this data to open and close the pack flow control and shutoff valve to regulate the flow of bleed air to the air conditioning pack.

Detailed Description

The functions of the flow control system are:

– Control the flow of bleed air to the air conditioning pack

– Supply pack inlet pressure and temperature signals to the ECU

– Supply a valve closed signal to the ECU.

Refer to Figure 1.

The flow of bleed air to the air conditioning pack is determined by three flow venturi differential pressure sensors. There is one differential pressure sensor located in each wing bleed air duct and one sensor located in the APU duct. The two wing bleed duct venturis are used when engine bleed air is selected. The APU duct venturi is used when APU bleed air is selected. Pack inlet pressure and temperature sensors, located upstream of the pack flow control and shutoff valve, are also used in regulating airflow to the air conditioning pack.

When both engines are operating and supplying bleed air, the ECU keeps the pack flow control and shutoff valve fully open. The ECU regulates the nacelle shutoff valves in the bleed air system to control the flow of bleed air. If the left or right side of the bleed air system is turned off (one nacelle shutoff valve is closed) the ECU regulates bleed airflow with the pack flow control and shutoff valve. The ECU also regulates bleed airflow with the pack flow control and shutoff valve if the APU is supplying the bleed air.

The pack inlet pressure and temperature sensor signals are supplied to the ECU. Both digital channels of the ECU use these signals to share control of the pack flow control and shutoff valve. During a flight, one digital channel gets full control of the pack flow control and shutoff valve (the other channel gets full control during the next flight). If the digital channel in control loses power or fails, the other digital channel takes control of the pack flow control and shutoff valve. The analog backup channels do not control the pack flow control and shutoff valve. The digital channel modulates the pack flow control and shutoff valve to provide either minimum, normal, or maximum pack flow. The MIN/NORM/MAX switch on the AIR CONDITIONING control panel sets the desired flow rate (refer to SDS 21–51–00).

Valve, Pack Flow Control and Shutoff

Refer to Figure 2.

The pack flow control and shutoff valve is a pneumatically operated, torque motor actuated butterfly valve. It has a position switch that indicates to the ECU when the valve is closed. The valve has a wire mesh screen upstream of the torque motor to protect the servo parts from contamination. The pack flow control and shutoff valve is installed in line with the air conditioning ducting in the aft fuselage. It controls bleed air flow to the air conditioning pack according to the demands of the Environmental Control System (ECS).

If a mechanical malfunction occurs in the pack flow control and shutoff valve, it defaults pneumatically to the open position to permit continued ECS operation. The pack flow control and shutoff valve defaults electrically to the closed position to stop ECS operation. This electrical default can occur only if both digital channels of the ECU lose power or fail. If this occurs, air is supplied to the cabin and flight compartments by emergency ram air (refer to SDS 21–24–00 Ram Air).

Sensor, Absolute Pressure

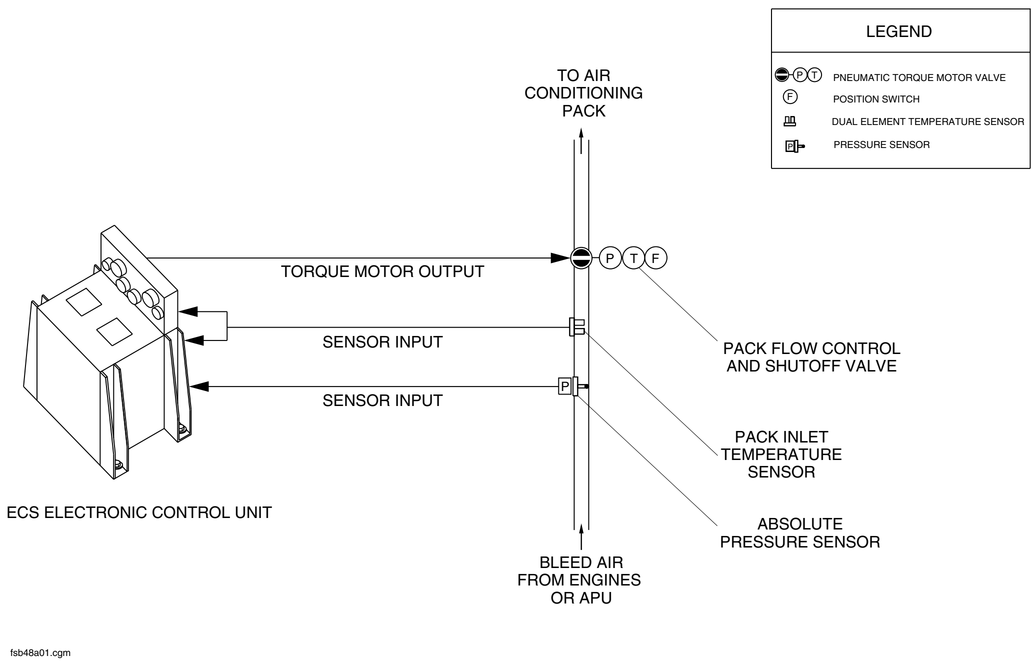
Refer to Figure 3.

The absolute pressure sensor is installed in the air conditioning ducting in the aft fuselage. It senses the pressure of the bleed air as it flows into the air conditioning pack. The sensor supplies a linear pressure signal to the ECU. The absolute pressure data is used by the ECU to determine the bleed air flow rate.

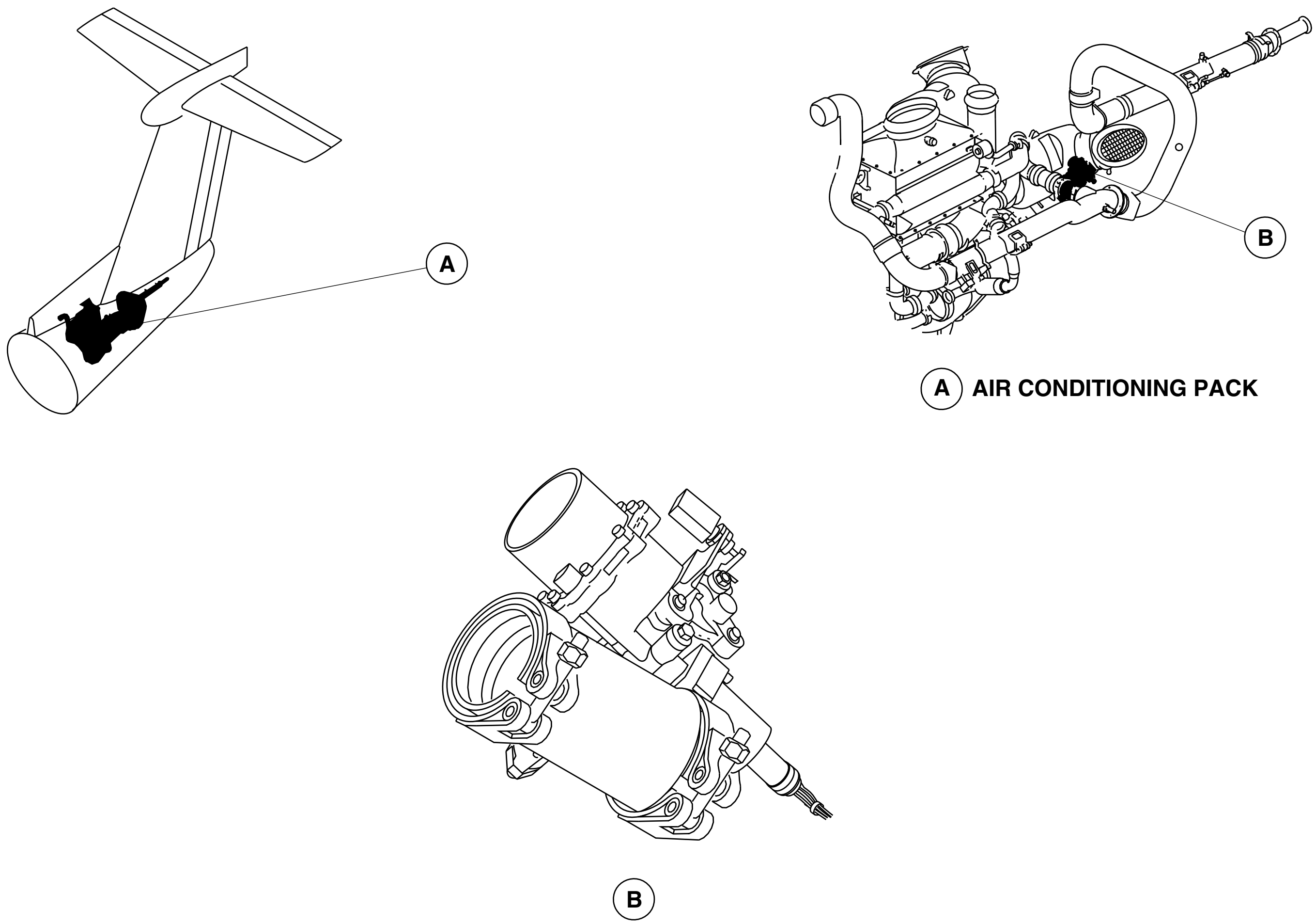
Sensor, Pack Inlet Temperature

Refer to Figure 4.

The pack inlet temperature sensor is a dual resistive temperature device (RTD) that supplies two independent temperature signals to the ECU (one to each digital channel). The pack inlet temperature sensor is installed in the air conditioning ducting in the aft fuselage. It senses the temperature of the bleed air as it flows into the air conditioning pack. The ECU uses this temperature data to determine the bleed air flow rate.



FLOW CONTROL SYSTEM
Figure 1



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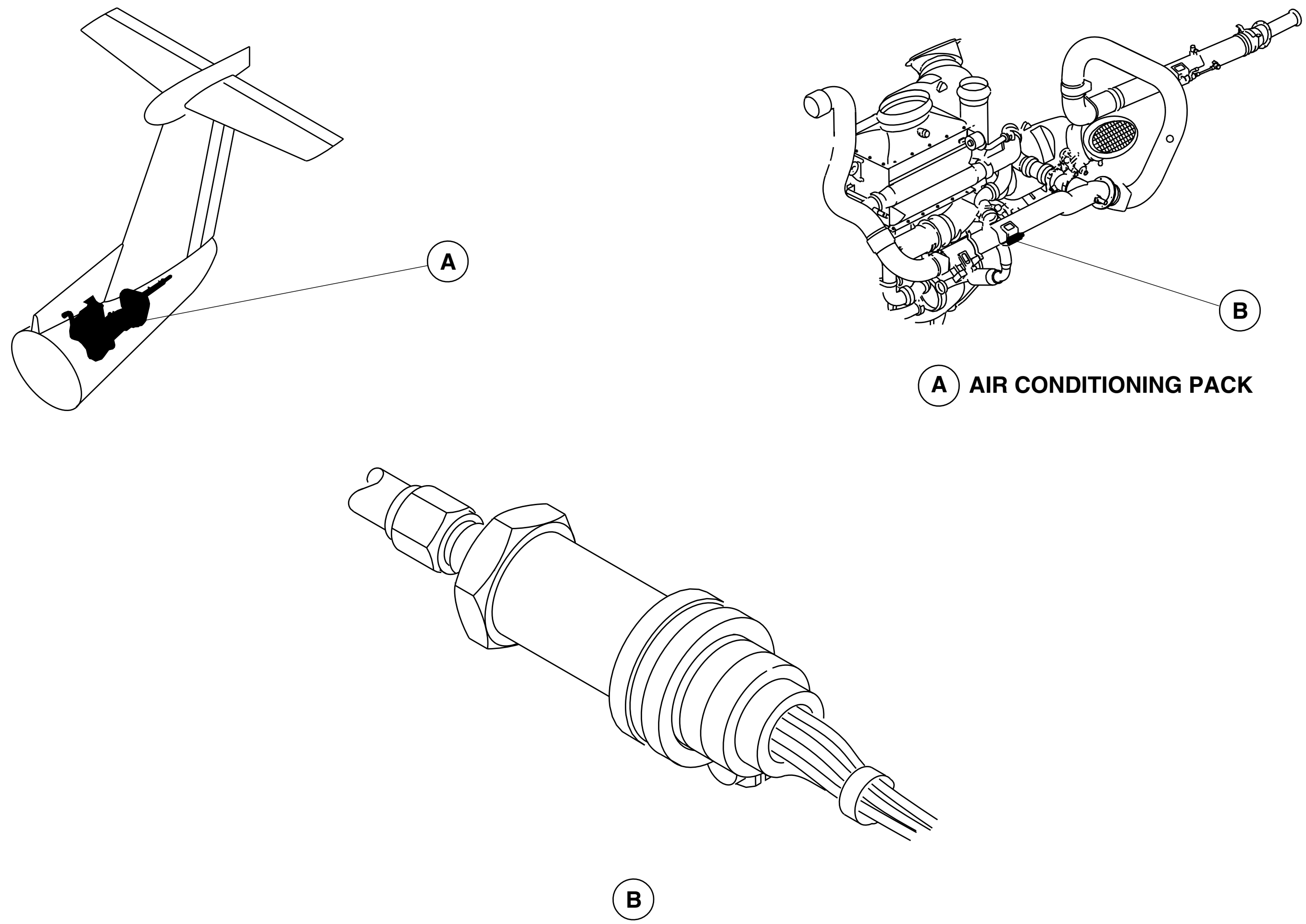
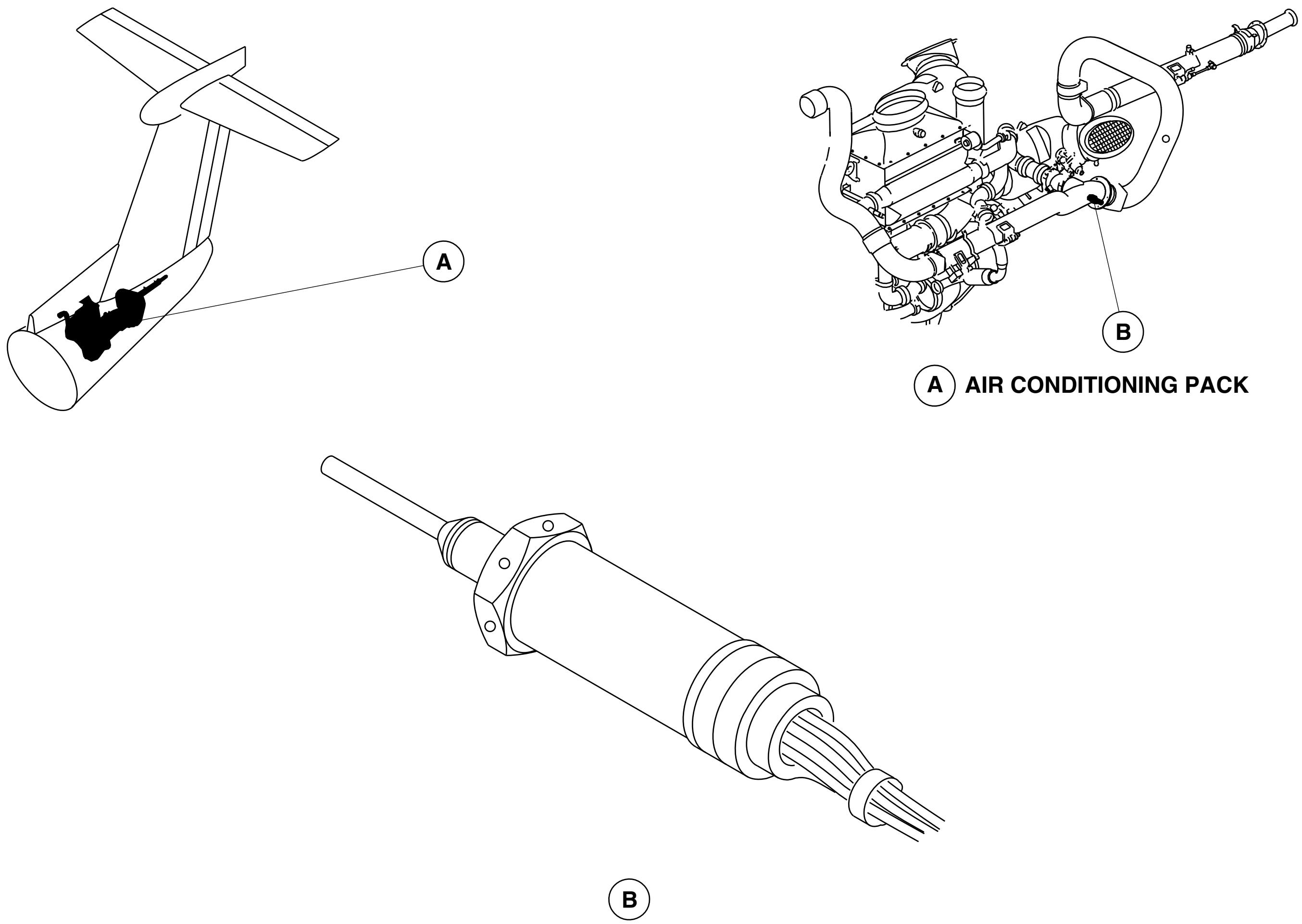
PACK FLOW CONTROL AND SHUT−OFF VALVE
Figure 2



fs893a01.cgm

ABSOLUTE PRESSURE SENSOR
Figure 3



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PACK INLET TEMPERATURE SENSOR
Figure 4

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21–61–00–001

TEMPERATURE CONTROL AND INDICATION

Introduction

The temperature control and indication system is part of the Environmental Control System (ECS), which supplies conditioned air to the flight and cabin compartments of the aircraft.

General Description

The main functions of the temperature control and indication system are to:

- supply temperature control of the Environmental Control System
- supply temperature indication to the flight crew
- sense malfunctions in the system and send fault codes to the fault indication system.

Temperature control starts with setting the desired temperature using the appropriate switches on the AIR CONDITIONING control panel in the overhead console in the flight compartment. The Electronic Control Unit (ECU) monitors the temperature sensors and switches located throughout the aircraft. The ECU then operates the appropriate valves to maintain the selected temperature setting.

The ECU uses sensors and switches to monitor temperature, pressure, flow rates, and valve positions. The ECU uses the pilot and crew commands to control the air conditioning system in manual or automatic operation, one or two ACM operation, bleed source selection, and distribution of conditioned air to the flight and cabin compartments. The ECU and the air conditioning system have redundant control systems for continued ECS operation with mechanical and electrical component malfunctions.

The temperature control and indication system has these components:

- Unit, Electronic Control — Environmental Control System (21–61–01)
- Sensors, Zone Supply Indication (21–61–06)
- Sensors, Temperature (21–61–09)
- Sensors, Zone Supply Temperature (21–61–11)
- Switches, Zone Supply Overtemperature (21–61–16)
- Panel, Air Conditioning Control (21–61–21)
- Selector, Temperature — Cabin (21–61–26)
- Selector, Temperature — Flight Compartment (21–61–31)
- Switches, Control — Packs (21–61–36)
- Gauge, Ducts Temperature (21–61–41)
- Switch, Ducts Temperature (21–61–46).

Detailed Description

The temperature control and indication system uses the following components to sense the temperature of the conditioned air entering the cabin and flight compartments:

- Two zone supply temperature sensors

- Three zone temperature sensors
- Two zone supply indication sensors
- Two zone supply overtemperature switches.

The two zone supply temperature sensors measure the temperature of the air in the cabin and flight compartment supply ducts. The ECU uses these signals to control the temperature of the air leaving the air conditioning pack. The ECU keeps the temperature in the supply ducts between 37 and 160 °F (2.8 and 71 °C). The actual temperature in the supply ducts depends on the settings of the CABIN and FLT COMP temperature selectors on the AIR CONDITIONING control panel. The minimum temperature of 37 °F (2.8 °C) makes sure that there is no ice formation on the condenser. The zone supply temperature sensors are located in the cabin and flight compartment supply ducts below the aft baggage compartment floor (refer to 21–61–11).

The three zone temperature sensors measure the temperature of the air in the cabin and in the flight compartment zones. The ECU uses two of the three zone temperature sensor signals to control the conditioned air temperature. The two zone temperature sensors measure the flight and cabin compartment temperatures and feed the data to the ECU as part of the control loop. The cabin zone sensor also feeds the data to the temperature gauge on the flight attendant's panel. The ECU keeps the temperature in the cabin and flight compartments between 59 and 80.6 °F (15 and 27.0 °C). The third zone temperature sensor (in the cabin compartment) is an indication sensor. It supplies the cabin temperature signal to the DUCT TEMP gauge on the AIR CONDITIONING control panel (refer to 21–61–06).

The two zone supply indication sensors are located in the flight compartment and cabin supply ducts near the zone supply temperature sensors. These indication sensors send the supply duct temperature signals to the DUCT TEMP gauge on the AIR CONDITIONING control panel (refer to 21–61–06). These temperature signals are supplied to the gauge independent of the Environmental Control Unit (ECU). The flight crew can select the two temperatures for display on the DUCT TEMP gauge:

- CABIN (supply) DUCT
- FLT COMP (supply) DUCT.

The two zone supply overtemperature switches are protective devices that turn off the air conditioning system if it overheats. They send discrete signals to the ECU if the temperature of conditioned air in the supply ducts is greater than 190 °F (87.8 °C). These signals cause the ECU to turn off the applicable (left or right) air cycle machine in the air conditioning system. The ECU also turns on the applicable FLT COMPT DUCT HOT or CABIN DUCT HOT caution lights. These switches are located in the cabin and flight compartment supply ducts near the zone supply indication and temperature sensors (refer to 21–61–16).

Refer to Figures 1 and 2.

The ECU is the interface between the AIR CONDITIONING control panel and the mechanical and electrical components of the air conditioning system. It is divided into a left digital channel with a left backup analog channel and a right digital channel with a right backup analog channel. The left digital channel and right analog channel receive power from the left 28 Vdc bus. The right digital channel and left analog channel receive power from the right 28 Vdc bus. This permits continued left and right channel control if one 28 Vdc bus loses power. For example, if the left digital channel loses power, the left analog channel continues operation.

Inside the ECU, the digital channels use an RS 422 digital data bus to communicate with each other. The ECU uses the ARINC 429 digital data bus to communicate with the other aircraft systems. The ARINC 429 receivers of both digital channels are connected to the aircraft wiring. Only the ARINC 429 transmitter of the left digital channel is connected to the aircraft wiring.

The left and right digital channels share control of the pack flow control and shutoff valve, the temperature reduction valve, and the recirculation fan (refer to SDS 21–51–00). One channel is selected to be the digital channel in control. The digital channel in control is responsible for all functions applicable to the shared components. Loss of the digital channel in control causes the other digital channel to take control of the shared components. The digital channel in control automatically changes (between the left and right channels) after every landing.

The digital channel in control of shared functions has air conditioning system ON/OFF control. For normal operation (both packs operating) the digital channel in control of shared functions opens the pack flow control and shutoff valve to enable bleed air flow.

Refer to Figures 3 and 4.

The digital channel in control will shut off the pack flow control and shutoff valve (and stop pack operation) for these conditions:

- Both PACKS switches on the AIR CONDITIONING control panel are set to OFF
- One or more of the three pack overtemperature protection switches (compressor outlet temperature switch (refer to SDS 21–51–00), two zone supply overtemperature switches) are set and turning off the applicable ACM does not remove the overtemperature condition
- The Built In Test (BIT) function of the ECU detects an unacceptable condition.

If the compressor outlet temperature switch causes the pack to shut down, the digital channel in control keeps the pack off until the condition goes away and the applicable PACKS OFF/MAN/AUTO switch is set to OFF. If a zone supply overtemperature switch causes the pack to shut down, the digital channel in control keeps the pack off until the condition goes away and the applicable PACKS OFF/MAN/AUTO switch is set to OFF or MAN.

The left digital channel has on/off control of the left ACM. The right digital channel has on/off control of the right ACM. To shut off an ACM, the digital channel gives sufficient torque motor current (50 m A) to close the applicable turbine shutoff valve. Closure of an overtemperature protection switch or BIT logic can turn off an individual ACM. The torque motor current (to the turbine shutoff valve) has rate limits in the open and closed directions to prevent ACM surge.

Refer to Figure 5.

The two digital channels also share control of the four ECS caution lights.

- FLT COMP DUCT HOT
- CABIN DUCT HOT
- CABIN PACK HOT
- FLT COMPT PACK HOT.

Both digital channels generate discrete outputs to turn on the four ECS caution lamps according to the pack overtemperature protection switches or the BIT logic. Either digital channel turns on the applicable caution light if the channel detects a condition which causes pack shutdown. The caution light stays on until the overtemperature condition or the BIT fault that caused the pack to turn off no longer exists.

The digital channel in control sets pack air flow. Differential pressure sensors in the nacelle bleed ducts and the APU bleed duct (refer to SDS 36–11–00) measure pack air flow. The digital channel in control uses the differential pressure sensors in the nacelle bleed ducts when bleed air is supplied by the engines. The sensors in the APU bleed duct are used when bleed air is supplied by the APU. The pack inlet (absolute) pressure and temperature sensors (refer to SDS 21–52–00) are used to determine pack air flow. All of these sensor inputs go to both digital channels.

Refer to Figure 4.

For operation with both engines and both packs operating, the digital channels regulate the nacelle shutoff valves (in the bleed air system refer to SDS 36–11–00) for minimum, normal, or maximum pack air flow. The pack air flow is selected on the BLEED MIN/NORM/MAX switch on the AIR CONDITIONING control panel. The flight condition determines the amount of flow for each of these levels.

With digital channel failure or APU bleed selection, the pack flow control and shutoff valve regulates the pack air flow. To limit compressor outlet temperature, pack outlet temperature or to prevent ACM overspeed, the digital channel decreases the pack air flow below the setting of the BLEED switch. This lets system operation continue but with reduced heat exchanger performance.

The left digital channel uses approximately 50% of the air from the left ACM to control the flight compartment temperature. The right digital channel uses the other 50% of the air from the left ACM and all the air from the right ACM to control the cabin compartment temperature.

Each digital channel has its own compressor outlet temperature sensor and pack outlet sensor, and receives the other channel's compressor outlet temperature sensor and pack outlet sensor data by the internal RS 422 digital data bus. The digital channel in control uses all four temperature sensors for temperature limiting control.

Refer to Figure 6.

The right digital channel receives two cabin compartment temperature reference inputs. One comes from the AIR CONDITIONING control panel and the other comes from the flight attendant's control panel. The right digital channel also receives a discrete input that enables or disables the flight attendant's control panel. If the flight attendant's control panel is enabled, it supplies the cabin compartment temperature reference. If the flight attendant's control panel is disabled, the AIR CONDITIONING control panel gives the cabin compartment temperature reference.

Refer to Figure 4.

The CABIN temperature selector on the AIR CONDITIONING control panel has a switch at the full counter clockwise position. Turning the selector knob to full COOL activates the switch which sends the discrete signal to the right digital channel to enable the flight attendant's control panel. This switch also turns on the ENABLED light on the flight attendant's control panel. This indicates the flight attendant has control of the cabin temperature selection. The temperature gauge on the panel receives its signal from the cabin zone temperature indication sensor. Both the ENABLED light and the cabin temperature gauge on the flight attendant's control panel operate independently from the ECU.

When the PACKS switch is set to AUTO, the digital channel in control opens the pack flow control and shutoff valve and the turbine shutoff valves. This starts the ACMs which supply cold air. The digital channel in control (ECU) opens and closes the ACM bypass valves to add warm air to the cool air coming out of the turbines. The ECU controls the pack outlet temperature based on the settings of the CABIN and FLT COMP temperature selectors on the AIR CONDITIONING control panel. The digital channel uses the two zone supply temperature sensors (one in the cabin supply duct and one in the flight compartment duct) for temperature feedback. The ECU can vary the cabin and flight compartment temperatures from 59 and 80.6 °F (15 to 27.0 °C). To prevent the possible formation of ice in the condenser, the ECU also keeps the supply duct temperature at or above 37 °F (2.8 °C). This removes the need for ice detection devices and deice controls.

When the PACKS switch is set to MAN, or if the digital channel fails, the analog channel supplies temperature control. The analog channel controls the temperature of the supply ducts by opening and closing the pack bypass valves. The analog channel uses the settings of the CABIN and FLT COMP temperature selectors as a reference for controlling the temperature.

If the temperature reduction valve switch identifies that the bleed air temperature to the pack is too hot, the digital channel in control opens the temperature reduction shutoff valve. This lets the cooler air from the secondary heat exchanger outlet mix with the hot bleed air. The temperature reduction shutoff valve stays open until the pack inlet temperature sensor identifies that the bleed air temperature has decreased to normal values. Each digital channel has its own discrete input for the temperature reduction valve switch and its own temperature sensor input for the pack inlet temperature. Both digital channels share control of the temperature reduction shutoff valve.

Refer to Figure 4.

The digital channels share control of the recirculation fan speed. The digital channel in control generates the discrete signal that sets the recirculation fan speed to high or low. The RECIRC switch on the AIR CONDITIONING control panel commands the recirculation fan on or off independently of the ECU. The digital channels receive an on/off discrete to identify the on/off status of the recirculation fan.

The ARINC 429 digital data bus gives hot/cold day operation condition information to the digital channels. Control logic uses the operation conditions and the length of time the fan is on to determine fan speed. The recirculation fan starts operation at low speed and runs at low speed when the air conditioning system operates with one ACM turned off (limited flow capacity). The recirculation fan runs at high speed for all other conditions.

The ECU performs system level BUILT IN TESTS (BIT) for both the bleed air system and the air conditioning system. These tests are performed on a continuous basis. For the air conditioning system, the BIT checks that pack flow and pack discharge temperature are within limits. If this system level check detects a fault, BIT monitors individual Line Replaceable Units (LRUs).

To ensure fail safe operation, each digital control channel does self test on itself and its interfaces. These BIT tests consist of power on diagnostics and continuous BIT tests. Each digital channel also monitors and performs BIT on the associated analog channel.

The ECU stores information about system and component faults in its non–volatile memory. These faults can then be accessed by maintenance personnel through the Central Diagnostic System (CDS). The fault information is shown separately for the left and right digital channels in the form of fault codes on a series of pages. The ECU also supplies flight history information and a table of the status of the individual LRUs.

If a fault is logged by the ECU, the ECS AUTO FAIL light on the maintenance panel comes on. When the faults have been corrected, the non–volatile memory (NVM) must be cleared to erase the fault information. Pushing the NVM RESET switch with the system switch set to ECS, clears the NVM.

Electronic Control Unit (ECU) (21–61–01)

Refer to Figures 1 and 2.

The ECU is installed in the avionics bay. It uses the temperature signals from the two zone supply temperature sensors and two of the zone temperature sensors (one each in the flight and cabin compartments) as feedback signals. These feedback signals are used to regulate the air temperatures in the flight and cabin compartments. The ECU has two channels (left and right). Each channel has a digital (automatic) part that has a power supply, processor, low level Input/Output (I/O) and motor driver board. Each channel also has an analog (manual) part that has an analog backup board.

The power supply provides all the internal voltages for the digital system. It supplies the power for the solenoid drivers and the pressure sensor excitations.

The processor contains the microprocessor RAM and EEPROM memory. It also has the ARINC 429 and RS422 communication circuitry.

The low level I/O board has the Analog/Digital interface for the thermistors, RTDs, potentiometer and pressure signal inputs.

The motor driver has output interfaces for the torque motors and the discrete outputs.

The analog backup has its own power regulation circuitry, thermistor, potentiometer and discrete inputs. It also has torque motor and discrete outputs. The analog backup provides duct temperature control, ECS (pack) shutdown logic and bleed system on/off control.

The ECU has monitors to provide automatic protection, fault recording and fault indications. The monitors are described as follows:

- The cabin outlet overtemperature monitor 1 will stop the cabin ACM in response to a first level pack outlet overtemperature fault. This fault is monitored by the dual thermistors of the cabin zone supply temperature sensor and the cabin zone supply overtemperature switch. This monitor will command the cabin ACM turbine shut off valve to close and set the CABIN DUCT HOT caution light. If both channels of the cabin zone supply temperature sensor are invalid and the cabin zone supply overtemperature switch indicates an overtemperature condition then the CABIN DUCT HOT caution light will come on.
- The cabin outlet overtemperature monitor 2 will stop the cabin ACM in response to a second level pack outlet overtemperature fault. This fault is monitored by both dual thermistors of the cabin zone supply temperature sensor. This monitor will command the cabin ACM turbine shutoff valve to close and the CABIN DUCT HOT caution light will come on.

- The cabin outlet overtemperature monitor 3 will stop the cabin ACM in response to a third level pack outlet overtemperature fault. This fault is monitored by either dual thermistor of the cabin zone supply temperature sensor. This monitor will command the cabin ACM turbine shutoff valve to close and the CABIN DUCT HOT caution light will come on.
- The cabin outlet overtemperature monitor 4 will stop the pack in response to fourth level pack outlet overtemperature fault. This fault is monitored by the persistence of monitors 1, 2 and 3 for either dual thermistor of the cabin zone supply temperature sensor or the cabin zone supply overtemperature switch. This monitor will command the pack flow control shutoff valve to close and the CABIN DUCT HOT and FLT COMP PACK HOT caution lights will come on.
- The flight compartment outlet overtemperature monitor 1 will stop the flight compartment ACM in response to a first level pack outlet overtemperature fault. This fault is monitored by either dual thermistor of the flight compartment zone supply temperature sensor and the flight compartment zone supply overtemperature switch. If both sensors are invalid and the overtemperature switch indicates an overtemperature condition the monitor will command the flight compartment ACM shutoff valve to close and the FLT COMP DUCT HOT caution light will come on.
- The flight compartment outlet overtemperature monitor 2 will stop the flight compartment ACM in response to a second level pack outlet overtemperature fault. This fault is monitored by both dual thermistors of the flight compartment zone supply temperature sensor. This monitor will command the flight compartment ACM turbine shutoff valve to close and set the FLT COMP DUCT HOT caution light.
- The flight compartment outlet overtemperature monitor 3 will stop the flight compartment ACM in response to a third level pack outlet overtemperature fault. This fault is monitored by both dual thermistors of the flight compartment zone supply temperature sensor. If both sensors are invalid and the overtemperature switch indicates an overtemperature condition. This monitor will command the flight compartment ACM turbine shutoff valve closed and set the FLT COMP DUCT HOT caution light.
- The flight compartment outlet overtemperature monitor 4 will stop the flight compartment ACM in response to a fourth level pack outlet overtemperature fault. This fault is monitored by the persistence of monitors 1, 2 and 3 for the dual thermistors of the flight compartment zone supply temperature sensor. This monitor will command the pack flow control shutoff valve to close and set the FLT COMP DUCT HOT and the CABIN PACK HOT caution lights.

– The ACM overtemperature shutdown monitor 1 will stop the related ACM in response to a first level ACM overtemperature fault. This fault is monitored by the compressor outlet temperature sensor and the compressor overtemperature switch. This monitor will command the related ACM turbine shut–off valve to close and set the CABIN PACK HOT or FLT COMP PACK HOT caution light.

– The ACM overtemperature shutdown monitor 2 will stop the related ACM in response to a second level ACM overtemperature fault. This fault is monitored by the compressor outlet temperature sensor. This monitor will command the related ACM turbine shutoff valve to close and set the CABIN PACK HOT or FLT COMP PACK HOT caution light.

– The cabin pack temperature error hot fault monitor will stop the cabin ACM in response to a cabin duct overtemperature fault. This fault is monitored by the cabin zone supply reference temperature and the cabin zone supply temperature sensor. This monitor will command the cabin ACM turbine shutoff valve to close and set the CABIN PACK HOT caution light.

– The flight compartment pack temperature error hot fault monitor will stop the flight compartment ACM in response to a flight compartment duct overtemperature fault. This fault is monitored by the flight compartment zone supply reference temperature and the flight compartment zone supply temperature sensor. This monitor will command the flight compartment ACM turbine shutoff valve to close and set the FLT COMP PACK HOT caution light.

– The cabin ACM performance degrade monitor will stop the cabin ACM in response to a cabin ACM overtemperature fault. This fault is monitored by the cabin zone supply temperature sensor and both compressor outlet temperature sensors. This monitor will command the cabin ACM turbine shutoff valve to close and set the CABIN PACK HOT caution light.

– The flight compartment ACM performance degrade monitor will stop the flight compartment ACM in response to a flight compartment ACM overtemperature fault. This fault is monitored by the flight compartment zone supply temperature sensor and both compressor outlet temperature sensors. This monitor will command the flight compartment ACM turbine shutoff valve to close and set the FLT COMP PACK HOT caution light.

– If the cabin turbine shutoff valve stay closed the fault monitor will stop the cabin ACM in response to a degraded cabin ACM fault. This fault is monitored by the cabin zone supply temperature sensor and cabin compressor outlet temperature sensor. This monitor will command the cabin ACM turbine shutoff valve to close and set the CABIN PACK HOT caution light.

– If the flight compartment turbine shutoff valve stay closed the fault monitor will stop the flight compartment ACM in response to a degraded flight compartment ACM fault. This fault is monitored by the flight compartment zone supply temperature sensor and flight compartment compressor outlet temperature sensor. This monitor will command the flight compartment ACM turbine shutoff valve to close and set the FLT COMP PACK HOT caution light.

- The cabin ACM fault monitor will stop the cabin ACM in response to a cabin ACM overtemperature fault. This fault is monitored by the cabin zone supply temperature sensor and cabin compressor outlet temperature sensor. This monitor will command the cabin ACM turbine shutoff valve to close and set the CABIN PACK HOT caution light.
- The flight compartment ACM fault monitor will stop the flight compartment ACM in response to flight compartment ACM overtemperature fault. This fault is monitored by the flight compartment zone supply temperature sensor and flight compartment compressor outlet temperature sensor. This monitor will command the flight compartment ACM turbine shutoff valve to close and set the FLT COMP PACK HOT caution light.
- The pack overtemperature shutdown monitor will stop the ACM pack in response to a pack overtemperature fault. This fault is monitored by the compressor outlet temperature sensor and the compressor overtemperature switch. This monitor will command the pack flow control valve to close and set the CABIN PACK HOT and FLT COMP PACK HOT caution lights.
- The pack overtemperature shutdown monitor 1 will stop the ACM pack in response to a pack overtemperature fault. This fault is monitored by the compressor outlet overtemperature switch when a channel failure occurs. This monitor will command the pack flow control valve to close and set the CABIN PACK HOT and FLT COMP PACK HOT caution lights.

Zone Supply Indication Sensors (21–61–06)

Refer to Figure 7.

The zone supply indication sensors are resistance temperature devices (RTD). An RTD gives a resistance which is proportional to the temperature it senses. These sensors supply the signals (flight compartment supply duct temperature, cabin supply duct temperature) for the DUCT TEMP gauge on the AIR CONDITIONING control panel. The wiring from the sensors to the gauge is independent from the ECU and the gauge has its own circuit breaker. One sensor is located at Stn. X765 in the flight compartment supply duct. The other sensor is located at the same station in the cabin supply duct.

Temperature Sensors (21–61–09)

Refer to Figures 8 , 9 , 10 and 11.

There are three temperature sensors (one in the flight and two in the cabin compartments) that sense the flight and cabin compartment temperatures. The ECU uses the feedback temperature signals from the temperature sensors to regulate the conditioned air temperature. The temperature sensors are dual thermistor sensors.

The flight compartment temperature sensor is mounted on the inlet housing assembly installed on the co–pilot's rudder–pedal cover assembly. The flow of the air past duct inlet draws flight compartment air over the temperature sensor. This makes the temperature sensor to measure accurate ambient air temperature in the flight compartment.

The No. 1 cabin temperature sensor is located on the cabin ejector. This sensor acts as a temperature indication sensor. It supplies a signal (cabin temperature) to the temperature gauge on the AIR CONDITIONING control panel.

The No. 2 cabin temperature sensor is also located on the cabin temperature ejector by the fourth dado panel (Stn. X 352) on the left side of the aircraft. Cabin supply air flows through a restriction in the dado duct. This accelerates the flow of air and causes a low pressure air which draws cabin air over the temperature sensor. This ensures an accurate measurement of the cabin ambient air temperature. It supplies the signal (cabin temperature) to the temperature gauge on the flight attendant's control panel.

Zone Supply Temperature Sensors (21–61–11)

Refer to Figure 12.

The zone supply temperature sensors are dual thermistor sensors that supply independent signals to the digital and analog channels of the ECU. These sensors send supply duct temperature information to the ECU so it can control the temperature of the cabin and flight compartment.

The zone supply temperature sensors are installed in the cabin and flight compartment supply ducts under the aft baggage compartment floor (Stn X 765). These ducts supply conditioned air from the air conditioning system to the flight and cabin compartments (refer to SDS 21–21–00 and 21–22–00).

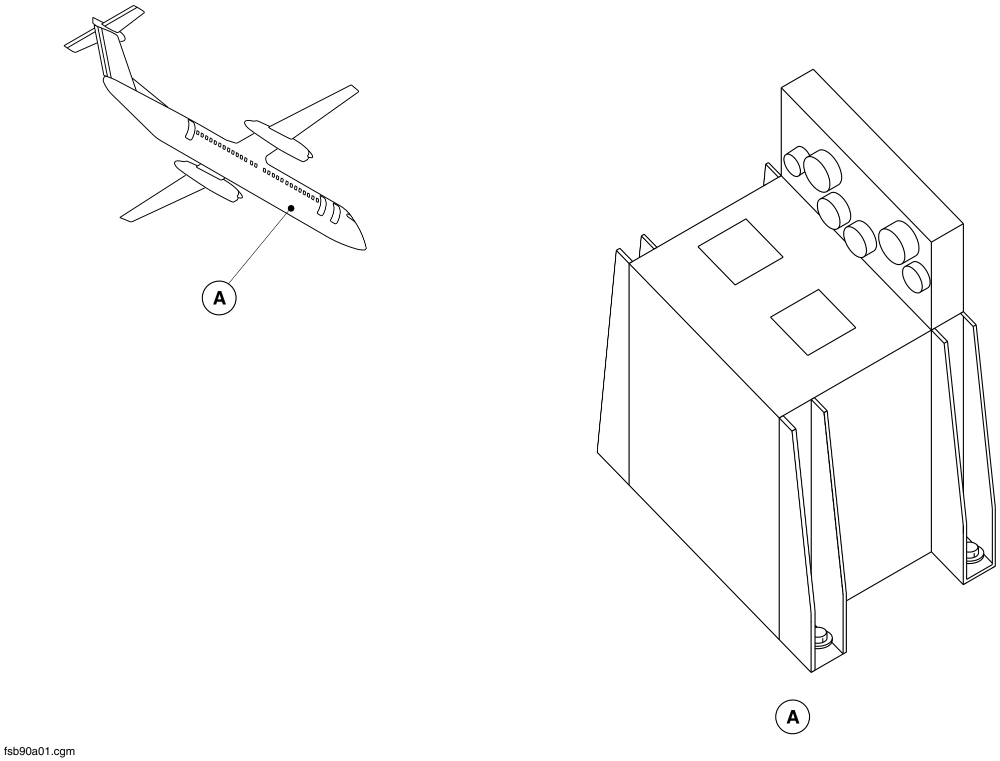
Zone Supply Overtemperature Switches (21–61–16)

Refer to Figures 13 and 14.

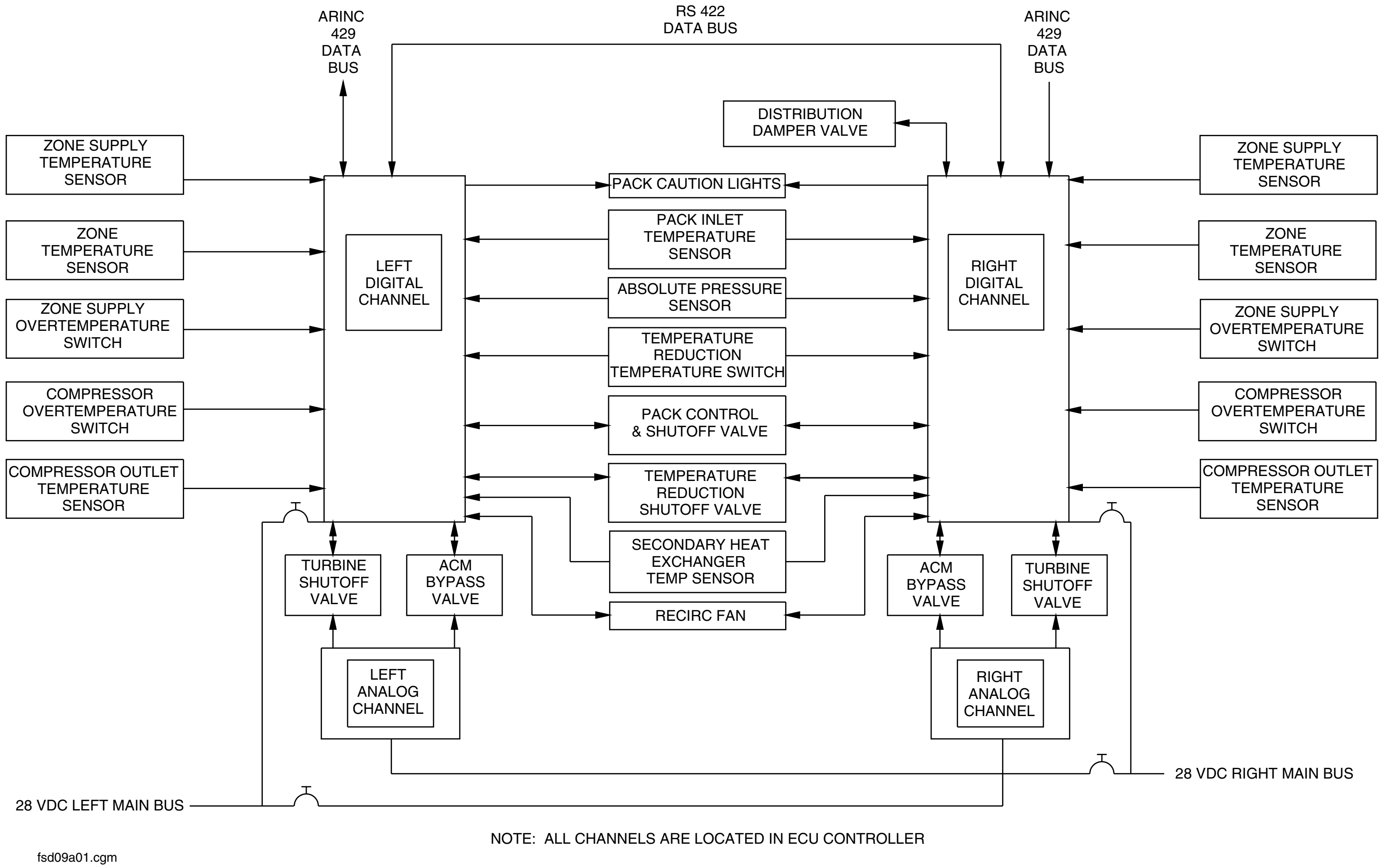
The zone supply overtemperature switches are installed in the cabin and flight compartment supply ducts under the aft baggage compartment floor (Stn X 765). These ducts supply conditioned air from the air conditioning system to the flight and cabin compartments (refer to SDS 21–21–00 and 21–22–00).

Refer to Figure 5.

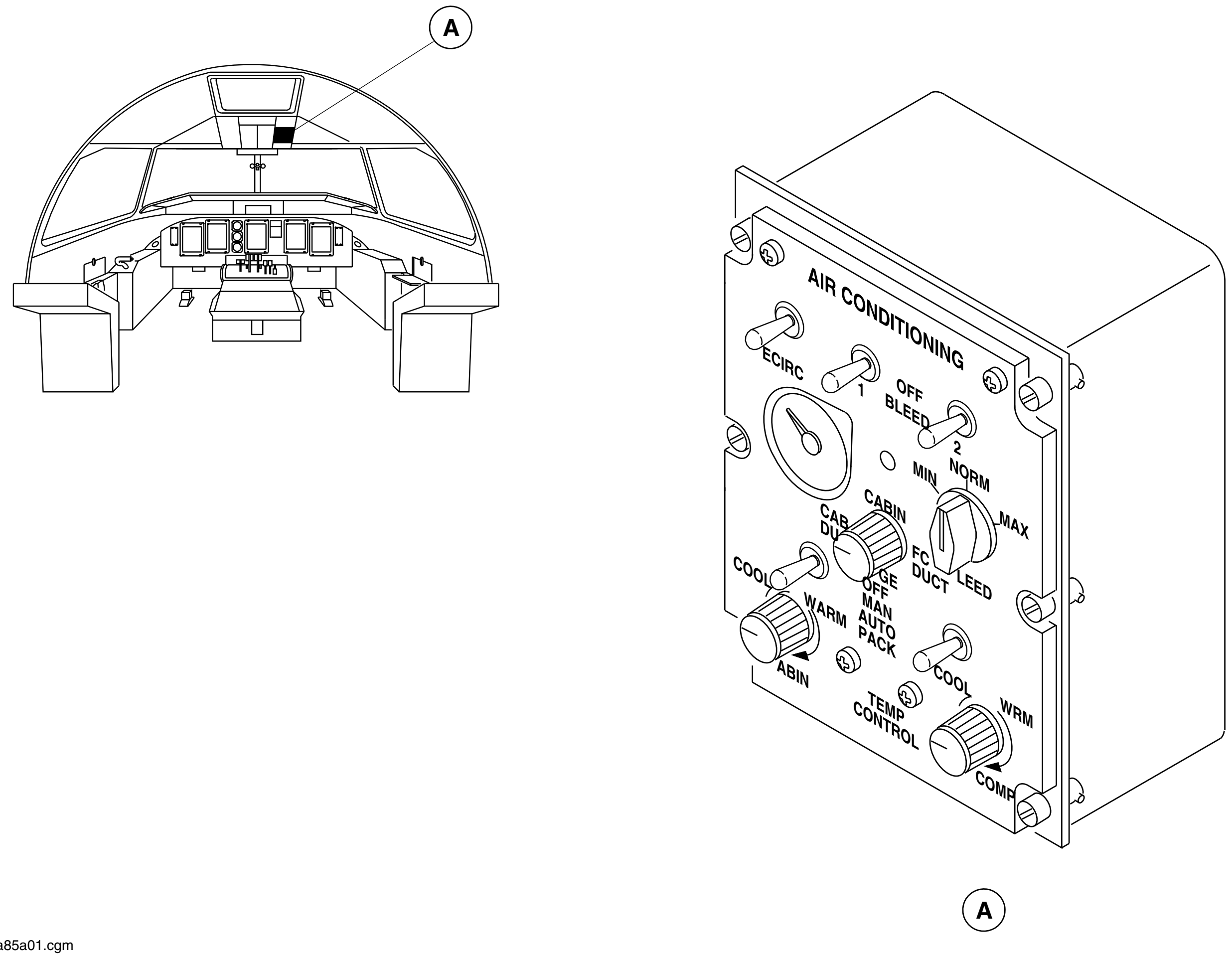
The zone supply overtemperature switches send a discrete signal to the ECU if the temperature in either supply duct is greater than 190 °F (87.8 °C). This signal causes the ECU to turn on the related CABIN DUCT HOT or FLT COMP DUCT HOT caution light on the caution and warning panel. The ECU also shuts off the related air conditioning pack (by closing the turbine and ACM bypass shutoff valves).



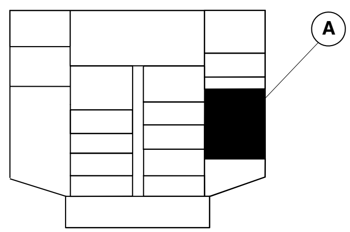
ELECTRONIC CONTROL UNIT (ECU) LOCATOR
Figure 1



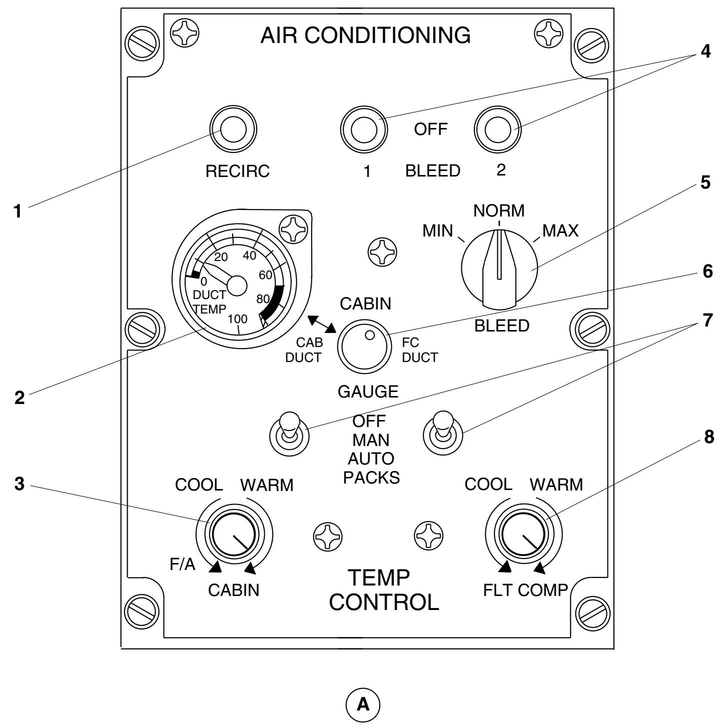
ELECTRONIC CONTROL UNIT (ECU) BLOCK DIAGRAM / AIR CONDITIONING
Figure 2



AIR CONDITIONING CONTROL PANEL LOCATOR
Figure 3



OVERHEAD CONSOLE



LEGEND

1. Recirculation Fan Switch.
2. Ducts Temperature Gauge.
3. Cabin Temperature Selector.
4. Bleed Control Switches.
5. Bleed Control Selector.
6. Ducts Temperature Gauge Switch.
7. Packs Control Switches.
8. Flight Compartment Temperature Selector.

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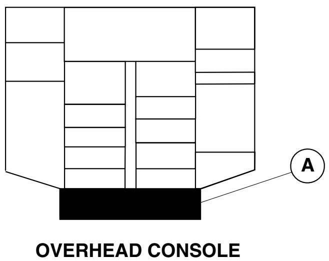
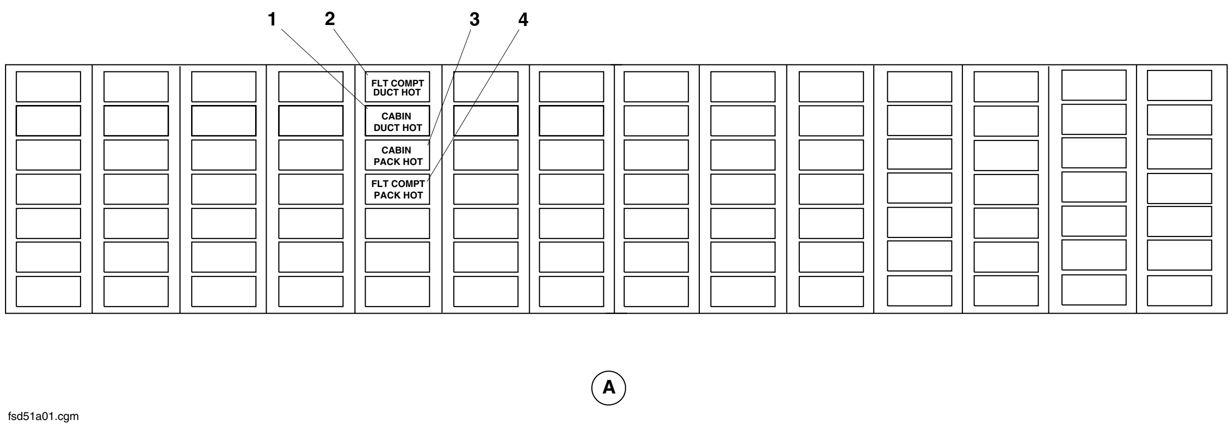
AIR CONDITIONING CONTROL PANEL DETAIL
Figure 4

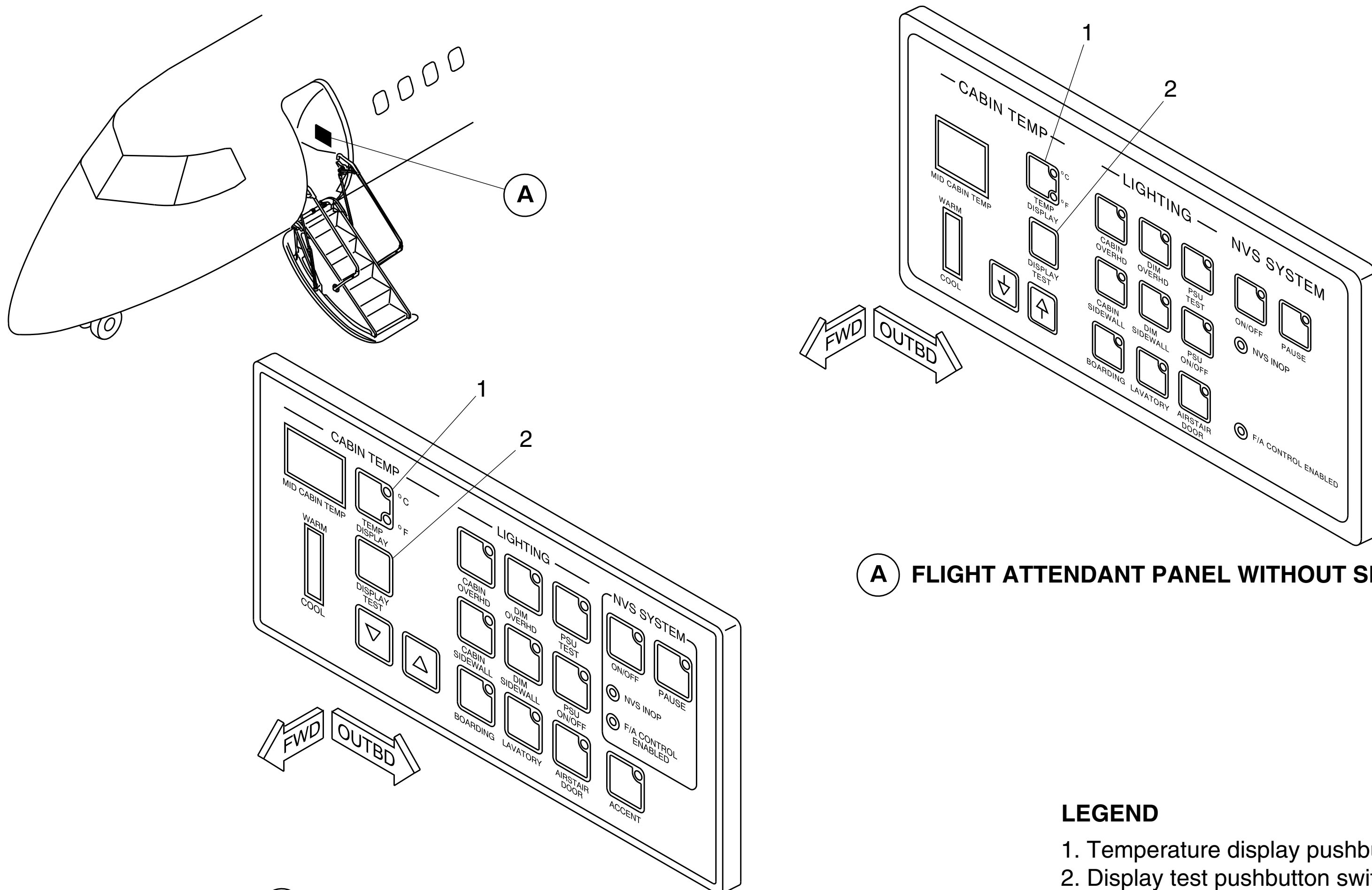


LEGEND

1. Cabin Supply Duct Hot (Amber).
2. Flight Compartment Supply Duct Hot (Amber).
3. Cabin Air Conditioning Pack Hot (Amber).
4. Flight Compartment Air Conditioning Pack Hot (Amber).



ECS CAUTION LIGHTS
Figure 5



A FLIGHT ATTENDANT PANEL WITHOUT SLEEP MODE

A FLIGHT ATTENDANT PANEL WITH SLEEP MODE

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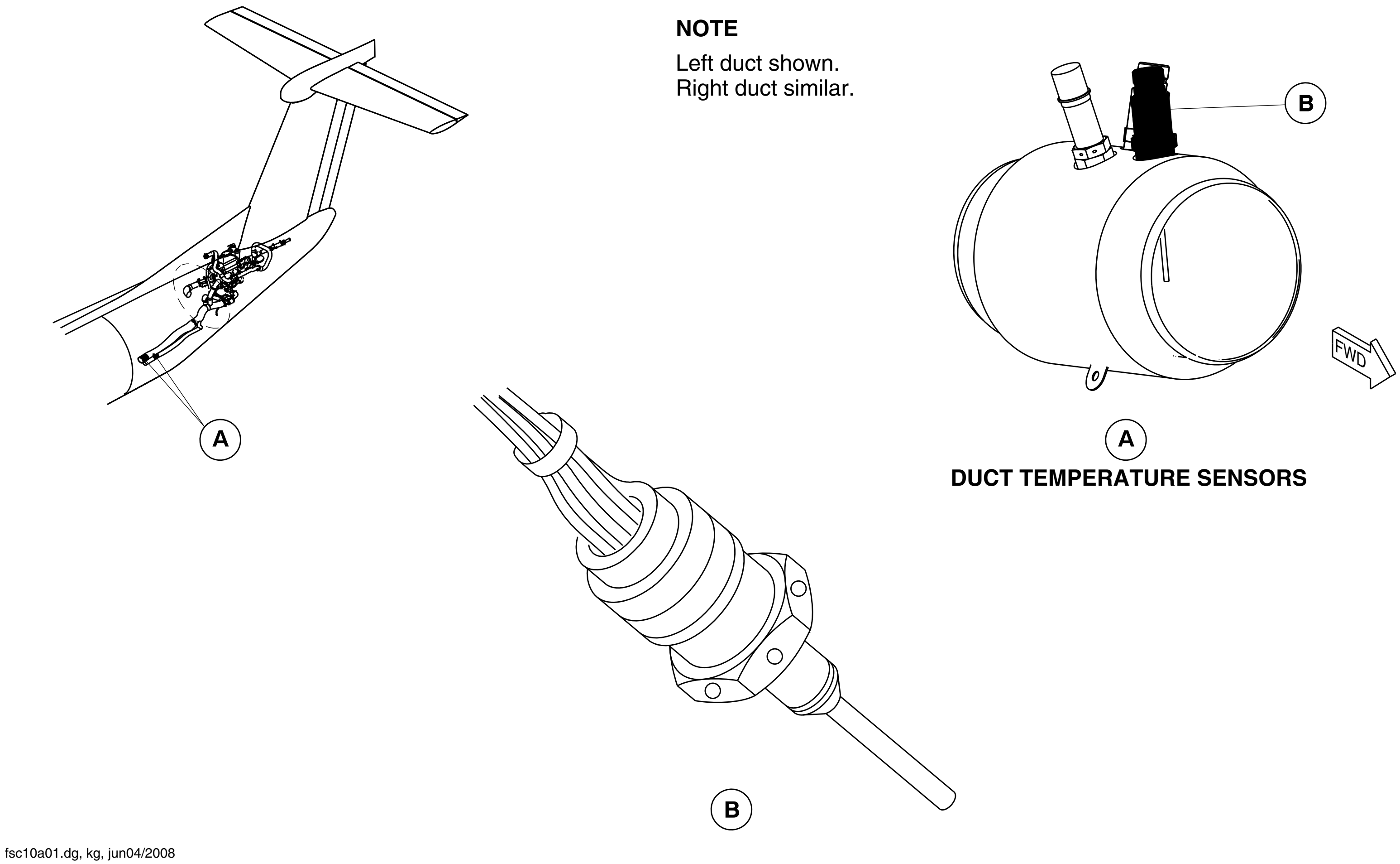
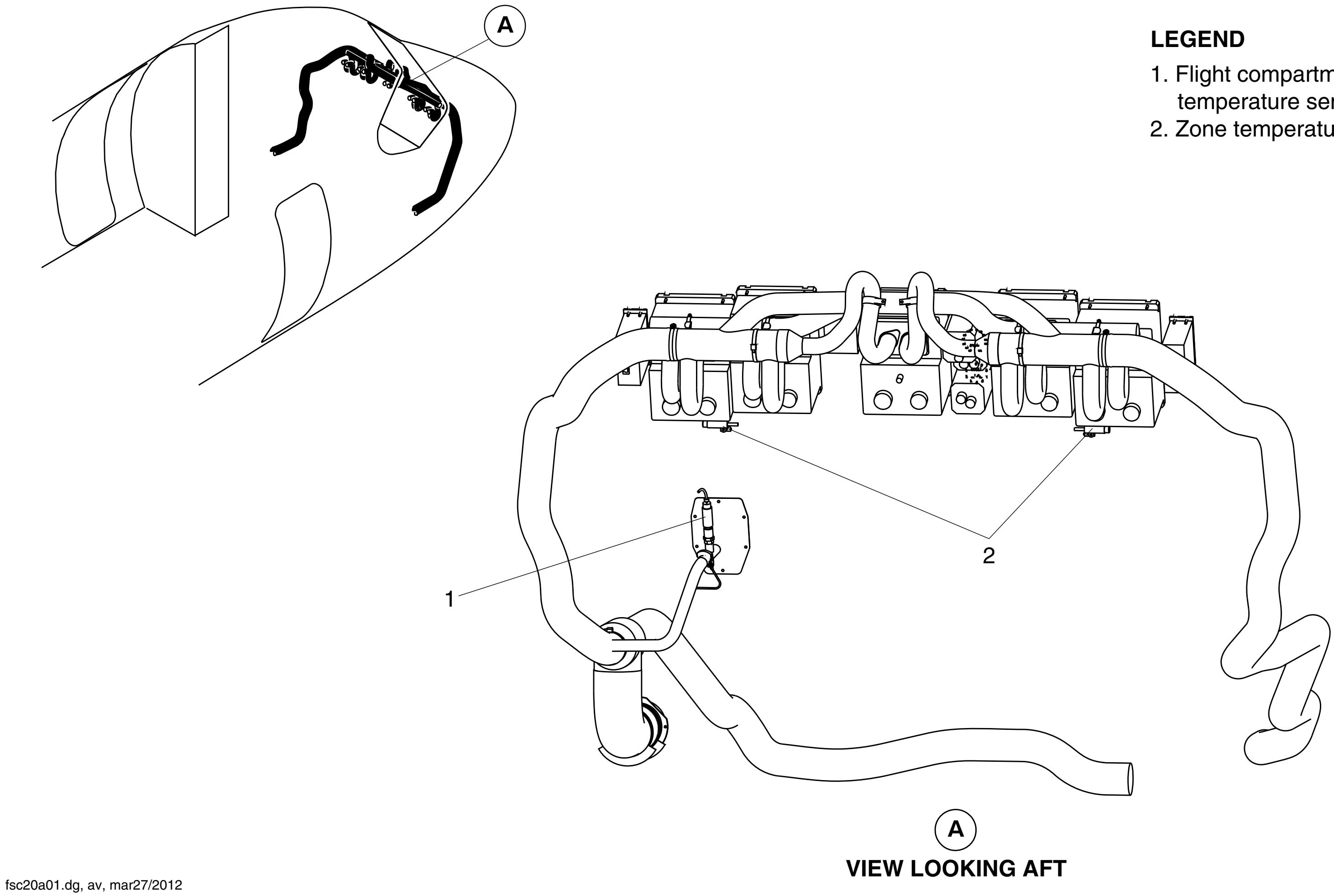
LEGEND

1. Temperature display pushbutton switch.
2. Display test pushbutton switch.

FLIGHT ATTENDANT CONTROL PANEL
Figure 6



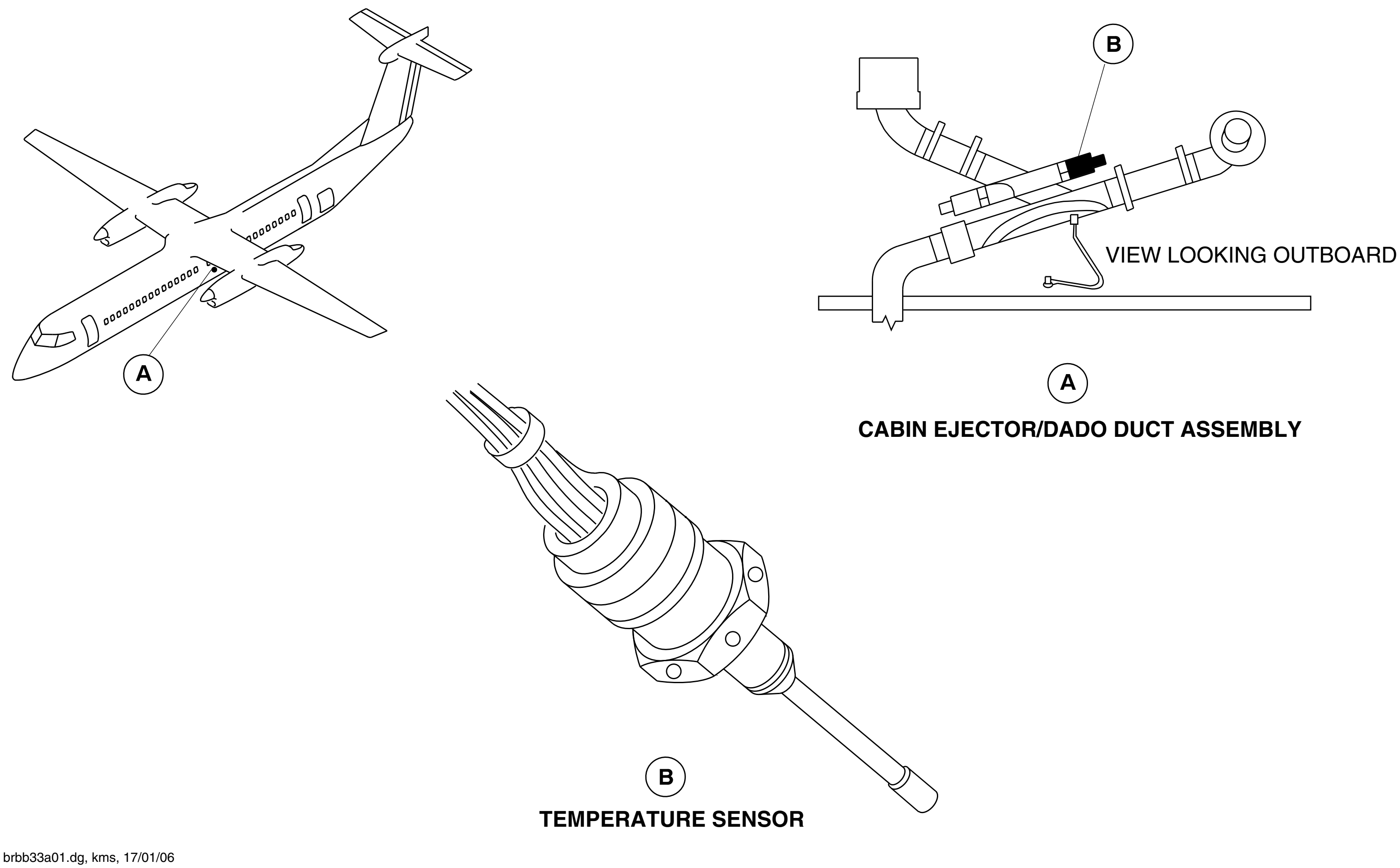
ZONE SUPPLY INDICATION SENSOR
Figure 7



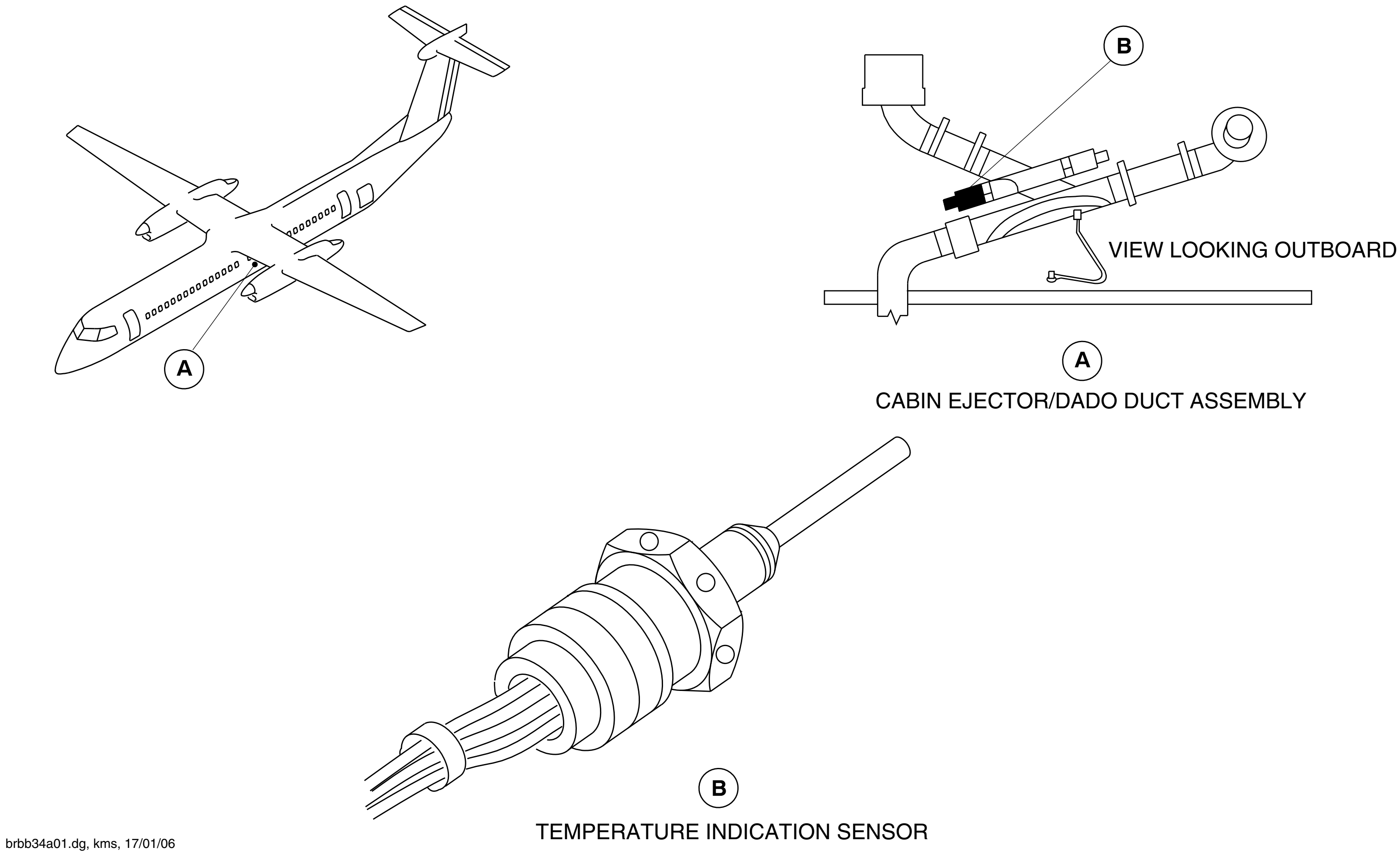
LEGEND

1. Flight compartment zone temperature sensor.
2. Zone temperature switches.

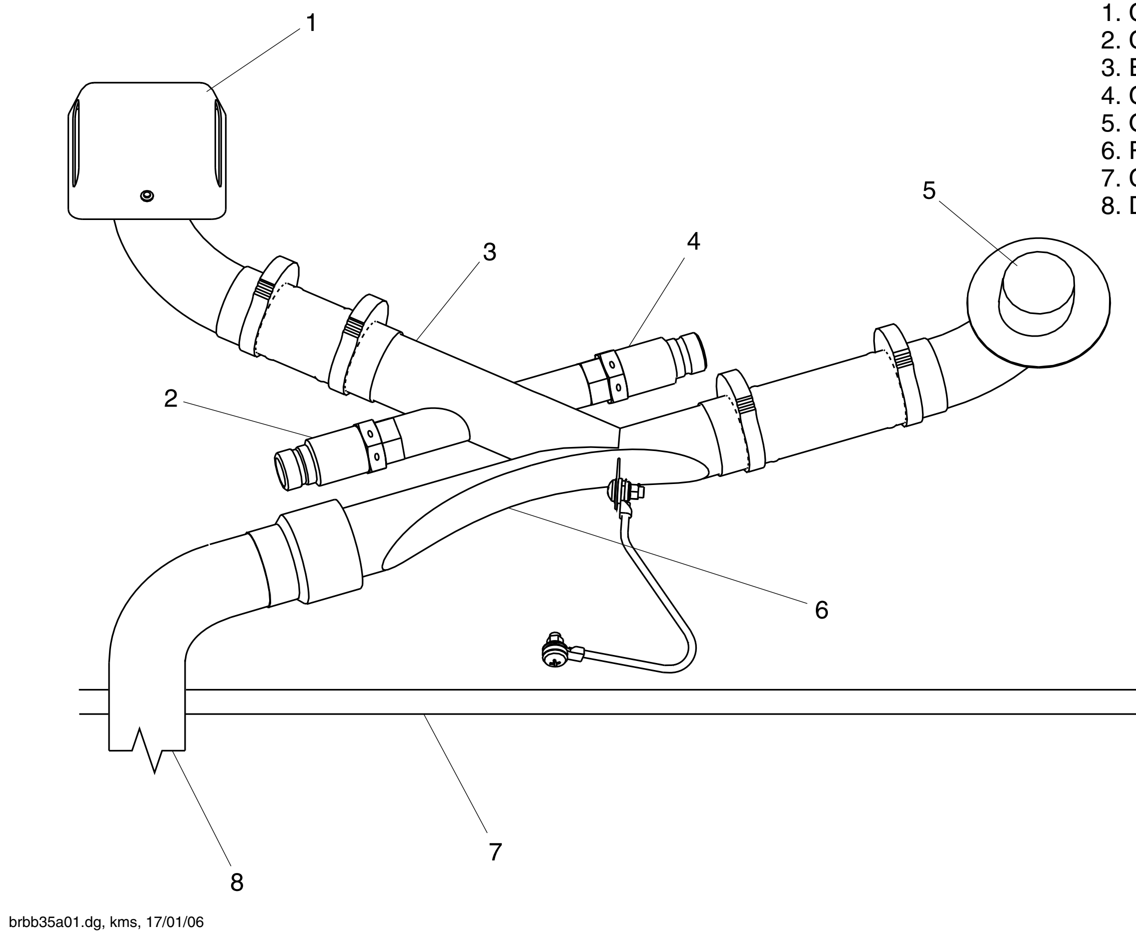
Flight Compartment Zone Temperature Sensor
Figure 8



Cabin Zone Temperature Sensor
Figure 9



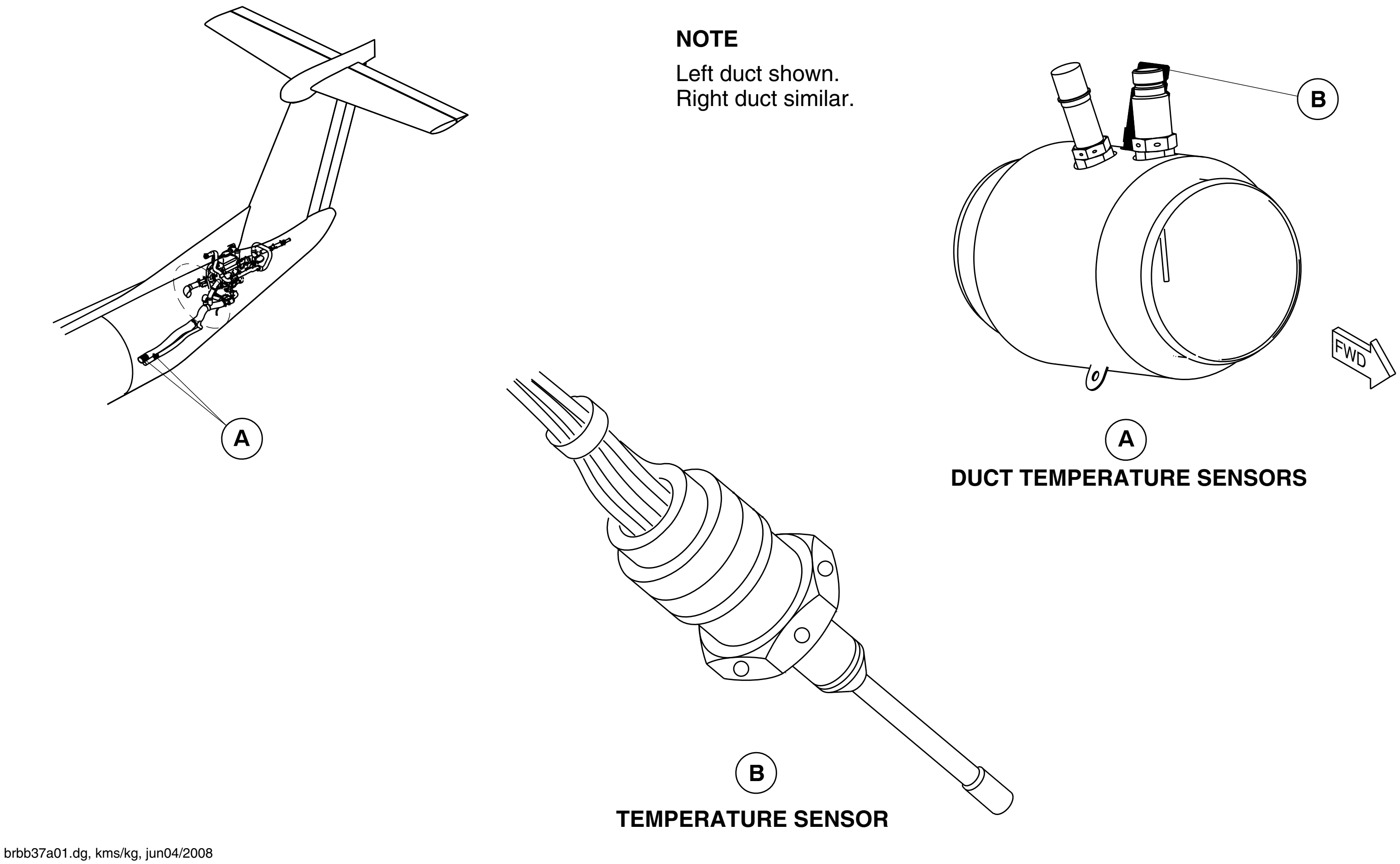
Cabin Zone Temperature Sensor
Figure 10



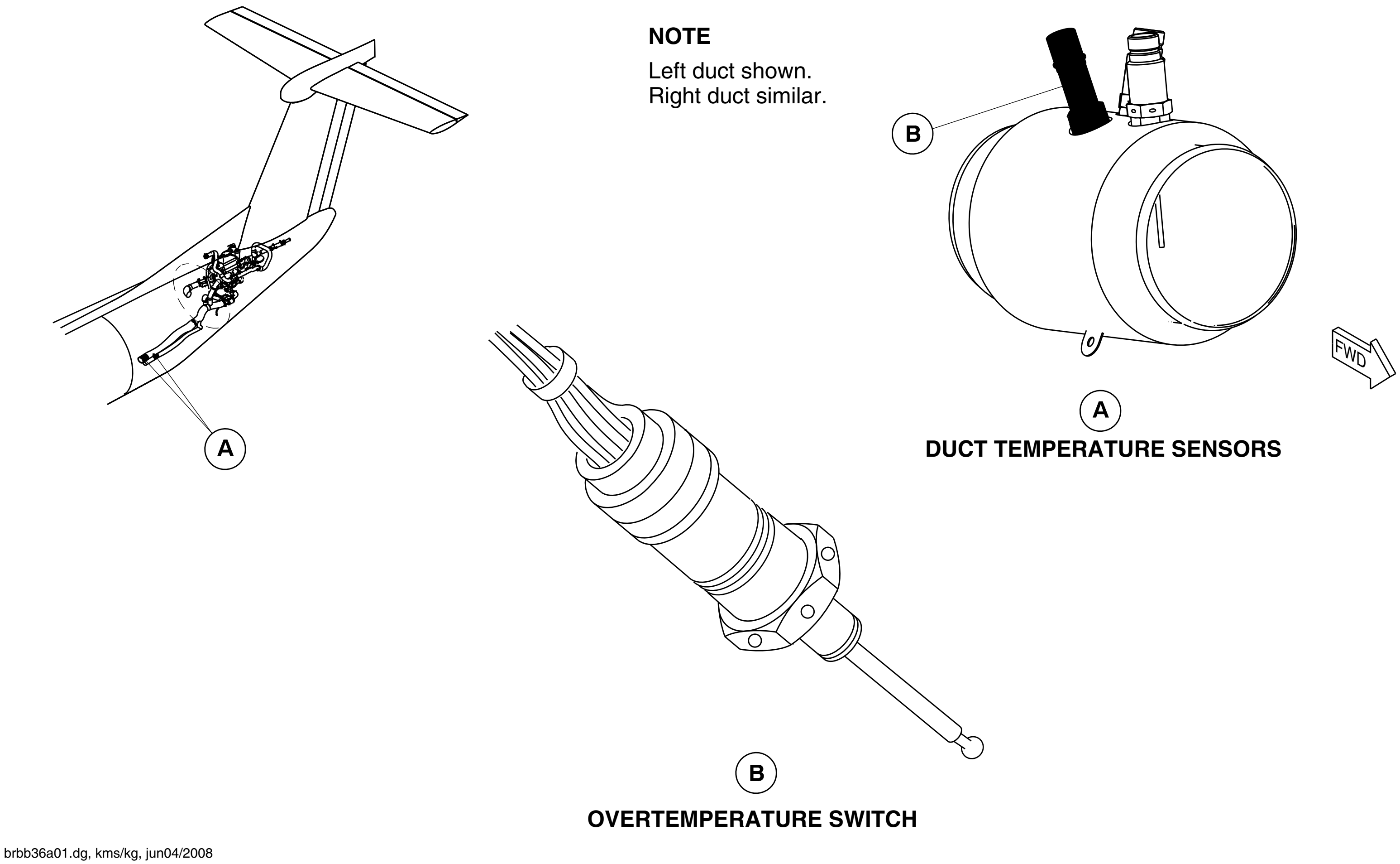
LEGEND

1. Cover (on dado panel).
2. Cabin zone temperature indication sensor.
3. Ejector.
4. Cabin zone temperature sensor.
5. Outlet to dado panel.
6. Restriction.
7. Cabin floor.
8. Dado supply air.

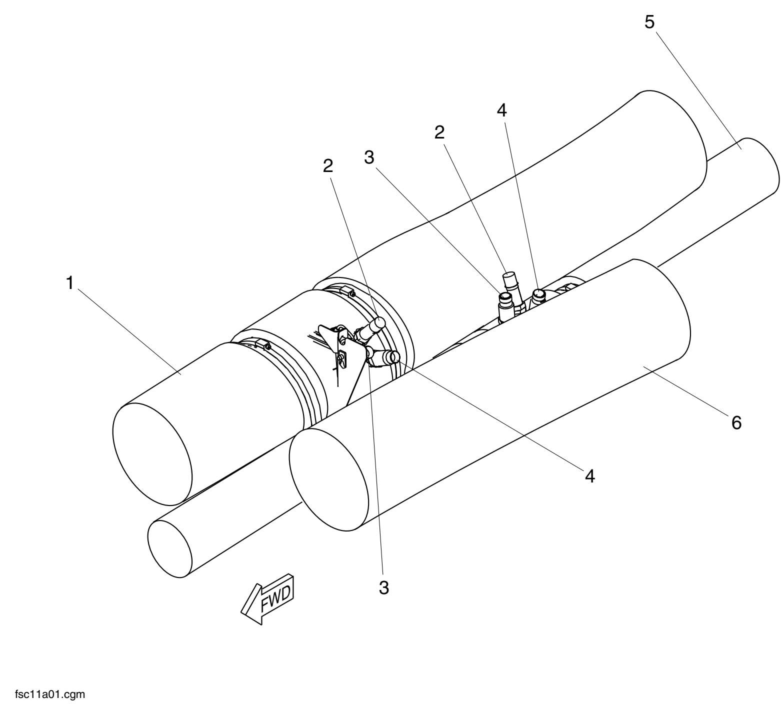
Cabin Zone Temperature Sensor/Indication Sensor Locator
Figure 11



Zone Supply Temperature Sensor
Figure 12



Zone Supply Overtemperature Switch
Figure 13



LEGEND

1. Cabin Supply Duct.
2. Zone Supply Overtemperature Switch.
3. Zone Supply Indication Sensor.
4. Zone Supply Temperature Sensor.
5. Flight Compartment Supply Duct.
6. Recirculation Duct.

ZONE SUPPLY TEMPERATURE SENSORS/SWITCHES LOCATOR
Figure 14

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